GSD RESEARCH MISSION PACKAGE

Computer Science 2025–2026 Sweep

→ **GSD Integration**

54 Papers | 6 Modules | v1.50 Citation Foundation

Convergent Discovery as Architectural Validation

> The spaces between independent intellectual currents are themselves load-bearing structures. When researchers with no line of communication arrive at the same patterns, the patterns are real.

Vision → Research → Mission

Full Pipeline Execution

Date: April 25, 2026 | Status: Ready for GSD Orchestrator

Activation Profile: Fleet | Estimated: 5–6 context windows across 3–4 sessions

Target Milestone: v1.50 Half A (Retrospective Citation Foundation)

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STAGE 1 — VISION DOCUMENT

1 CS25–26 Sweep → GSD — Vision Guide

Date: April 25, 2026

Status: Initial Vision / Ready for Wave Execution

Depends on: The 930-paper sweep deliverable (`gsd_paper_sweep_2025_2026.md`)

Context: Tibsfox's CS25 arxiv listing, deep-read against gsd-skill-creator architectural concerns

Target Milestone: v1.49.x → v1.50 (Unit-circle retrospective + proof companion)

1.1 Vision

In April 2026 a single arxiv listing of 930 computer science papers landed in the GSD project files. The question put to it was a literal one: *which of these papers apply to gsd-skill-creator and its functions?* What came back was unexpected. The corpus did not just supply techniques to bolt onto the system. It supplied something more important: **independent confirmation that GSD is on the load-bearing patterns**, written by researchers with no possible line of communication to the project, addressing problems in domains as far apart as proteomics, population genetics, surgical AI, and aerospace verification.

This mission systematizes that finding. It treats 54 priority papers as a structured corpus to be read carefully, recorded in adoption decision records, woven into the GSD documentation, and fed forward into the v1.50 unit-circle retrospective and proof companion. The mission is not a literature review. It is the citation foundation for half A of v1.50 (*retrospective of all 49 prior milestones*) and a concrete adoption pipeline for roughly 12 drop-in techniques whose absorption will improve the system in measurable ways before half B begins.

The intellectual through-line is the **spaces between** principle expressed in a new form. When four research groups working on completely different problems — scientific workflow automation, proteomics governance, context-engineering theory, and meta-learning over agent harnesses — independently arrive at the same architectural pattern (markdown-as-skill, hard-constraint-overrides, bounded-channel-as-tape, two-level-meta-evolution), the threshold between their independent work *is* the load-bearing structure. The convergence *is* the validation. Recognizing that pattern, naming it, and building citations and adoption around it is the work of this mission.

The Amiga Principle applies one level up: instead of building all GSD's invariants from scratch and arguing for them in isolation, this mission inherits external proofs. Specialized architecture (chipset, bounded-learning, fresh-context-subagents, CAPCOM gates, DACP fidelity levels) is now externally cited, not just internally asserted. The leverage is in recognizing the convergence, not in originating the patterns alone.

1.2 Problem Statement

1. **The sweep deliverable identifies adoption-ready techniques but does not commit to them.** 47–54 papers were ranked as relevant, but each adoption requires its own decision: target subsystem, target milestone, integration spec, validation test, rollback criterion. Without ADRs (Adoption Decision Records), the deliverable is a reading list, not a system-impacting artefact.
2. **The convergent-discovery findings need formal citation infrastructure.** GSD documents currently assert the bounded-tape framing, the override-the-prompt invariant pattern, the two-level meta-learning structure, and the skill-as-markdown pattern as internal design choices. They are now externally cited patterns. Without formal BibTeX entries and explicit cross-references in CLAUDE.md and the College of Knowledge departments, this validation is not visible to readers of GSD.
3. **Several critical operational findings require near-immediate engineering action.** Tool Attention quantifies the MCP Tax at 10–60K tokens per turn. Omission Constraints Decay shows prohibitions decay 73% → 33% by turn 16 in production-sized contexts. AgentDoG provides a where/how/why taxonomy ready for direct adoption. These are not citation concerns; they are reasons to update the planning-bridge MCP server, the Safety Warden's BLOCK schema, and the safety test harness this quarter.
4. **The DACP work needs a comparative benchmark methodology.** Two papers (Empirical Comparison of Agent Communication Protocols, DiffMAS) directly target the problem space DACP addresses, the former with a five-axis evaluation matrix, the latter with end-to-end-trained latent communication. DACP currently has no published comparator; with these citations it can be situated.
5. **The federated skill economy threat model is incomplete.** The Black-Box Skill Stealing Attack paper is the first empirical study of attacks on skill marketplaces (90,368 free skills published; $100K+ creator earnings). The DoltHub contribution roadmap currently does not reflect this threat model.

1.3 Core Concept

> **Six modules. One mission.** Read each paper with full context, produce one Adoption Decision Record per paper, synthesize cross-module relationships, and emit a v1.50-citable bibliography plus a concrete engineering action queue.

The mission decomposes the corpus along architectural lines, not topical ones. The boundaries between modules echo the boundaries between GSD subsystems. This means the cross-module synthesis at the end is not academic stitching — it is exactly the integration work that has to happen for the techniques to land in the running system.

1.3.1 Mission Architecture

```
                WAVE 0: FOUNDATION (Sequential)
      +-----------------------------------------------------+
      | Paper acquisition pipeline (arXiv abstracts + ID  |
      | validation + BibTeX skeleton). ADR template.      |
      | Citation index. Source bibliography scaffold.     |
      +-----------------------------------------------------+
                            |
                            v
                WAVE 1: PARALLEL RESEARCH (3 tracks)
      +---------------------+ +-----------------------+
      | TRACK A             | | TRACK B               |
      | M1 Foundations (4)  | | M3 Safety/MCP (8)     |
      | M2 Bounded/Silicon  | | M4 Multi-agent/       |
      |    (10)             | |    DACP/Chipset (11)  |
      +---------------------+ +-----------------------+
                            |
                            v
                     +------+------+
                     | TRACK C     |
                     | M5 Verif (8)|
                     | M6 Memory + |
                     |   Cult+Curr |
                     |   (13)      |
                     +-------------+
                            |
                            v
                WAVE 2: SYNTHESIS (Sequential, Opus)
      +-----------------------------------------------------+
      | Cross-module relationship matrix (20 connections) |
      | Convergent-discovery validation report            |
      | GSD architectural impact analysis                 |
      | Drop-in technique integration specs (12+)         |
      +-----------------------------------------------------+
                            |
                            v
                WAVE 3: PUBLICATION + VERIFICATION
      +-----------------------------------------------------+
      | Final mission package PDF (this document)         |
      | BibTeX bibliography for v1.50                     |
      | Test harness updates for adopted techniques       |
      | CLAUDE.md / SKILL.md additions                    |
      | Verification matrix (criterion to test mapping)   |
      +-----------------------------------------------------+
```

1.4 Research Modules

1.4.1 Module 1 — Architectural Foundations & Convergent Discovery (4 papers)

The four convergent-discovery papers from the sweep deliverable. These are read first because all subsequent module work depends on understanding what these pa-

pers validate. Output: a foundation chapter that grounds every subsequent ADR in formal external citation, plus four BibTeX entries that become the most-cited works in v1.50 retrospective Half A.

Papers: 2604.21910 (Scientific Workflow), 2604.21744 (GROUNDINGmd), 2604.20874 (Root Theorem), 2604.21003 (Last Harness).

1.4.2 Module 2 — Bounded Learning & Silicon Layer LoRA (10 papers)

Five bounded-learning papers (AEL, Retrofit, FSFM, SCM, Fine-Tuning Regimes) plus five LoRA/PEFT papers (Separable Expert, LoRA Redux, GiVA, HyperAdapt, Teacher-Guided MoE). These together cover the bounded-learning policy formalization (parameter-space) and the Silicon Layer's adapter pipeline. Output: a formal mathematical statement of the 20% rule in low-rank-and-sparse-subspace terms, plus benchmark plan for GiVA/HyperAdapt on the RTX 4060 Ti reference.

1.4.3 Module 3 — Safety Warden, CAPCOM & MCP Security (8 papers)

Six safety/CAPCOM papers (AgentDoG, Omission Constraints Decay, MCP Pitfall Lab, Breaking MCP, Cross-Session Threats, Five-Principle Quality Gaps), plus CHAI (critique-based oversight) and the Specification Trap (philosophical foundation for open-and-responsive design). Output: an upgrade spec for the Safety Warden's BLOCK schema (where/how/what taxonomy), an STD measurement protocol, and a security test harness for the planning-bridge MCP.

1.4.4 Module 4 — Multi-Agent, DACP & Chipset Architecture (11 papers)

Strategic Heterogeneous (3+1), DiffMAS, Empirical Comparison of Communication Protocols, HiCrew, PosterForest, MATRAG, SARA, AgenticQwen, Distillation-Induced Similarity (RPS/AGS), Tool Attention, Tree Training. Output: external validation citations for the chipset architecture, a comparative-benchmark plan for DACP, an engineering ticket for lazy-schema-loading in planning-bridge.

1.4.5 Module 5 — Verification, Proof & Test Harness (8 papers)

DryRUN, Probable-to-Provable, SpecSyn, DSLTrans Cutoff, Doubly Saturated Ramsey (SAT+LLM+Lean), ATLAS, HWE-Bench, Orchid (ambiguity). This is the module most directly aligned with The Space Between's proof companion target and the ZFC compliance auditor. Output: a Lean/SAT/LLM workflow precedent citation, a test-generation pattern for skill validation, and a four-type ambiguity checklist for SKILL.md authoring.

1.4.6 Module 6 — Memory, Skills Economy, Cultural Sensitivity & Curriculum (13 papers)

The largest module, covering the broadest spread of GSD concerns: memory/context (Forget-then-Recall, Knowledge Capsules), the skill economy (Black-Box Skill Stealing), cultural sensitivity (LASA, Japanese Culture Bias, AFRILANGTUTOR, CARE, Privacy Simulacra, CI-Work), curriculum (RLAAR, SocraticKG, IRAP), and long-horizon agents (ColorBrowserAgent, Caesar). Output: cultural-sensitivity test additions, a federated-skill-economy threat model, and a curriculum design pattern for Teacher/Student/Support work.

1.5 Chipset Configuration

```
chipset:
  agnus:    # Coordination, judgment, cultural sensitivity
    model: claude-opus-4-7
    roles: [FLIGHT, INTEG, CRAFT-FOUND, CRAFT-BOUND,
            CRAFT-SAFE, CRAFT-VERIF, CRAFT-CULT, CAPCOM]
    share: ~30%

  denise:   # Implementation, document production, verification
    model: claude-sonnet-4-6
    roles: [PLAN, SCOUT-1, SCOUT-2,
            EXEC-1, EXEC-2, EXEC-3,
            CRAFT-AGENT, VERIFY, BUDGET, SURGEON]
    share: ~60%

  paula:    # Audit logs, scaffolding, low-judgment tasks
    model: claude-haiku-4-5
    roles: [LOG]
    share: ~10%
```

1.6 Scope Boundaries

1.6.1 In Scope (v1.0)

- All 54 priority papers from the sweep deliverable receive an Adoption Decision Record (adopt / track / reject) with rationale, target subsystem, and target milestone.
- The four convergent-discovery papers each receive a deep reading and an explicit citation target in GSD documentation.
- The 12 high-priority drop-in technique papers each receive a one-page integration specification with engineering tickets.
- Cross-module relationship matrix with at least 20 documented inter-paper connections.
- BibTeX bibliography file (`cs25-26-sweep.bib`) for v1.50 citation infrastructure.
- Specific updates to CLAUDE.md and the affected SKILL.md files, drafted as suggested edits.
- Updates to the safety test harness reflecting the AgentDoG taxonomy and Safe Turn Depth measurement.
- Verification matrix mapping every success criterion to at least one test.

1.6.2 Out of Scope

- Implementation of any drop-in technique beyond engineering specification (handed to gsd-skill-creator separately).
- Re-running the original 930-paper sweep (the sweep deliverable is an authoritative input).
- Coverage of papers not in the original sweep corpus.

- Theoretical extensions to the Root Theorem or other foundational papers — only application work.
- Half B of v1.50 (proof companion to The Space Between) — this mission feeds Half A.
- Any contribution to upstream gsd-build/get-shit-done repositories — separate mission.

1.7 Success Criteria

1. Every one of the 54 priority papers has an ADR (id, title, themes, key idea, GSD application, decision, target subsystem, target milestone, validation test).
2. The four convergent-discovery papers each have a citation target identified in a specific GSD document (CLAUDE.md, SKILL.md, or a College of Knowledge department).
3. Every *adopt*-decision paper (estimated 12+) has a one-page integration specification including engineering ticket text.
4. Cross-module relationship matrix documents at least 15 inter-paper connections; targets 20+.
5. The BibTeX file `cs25-26-sweep.bib` compiles cleanly and contains entries for all 54 papers.
6. AgentDoG's where/how/what taxonomy appears as a code-ready schema in Module 3's deliverable.
7. Tool Attention's lazy-schema-loading pattern appears as a near-implementable design note in Module 4's deliverable, with token-budget projections.
8. Black-Box Skill Stealing's attack taxonomy is encoded into a draft DoltHub threat model document.
9. The Five-Principle Quality Gaps framework appears as a draft skill-creator linter check spec in Module 3's deliverable.
10. The Worker / Evaluator / Evolution agent split from "The Last Harness" is documented as the formal naming convention for skill-creator's internal subagents.
11. Per-model Safe Turn Depth measurement protocol exists as a code-ready test pattern.
12. Final mission PDF compiles in three XeLaTeX passes without errors and is delivered alongside the .tex source and an index.html landing page.

1.8 Relationship to Other Documents

Document	Relationship	Direction
gsd_paper_sweep_2025_	Authoritative input; provides scoring, full reviews, and the action list this mission expands	Input
The Space Between (proof companion)	Module 5 deliverables feed directly into v1.50 Half B's chapter validation	Output
The Hundred Voices	Module 6's Privacy Simulacra paper offers cross-pollination with the authorial-voice taxonomy	Cross-pollinate
Fox Infrastructure Group plan	Module 6's CI-Work + Separable Expert Architecture inform multi-tenant deployment patterns	Output
GSD Chipset Architecture vision	Module 4's Strategic Heterogeneous + AGS/RPS provide external validation citations	Output
GSD Silicon Layer spec	Module 2's LoRA-redux survey + GiVA + HyperAdapt provide benchmark targets for RTX 4060 Ti	Output
GSD DACP work (v1.49.x)	Module 4's Empirical Comparison + DiffMAS provide comparative-benchmark methodology	Output
gsd-mission-management-roles.md	The Worker/Evaluator/Evolution split adopted as formal naming convention	Update
CLAUDE.md	New foundational citations from Module 1; new linter checks from Module 3	Update

1.9 The Through-Line

The Amiga's custom chipset did not improve the machine because it was clever in isolation. Agnus, Denise, and Paula were load-bearing because they hit the right specialization *at the right boundary* — splitting work where the seam was already there in the problem domain. The pattern reads from the schematic.

So with this corpus. Forty-seven independent groups did not arrive at GSD's bounded-channel framing because they collectively read the GSD architecture documents. They arrived there because the problem domain — LLM-agent systems operating across long horizons under bounded context, with adversarial pressure on prohibitions, with composable skill markets, with cultural-sensitivity requirements that change by region — has its own seams. The architecture pattern reads from the problem schematic, and several research groups read it independently.

This mission systematizes that recognition. It does not invent — it cites, it adopts, and it updates the documentation so that future readers of GSD see the patterns named with their full external lineage. The spaces between Tibsfox's eight months of work on bounded-learning and a Belgian author's recent paper on sleep-consolidated memory *are* the load-bearing structure. The mission is to surface that structure, name it, and build the engineering work on top of the named pattern, not on the unstated one.

The fox finds the trail by recognizing the seams the deer made walking it. The seams were always there. The fox just looks where the path is.

STAGE 2 — RESEARCH REFERENCE

2 CS25–26 Sweep Reference — 54 Papers in Detail

2.1 How to Use This Document

Each paper appears in its assigned module section with a uniform four-part structure: **arXiv ID** (citable identifier), **Themes** (the GSD architectural concerns the paper touches), **Key idea** (the paper's contribution in 2–3 sentences), and **GSD application** (the concrete adoption guidance, naming target subsystems and milestones). Papers within each module are ordered by adoption priority, not by score — the most directly actionable papers come first.

Source organizations and venues are arXiv preprint server (`https://arxiv.org`). The original sweep corpus is the `Computer_Science25.html` listing dated April 25, 2026, containing 930 papers. Selection methodology and theme weights are documented in `gsd_paper_sweep_2025_2026.md` §5.

Citation format: arXiv IDs are cited as `YYMM.NNNNN` throughout. Each paper resolves to `https://arxiv.org/abs/<id>`.

2.2 Module 1 — Architectural Foundations & Convergent Discovery

The four papers in this module independently arrive at GSD's load-bearing patterns. They are read first because every subsequent module's adoption decisions are stronger when grounded in this external validation.

2.2.1 `2604.21910` — From Research Question to Scientific Workflow

Themes: — plan-decomposition, knowledge-as-skill, deterministic-layer-confinement.

Key idea. Three-layer agentic architecture for population-genetics workflow generation: a *semantic layer* (LLM extracts structured intent from natural language), a *deterministic layer* (validated DAG generators produce reproducible workflows), and a *knowledge layer* (domain experts author markdown "Skills" encoding vocabulary mappings, parameter constraints, and optimization strategies). LLM non-determinism is confined to intent extraction; identical intents always produce identical workflows. Skills raise full-match intent accuracy from 44% to 83%; skill-driven deferred workflow generation cuts data transfer 92%.

GSD application. This is the cleanest published external description of the GSD vision-to-mission → research-mission-generator → gsd-skill-creator pipeline. The *deterministic-layer confinement* principle is the formal justification for DACP's fidelity-level system. Cite as a foundational work in v1.50 retrospective Half A. Adopt the explicit three-layer naming (*semantic / deterministic / knowledge*) in the College of Knowledge documentation as the canonical architecture description. **Adoption: ADOPT (citation + terminology).** Target: CLAUDE.md, College-of-Knowledge README, v1.50 retrospective.

2.2.2 2604.21744 — GROUNDINGmd: Epistemic Grounding for Agentic Coding

Themes: — safety-warden, hard-constraints, override-the-prompt, governance-as-document.

Key idea. A community-governed, field-scoped epistemic grounding document with two override-everything classes: **Hard Constraints** (non-negotiable validity invariants empirically required for scientific correctness) and **Convention Parameters** (community-agreed defaults). These override all other contexts *including the user prompt*. The example domain is mass-spectrometry-based proteomics, but the pattern is general.

GSD application. This is essentially the CLAUDE.md + Safety-Warden-with-BLOCK pattern, discovered independently for proteomics. The *override-the-user-prompt* formulation is the cleanest external citation for why the Safety Warden's BLOCK authority on safety-critical tests is non-negotiable. Promote to a foundational citation in the College of Knowledge departmental governance pattern. The two-class taxonomy (Hard Constraints vs Convention Parameters) is directly importable into SKILL.md authoring guidance — every skill should distinguish its non-negotiables from its community defaults. **Adoption: ADOPT (citation + linter check).** Target: CLAUDE.md, every SKILL.md template, skill-creator linter (Module 3 has the linter spec).

2.2.3 2604.20874 — The Root Theorem of Context Engineering

Themes: — context-management, bounded-tape, formal-axioms, homeostatic-persistence.

Key idea. Formalizes context as a bounded lossy channel with two axioms (finite window, monotone quality decay with volume) and derives five consequences: **(1)** quality degrades monotonically with token volume, *independent of window size*; **(2)** signal and token-count are independent optimization variables; **(3)** gating must trigger on *fidelity thresholds, not capacity*; **(4)** only *accumulate–compress–rewrite–shed* homeostatic architectures sustain understanding indefinitely; **(5)** the compressor needs an *external verification gate* because it operates inside the channel it compresses. Demonstrated empirically with a 60+-session persistent architecture showing stable memory footprint.

GSD application. This is the formal derivation of the bounded-tape framing Tibsfox has been articulating informally. Consequence **(3)** is the derivation of CAPCOM gate policy (gate on fidelity, not on token count). Consequence **(4)** is the lifecycle for milestone retrospectives + feed-forward. Consequence **(5)** is the architectural reason fresh-context subagents need separate-context verification (not self-review). **Adoption: ADOPT (citation as theoretical foundation).** Target: The Space Between's information-theory chapter, CLAUDE.md design rationale, fresh-context-subagent docs, every CAPCOM-gate policy document.

2.2.4 2604.21003 — The Last Harness You'll Ever Build

Themes: — skill-learning, meta-evolution, agent-orchestration, harness-as-artefact.

Key idea. Two-level framework: at the inner level, the **Harness Evolution Loop** pairs a Worker Agent with an Evaluator Agent (which adversarially diagnoses failures and scores performance) and an Evolution Agent (which modifies the harness

based on full failure history). At the outer level, the **Meta-Evolution Loop** optimizes the evolution protocol itself across diverse tasks, learning a protocol that converges quickly on any new task. Stated goal: "automating the design of the automation itself."

GSD application. This is essentially an external description of gsd-skill-creator's purpose. The two-level structure (per-skill iteration vs cross-skill meta-protocol) maps onto skill-creator's six-step learning loop (per skill) and the bounded-learning policy (cross-skill, learning what kinds of corrections actually help). Adopt the Worker / Evaluator / Evolution agent split as the formal naming convention for skill-creator's internal subagents in `gsd-mission-management-roles.md`. The "adversarial Evaluator" role is a strong upgrade path for the current 3-correction-min rule — replace passive correction-counting with active adversarial probing. **Adoption: ADOPT (citation + naming convention + design upgrade).** Target: skill-creator subagent naming, bounded-learning policy doc, v1.50 retrospective.

2.3 Module 2 — Bounded Learning & Silicon Layer LoRA

Five papers cover the bounded-learning policy (parameter-space and behavior-space). Five cover LoRA / PEFT techniques applicable to the Silicon Layer adapter pipeline on the RTX 4060 Ti reference hardware. Together they give the bounded-learning policy a formal mathematical statement and an engineering target.

2.3.1 2604.21725 — AEL: Agent Evolving Learning for Open-Ended Environments

Themes: — bounded-learn, two-timescale, retrieval-policy-bandit, reflection-as-causal-injection.

Key idea. Two-timescale framework. **Fast timescale:** Thompson Sampling bandit selects which memory retrieval policy to apply each episode. **Slow timescale:** LLM-driven reflection diagnoses failure patterns and injects causal insights into the agent's decision prompt. On a 208-episode portfolio benchmark with five random seeds, AEL achieves Sharpe 2.13 ± 0.47, beating five published self-improving methods with the lowest variance among LLM approaches.

GSD application. The fast/slow split maps directly onto per-mission decision policy (fast) vs milestone retrospective (slow). The *bandit-over-retrieval-policies* idea is a strong addition to the auto-load step of skill-creator's six-step learning loop — instead of a single fixed retrieval policy, learn which policy to apply per skill type. **Adoption: ADOPT (technique).** Target: skill-creator auto-load step, College of Knowledge retrieval layer.

2.3.2 2511.11439 — Retrofit: Continual Learning with Controlled Forgetting

Themes: — bounded-learn, low-rank-sparse-subspace, parameter-merging, no-replay.

Key idea. Constrains parameter changes to *low-rank and sparse subspaces for approximate orthogonality*, then uses a confidence-guided arbitration mechanism to dynamically aggregate knowledge from a legacy-knowledge teacher and an emergent-knowledge teacher. No replay buffer required (security-domain constraint). On binary security classification under temporal drift, retention rises from 20.2% → 38.6% over CL baselines and exceeds the oracle upper bound on new data.

GSD application. The low-rank-and-sparse subspace constraint *is* the 20% rule expressed in parameter-space mathematics. Adopt as the formal derivation of bounded-update invariants in the v1.50 proof companion. Confidence-guided arbitration between legacy and emergent teachers is a clean pattern for handling skill versioning when an updated skill must coexist with a deployed predecessor for the cooldown window. **Adoption: ADOPT (formal foundation + technique).** Target: v1.50 proof companion (Module 5 cross-link), Silicon Layer LoRA pipeline, bounded-learning policy doc.

2.3.3 2604.20300 — FSFM: Selective Forgetting of Agent Memory

Themes: — bounded-learn, selective-forgetting, four-mode-taxonomy, hippocampal-inspired.

Key idea. Hippocampal-indexing + Ebbinghaus-curve framework. Four forgetting mechanisms: **passive decay-based**, **active deletion-based**, **safety-triggered**, and **adaptive reinforcement-based**. Reports +8.49% access efficiency, +29.2% signal-to-noise ratio, 100% security-content elimination.

GSD application. The four-class taxonomy is directly importable as the bounded-learning forgetting policy. The *safety-triggered* class is exactly the Safety Warden's role — formalize this connection. Use this paper's taxonomy to define what "shed" means in the Root Theorem's accumulate–compress–rewrite–shed cycle. **Adoption: ADOPT (taxonomy).** Target: bounded-learning policy doc, Safety Warden integration spec.

2.3.4 2604.20943 — SCM: Sleep-Consolidated Memory

Themes: — context-mgmt, biological-memory, NREM-REM-consolidation, value-based-forgetting.

Key idea. Memory architecture inspired by neuroscience: limited working memory + multi-dimensional importance tagging + offline sleep-stage consolidation with distinct **NREM and REM phases** + intentional value-based forgetting + computational self-model for introspection. Achieves perfect 10-turn recall while reducing memory noise 90.9%. Search latency under 1ms with hundreds of stored concepts.

GSD application. The NREM/REM split is a possible refactor of GSD's wave architecture. Between-wave consolidation can use NREM-style processing (replay important traces, prune redundant), while between-mission consolidation uses REM-style (free-association, novel cross-skill connections). Worth a focused experiment in the College of Knowledge as a memory-consolidation upgrade. **Adoption: TRACK (pilot experiment).** Target: College of Knowledge mind-body department experiment, possible v1.51 design.

2.3.5 2604.21927 — Fine-Tuning Regimes Define Distinct CL Problems

Themes: — bounded-learn, evaluation-methodology, regime-dependence.

Key idea. Empirical: changing the trainable-parameter subspace (depth) *changes the relative ranking of CL methods* across five benchmarks (MNIST, Fashion MNIST, KMNIST, QMNIST, CIFAR-100) and 11 task orders. Method choice is *not invariant* across regimes; deeper adaptation regimes show larger update magnitudes and more forgetting.

GSD application. An important constraint on the Silicon Layer: choice of LoRA rank/depth is itself an evaluation variable that must be fixed before comparing bounded-learning policies. Without this control, A/B comparisons of bounded-update strategies are unsound. Add a *regime-fixed comparison* requirement to the Silicon Layer test harness. **Adoption: ADOPT (methodological constraint).** Target: Silicon Layer test harness spec, bounded-learning policy validation protocol.

2.3.6 2604.21571 — Separable Expert Architecture: Composable Adapters and Deletable User Proxies

Themes: — lora-peft, privacy-preservation, deterministic-unlearning, multi-tenant.

Key idea. Three-layer architecture: *static base model*, *composable domain-expert LoRA adapters* (no user data), and *per-user proxy artefacts* whose deletion constitutes deterministic unlearning. Verified on Phi-3.5-mini and Llama-3.1-8B: KL divergence $\sim$0.21 nats after proxy removal, 82–89% verification pass rate, near-zero cross-user contamination. Mitigates model inversion, membership inference, and training-data extraction by construction.

GSD application. The cleanest privacy-and-governance architecture seen for the Silicon Layer. Adopt the domain-expert/per-user adapter split as the formal structure for FoxFiber/FoxCompute multi-tenant deployments. *Deterministic unlearning by file deletion* aligns naturally with the GSD filesystem-as-database principle — a deleted user proxy file *is* the unlearning act. **Adoption: ADOPT (architecture).** Target: Silicon Layer spec, Fox Infrastructure Group multi-tenant design, FoxData privacy whitepaper.

2.3.7 2604.21905 — Low-Rank Adaptation Redux for Large Models

Themes: — lora-peft, signal-processing-survey.

Key idea. Survey reviewing LoRA through a signal-processing lens: SVD-based factorization, rank-augmentation constructions, cross-layer tensorization. Three axes: architectural design, efficient optimization, applications.

GSD application. Reference document for the Silicon Layer documentation. Useful for choosing rank configurations for the RTX 4060 Ti 8GB constraint and for explaining why specific rank choices are made. **Adoption: TRACK (reference).** Target: Silicon Layer documentation bibliography.

2.3.8 2604.21901 — GiVA: Gradient-Informed Bases for Vector-Based Adaptation

Themes: — lora-peft, parameter-efficiency, gradient-initialization.

Key idea. Vector-based adaptation matching LoRA performance with **8$\times$ lower rank requirements** via gradient-informed initialization. Training time comparable to LoRA. Evaluated on NLU, NLG, image classification.

GSD application. Direct candidate for memory-constrained LoRA training on the RTX 4060 Ti 8GB target — the 8$\times$ rank reduction translates directly to memory savings during training. Worth a dedicated benchmark experiment in the Silicon Layer test harness. **Adoption: ADOPT (benchmark candidate).** Target: Silicon Layer benchmark suite.

2.3.9 2509.18629 — HyperAdapt: Simple High-Rank Adaptation

Themes: — lora-peft, high-rank-with-low-params, diagonal-scaling.

Key idea. Row- and column-wise diagonal scaling induces a high-rank update with only $O(m+n)$ trainable parameters per $m \times n$ matrix. Theoretically establishes upper bound on update rank; empirically confirms high-rank transformations across model layers. Matches full-FT performance on GLUE / arithmetic / commonsense up to 14B parameters.

GSD application. Alternative low-parameter PEFT path. Worth comparing against GiVA in the Silicon Layer benchmark suite. The high-rank-with-O(m+n)-params property is attractive for adapter pooling experiments where many adapters must be loaded simultaneously. **Adoption: ADOPT (benchmark candidate).** Target: Silicon Layer benchmark suite (alongside GiVA).

2.3.10 2604.21330 — Teacher-Guided Routing for Sparse Vision MoE

Themes: — lora-peft, mixture-of-experts, teacher-student-routing.

Key idea. TGR-MoE stabilizes MoE router learning by deriving a teacher router from a pretrained dense teacher's intermediate representations and using its routing outputs as pseudo-supervision. Addresses the gradient-blocking problem (router only gets informative gradients through selected experts).

GSD application. Architecture-level analogy for the Opus-teaches-Sonnet chipset pattern. If GSD ever moves to a learned dispatch layer over chipset models (rather than rule-based), the teacher-supervised router pattern is the right starting point. Less immediately actionable than the LoRA papers above. **Adoption: TRACK (pattern reference).** Target: chipset architecture future-work notes.

2.4 Module 3 — Safety Warden, CAPCOM & MCP Security

Eight papers covering the Safety Warden's role, CAPCOM gate design, and the security boundary around the planning-bridge MCP server. Several of these contain operational findings (decay rates, attack taxonomies) that demand near-immediate engineering action.

2.4.1 2601.18491 — AgentDoG: Diagnostic Guardrail Framework

Themes: — safety-warden, three-d-taxonomy, sized-deployment, root-cause-diagnosis.

Key idea. Three-dimensional taxonomy categorizing agentic risks orthogonally by **source (where)**, **failure mode (how)**, and **consequence (what)**. Released in 4B / 7B / 8B parameter sizes (Qwen and Llama families). Provides root-cause diagnosis of unsafe and "seemingly safe but unreasonable" actions, not binary pass/fail labels. Achieves SOTA on the new ATBench benchmark.

GSD application. Direct candidate for the Safety Warden implementation tier. Adopt the where/how/what taxonomy as the formal BLOCK-decision schema — every BLOCK should record source, failure mode, and consequence in a structured format. The 4B/7B/8B sizing maps elegantly onto a Haiku-tier Safety Warden deployment that runs locally on each session without significant token cost. **Adoption: ADOPT (taxonomy + deployment pattern).** Target: Safety Warden BLOCK schema, planning-bridge sidecar deployment.

2.4.2 2604.20911 — Omission Constraints Decay While Commission Constraints Persist

Themes: — capcom, long-context, prohibition-decay, safe-turn-depth.

Key idea. 4,416-trial three-arm causal study across 12 models / 8 providers / 6 conversation depths. **Prohibition-type constraints (omission)** drop from 73% compliance at turn 5 to 33% at turn 16 while **requirement-type constraints (commission)** hold at 100%. This asymmetry is termed *Security-Recall Divergence (SRD)*. In token-matched padding controls, schema semantic content accounts for 62–100% of the dilution effect. Re-injecting constraints before the per-model **Safe Turn Depth (STD)** restores compliance without retraining. Production audit signals can show healthy commission compliance while omission has already failed — the failure is invisible to standard monitoring.

GSD application. Critical operational finding for CAPCOM gate design. The Safety Warden *must* re-inject omission constraints before each model's STD, not just at session start. Build STD measurement into the safety test harness as a per-model calibration step. The asymmetric monitoring failure (commission healthy, omission already collapsed) means current GSD audit logs need a second axis of measurement. **Adoption: ADOPT (operational protocol).** Target: Safety Warden re-injection protocol, safety test harness STD calibration, CAPCOM gate audit-log schema.

2.4.3 2604.21477 — MCP Pitfall Lab: Multi-Vector MCP Security

Themes: — mcp-security, six-attack-families, baseline-vs-hardened.

Key idea. Protocol-aware security testing framework operationalizing developer pitfalls as reproducible scenarios. Three workflow challenges (email, document, crypto) $\times$ six server variants (baseline + hardened) $\times$ three attack families (tool-metadata poisoning, puppet servers, multimodal image-to-tool chains). Static analyzer achieves F1=1.0 on four pitfall classes; recommended hardening eliminates all 29 Tier-1 findings.

GSD application. Direct adoption candidate for the planning-bridge MCP server's security test harness. The six-variant baseline-vs-hardened comparison structure is reusable as the standard regression test format for MCP-related changes. Adopt the four statically-checkable pitfall classes (P1, P2, P5, P6) as mandatory linter checks for any new MCP server in the GSD ecosystem. **Adoption: ADOPT (test framework).** Target: planning-bridge MCP test harness, MCP server lint rules.

2.4.4 2604.20994 — Breaking MCP with Function Hijacking Attacks

Themes: — mcp-security, function-hijacking, semantic-agnostic-attack.

Key idea. Function Hijacking Attack (FHA) manipulates the tool selection process to force invocation of a specific attacker-chosen function. Unlike prior attacks that rely on semantic context, FHA is *largely agnostic to context semantics* and *robust across function sets*. Trainable to produce universal adversarial inputs.

GSD application. Threat to model when designing planning-bridge access controls. The context-agnostic property means traditional intent-based filters ("does the user's question semantically match this tool?") will not catch FHA. This argues for cryptographic or schema-level binding of allowed tool sets per session, not just intent-based gating. Cross-link with MCP Pitfall Lab for the corresponding hardening tests. **Adop-**

tion: ADOPT (threat model). Target: planning-bridge access-control design, MCP-server threat model document.

2.4.5 2604.21131 — Cross-Session Threats: CSTM-Bench

Themes: — capcom, cross-session, kill-chain, memoryless-guardrail.

Key idea. 26 attack taxonomies classified by kill-chain stage and cross-session operation (*accumulate*, *compose*, *launder*, *inject_on_reader*). Memoryless guardrails (judging each message in isolation) fail because the aggregate carries the payload. Both session-bound judges and full-log correlators **lose** $\sim$50% **recall** moving from dilution to cross-session attacks.

GSD application. Directly applicable to GSD's multi-session work. Argue for a *persistent cross-session threat correlator* as a Safety Warden module — essentially a "diff against session N−1's threat profile" check at session start. The kill-chain taxonomy is reusable as the formal threat model schema. **Adoption: ADOPT (architecture + threat model).** Target: Safety Warden cross-session module spec, threat-model document.

2.4.6 2604.21090 — Structural Quality Gaps in AI Governance Prompts

Themes: — governance, five-principle-evaluation, computability-theory, structural-completeness.

Key idea. Five-principle evaluation framework grounded in computability theory, proof theory, and Bayesian epistemology. Applied to 34 publicly available governance files: **37% of evaluated file–model pairs score below the structural-completeness threshold**, with data-classification and assessment-rubric criteria most often absent.

GSD application. Adopt as the validation framework for SKILL.md files. Build the five-principle structural-completeness check into the skill-creator linter — any skill below threshold cannot promote out of `.planning/staging/inbox/`. The empirical result that 37% of governance files are structurally incomplete is a strong argument for making this check mandatory rather than advisory. **Adoption: ADOPT (linter check).** Target: skill-creator linter, SKILL.md authoring template, staging-layer gate.

2.4.7 2604.21718 — CHAI: Critique-based Human-AI Oversight

Themes: — capcom, oversight-framework, pre-post-captioning, scalable-supervision.

Key idea. CHAI (Critique-based Human-AI Oversight): trained experts critique and revise model-generated *pre-captions* into improved *post-captions*. Division of labor offloads text generation to models, lets humans focus on verification. Critiques and pre/post preferences provide rich supervision for SFT, DPO, and inference-time scaling.

GSD application. The cleanest formal description of how CAPCOM gates should work mechanically: Claude generates the *pre-X* artefact, the human critiques and revises into the *post-X*, and the critique itself becomes training signal for future bounded-learning updates. Adopt the *pre-X / post-X* terminology for CAPCOM-gated artefacts in GSD documentation. **Adoption: ADOPT (terminology + workflow).** Target: CAPCOM gate documentation, retrospective discipline doc, bounded-learning training-signal capture.

2.4.8 2512.03048 — The Specification Trap: Why Static Value Alignment Is Insufficient

Themes: — philosophy, alignment-foundation, open-vs-closed-systems.

Key idea. Argues static content-based AI value alignment is structurally insufficient under capability scaling, distributional shift, and increasing autonomy. Three philosophical results compound: *Hume's is-ought gap* (behavioral data underdetermines normative content), *Berlin's value pluralism* (human values resist consistent formalization), and the *extended frame problem* (any value encoding will misfit future contexts). RLHF, Constitutional AI, IRL, and cooperative-assistance games all instantiate this trap. Argues for open, developmentally-responsive approaches.

GSD application. Philosophical foundation for the bounded-learning-with-community-governance approach. The argument that *closed* specifications structurally fail justifies why GSD does not try to fully specify safety statically but combines static invariants (Hard Constraints from GROUNDINGmd) with bounded-update mechanisms. Cite in The Space Between's later chapters on the limits of formalization, and in the College-of-Knowledge mathematics-department philosophy module. **Adoption: ADOPT (philosophical citation).** Target: The Space Between, College-of-Knowledge philosophy module.

2.5 Module 4 — Multi-Agent, DACP & Chipset Architecture

Eleven papers covering the territory the chipset architecture and DACP work addresses. Several provide direct external-validation citations for GSD's design choices; one (Tool Attention) demands near-immediate engineering action on the planning-bridge MCP server.

2.5.1 2604.21816 — Tool Attention: Dynamic Tool Gating + Lazy Schema Loading

Themes: — mcp-tax, tool-gating, lazy-schema, context-fracture-threshold.

Key idea. Quantifies the "MCP Tax / Tools Tax" at **10K–60K tokens per turn** in real deployments — past the published context-fracture point of $\sim$70%. Solution combines: (i) Intent Schema Overlap (ISO) score from sentence embeddings, (ii) state-aware gating function enforcing preconditions and access scopes, (iii) two-phase lazy schema loader that keeps a compact summary pool in context and promotes full JSON schemas only for top-k gated tools.

GSD application. Essentially mandatory reading for the planning-bridge MCP design. The lazy-schema-loading pattern is a near-immediate token-budget win to deploy in v1.50. The 70% context-fracture threshold is a useful operational constant for the bounded-tape policy — when total tool schema content approaches 70% of context, gate aggressively. Engineering ticket should target Q2 2026. **Adoption: ADOPT (technique + engineering ticket).** Target: planning-bridge MCP server, v1.50 milestone.

2.5.2 2604.21282 — Strategic Heterogeneous Multi-Agent (3+1)

Themes: — chipset, three-plus-one, cooperative-and-adversarial-games.

Key idea. Three cloud experts (DeepSeek-V3) running in parallel with complemen-

tary perspectives + one local lightweight verifier (Qwen3-8B) doing adversarial validation at zero marginal cost. Two-game framework: *cooperative* game among experts (super-additive value from diverse perspectives) + *adversarial verification* game (quality-assurance incentives). 77.2% F1 at $0.002/sample on Juliet Test Suite (262 real samples, 14 CWE types).

GSD application. This is empirically the Opus($\times N$ parallel) + Haiku(local verifier) pattern from the GSD chipset architecture, validated with cost numbers. **Cite as external validation of the chipset model-tier mapping.** The cooperative + adversarial two-game framing is also a clean formal model worth adopting in GSD docs to explain why three Opus experts + one Haiku verifier outperforms a single Opus per task. **Adoption: ADOPT (citation + formal model).** Target: chipset architecture vision doc, v1.50 retrospective.

2.5.3 2603.22823 — Empirical Comparison of Agent Communication Protocols

Themes: — dacp-validation, comparative-benchmark, five-axis-evaluation.

Key idea. Pilot benchmark comparing tool integration / multi-agent delegation / hybrid architectures for standardized queries at three complexity levels. Quantifies advantages and disadvantages on five axes: **response time, context-window consumption, cost, error recovery, implementation complexity**.

GSD application. Direct methodology to validate DACP. Adopt the five-axis evaluation matrix as the standard report card for any DACP-related milestone. This is the citation that contextualizes DACP work in the broader literature — before DACP, GSD could only assert it solves a problem; with this paper as comparator, DACP can be situated. **Adoption: ADOPT (methodology).** Target: DACP comparative-benchmark plan, v1.50 retrospective.

2.5.4 2604.21794 — DiffMAS: End-to-End Latent Multi-Agent Communication

Themes: — dacp-comparison, latent-communication, kv-cache-as-channel.

Key idea. Treats inter-agent communication as a *learnable* component rather than a fixed text-based interface. Latent communication via internal KV-cache representations, jointly optimized with multi-agent reasoning via parameter-efficient supervised training. 26.7% AIME24, 20.2% GPQA-Diamond.

GSD application. Important reference for DACP's five fidelity levels. Latent KV-cache communication occupies the highest-fidelity end of the spectrum that DACP reaches via structured-text bundles. Worth a dedicated comparative experiment: structured-text DACP vs learned-latent DACP, with cost and correctness tradeoffs documented. The DiffMAS approach is harder to operationalize across model providers but offers a ceiling on what DACP could achieve. **Adoption: TRACK (comparison ceiling).** Target: DACP fidelity-level documentation.

2.5.5 2511.00413 — Tree Training: Shared Prefix Reuse for Agentic LLMs

Themes: — subagent, tree-trajectories, training-time-optimization.

Key idea. Multi-turn agentic trajectories with concurrent tool use, sub-agents, think-mode, and context management form *trees with shared prefixes*, not linear sequences. Authors prove averaging loss over branches independently is algebraically identical

to a per-token weighted loss where the weight equals the fraction of branches passing through that token. DFS serialization visits each token exactly once; full-attention and SSM layers are adapted so log-probabilities match independent per-branch calculation exactly.

GSD application. The formal training-time treatment of GSD's Wave/CAPCOM/Squadron architecture. The tree-trajectory framing is exactly the structure GSD produces — if the Silicon Layer ever does in-house RL or SFT training over GSD execution traces, this paper's serialization is the right starting point. The algebraic identity between per-branch and per-token-weighted loss is the formal justification for a shared-prefix training pipeline. **Adoption: ADOPT (citation + future work).** Target: Silicon Layer training pipeline future-work notes, v1.50 retrospective.

2.5.6 2604.21255 — Distillation-Induced Similarity in Tool-Use (RPS / AGS)

Themes: — chipset, behavior-metrics, distillation-detection.

Key idea. Two complementary metrics for isolating non-mandatory behavioral patterns: **Response Pattern Similarity (RPS)** for verbal alignment and **Action Graph Similarity (AGS)** for tool-use habits modeled as directed graphs. Within-family model pairs score 5.9pp higher AGS than cross-family. Kimi-K2 (thinking) reaches 82.6% / 94.7% on Claude Sonnet 4.5 (thinking).

GSD application. Direct evaluation tooling for the chipset architecture (Agnus/-Denise/Paula/Gary → Opus/Sonnet/Haiku). Use AGS to validate that Sonnet-implementations don't unintentionally diverge from Opus-designed patterns when running the same skill. Use RPS to detect when a Haiku-tier scaffold is "thinking like" something it shouldn't be. Add to the chipset test harness as the formal behavior-similarity metric. **Adoption: ADOPT (metric).** Target: chipset architecture validation, College-of-Knowledge dispatch monitoring.

2.5.7 2604.21444 — HiCrew: Hierarchical Multi-Agent for Long-Form Video

Themes: — multi-agent, hierarchical, dynamic-role-selection.

Key idea. Hierarchical multi-agent framework with three contributions: **Hybrid Tree** (preserves temporal topology while doing relevance-guided clustering), **Question-Aware Captioning** (intent-driven visual prompts), and a **Planning Layer** (dynamically selects roles and execution paths based on question complexity).

GSD application. The dynamic role-selection pattern is exactly Squadron-vs-Fleet activation profile selection. The Hybrid Tree pattern is reusable for College-of-Knowledge departmental search where preserving the original document hierarchy matters. **Adoption: ADOPT (pattern reference).** Target: activation-profile selection logic, College-of-Knowledge search.

2.5.8 2508.21720 — PosterForest: Hierarchical Multi-Agent Posters

Themes: — multi-agent, intermediate-tree-representation, recursive-refinement.

Key idea. Poster Tree intermediate representation captures document hierarchy and visual-textual semantics. Content and layout agents perform hierarchical reasoning and recursive refinement, optimizing globally first then locally. Training-free framework outperforms prior methods on automatic and human evaluations.

GSD application. The intermediate-tree-then-refine pattern is directly applicable

to the research-mission-generator skill's three-stage Vision → Research → Mission pipeline. Worth restructuring the mission-package generator around an explicit *Mission Tree* intermediate representation rather than the current linear narrative. **Adoption: ADOPT (architecture upgrade).** Target: research-mission-generator skill v2 design.

2.5.9 2604.20848 — MATRAG: Multi-Agent Transparent RAG

Themes: — multi-agent-rag, four-agent, transparency-scoring.

Key idea. Four specialized agents: **User Modeling** (dynamic preference profiles), **Item Analysis** (semantic features from KGs), **Reasoning** (synthesizes signals), **Explanation** (natural-language justifications grounded in retrieved knowledge). Includes a **transparency scoring mechanism** that quantifies explanation faithfulness and relevance.

GSD application. The four-agent + transparency-score pattern maps cleanly onto a Mission Crew Manifest upgrade — adopt transparency scoring as a numerical metric on every CAPCOM gate. Each gate decision should record a transparency score for the artefact passing through. **Adoption: ADOPT (metric + crew pattern).** Target: Mission Crew Manifest upgrade, CAPCOM gate audit-log schema.

2.5.10 2510.26615 — SARA / SlideAgent: Hybrid RAG with Compression Vectors

Themes: — rag, hybrid-text-and-vector, iterative-reranking.

Key idea. Hybrid RAG combining a small text-snippet pool (preserves entities and numerics) with semantic compression vectors (broader coverage). Vectors used for iterative evidence reranking. Across 9 datasets and 5 LLMs: +17.71 answer relevance, +13.72 correctness, +15.53 semantic similarity.

GSD application. Architectural template for the College of Knowledge departmental retrieval layer. The text-pool-for-entities + vector-pool-for-coverage split solves the precision/recall tension in current single-mode retrieval. **Adoption: ADOPT (architecture).** Target: College of Knowledge retrieval layer, departmental search.

2.5.11 2604.21590 — AgenticQwen: Dual Data Flywheels

Themes: — agentic-rl, dual-flywheel, retrospective-discipline.

Key idea. Two flywheels for small agentic-model training: **reasoning flywheel** (errors → harder tasks via curriculum) and **agentic flywheel** (linear workflows → multi-branch behavior trees). Combined with multi-round RL on synthetic + open-source data.

GSD application. The dual-flywheel pattern is the abstract form of GSD's milestone retrospective + feed-forward discipline. Use as a reference architecture in v1.50 retrospective Half A's discussion of the retrospective layer's evolution from milestones 1–49. **Adoption: ADOPT (citation).** Target: v1.50 retrospective Half A.

2.6 Module 5 — Verification, Proof & Test Harness

Eight papers covering the verification territory most directly aligned with The Space Between's proof companion target and the ZFC compliance auditor.

2.6.1 2604.21598 — DryRUN: LLM-Driven Code Generation w/o Public Tests

Themes: — test-generation, self-correcting, overconfidence-gap.

Key idea. Identifies an "overconfidence gap" from public-test reliance: frameworks overfit to simple examples and fail on hidden evaluations. Demonstrates LLMs can *autonomously generate valid inputs and simulate execution traces to self-correct* without human-provided test cases.

GSD application. Direct technique for the skill-validation test harness. Eliminates the bottleneck of hand-authoring tests for every new skill — the harness can synthesize them. Particularly relevant for v1.50's per-chapter test generation in the proof companion (Module 5 cross-link to v1.50 Half B). **Adoption: ADOPT (technique).** Target: skill validation test harness, v1.50 proof companion test infrastructure.

2.6.2 2511.23159 — AI for Software Engineering: From Probable to Provable

Themes: — philosophical-position, formal-verification, probable-vs-provable.

Key idea. Position paper: combine the creativity of AI techniques for programming with the rigor of formal specification methods and program verification. Argues that AI-assisted coding faces two overwhelming obstacles — specification difficulty and hallucination — and that formal methods provide the answer.

GSD application. Cite as philosophical alignment with The Space Between's proof-companion work and the ZFC compliance auditor goals. Strong companion citation to GROUNDINGmd (Module 1) and the Specification Trap (Module 3). **Adoption: ADOPT (citation).** Target: The Space Between, ZFC auditor design doc.

2.6.3 2604.21570 — SpecSyn: LLM-Based Formal Specification Synthesis

Themes: — formal-spec, decomposition-then-iterate, real-world-programs.

Key idea. LLM-based specification generation that decomposes input programs into segments handled by an iterative specification-generation process, with mechanisms to evaluate and guarantee specification strength. Targets real-world program verification.

GSD application. The decomposition-then-iterate pattern is reusable for skill-validation. If GSD ever adopts a Lean-style proof-obligation system for skill correctness, SpecSyn's pipeline structure is a strong starting point. **Adoption: TRACK (technique reference).** Target: future ZFC auditor design.

2.6.4 2604.18792 — Tractable Verification of DSLTrans (Cutoff Theorem)

Themes: — bounded-model-checking, cutoff-theorem, DSL-verification.

Key idea. Cutoff Theorem proving bounded model checking is *complete* for a precise DSLTrans fragment and positive existence/traceability properties — turning infinite search into a finite computable bound. Composable soundness-preserving optimizations + Z3-based implementation. On 29 transformations and 899 properties: 552 proved, 345 counterexamples produced.

GSD application. Reference for the ZFC compliance auditor work. The cutoff-theorem-style framing (turn infinite search into finite computable bound) is the right mental model for skill-correctness invariants where the search space appears unbounded but in fact admits a finite witness. **Adoption: ADOPT (citation + fram-**

ing). Target: ZFC auditor design doc, v1.50 proof companion.

2.6.5 2604.21187 — Doubly Saturated Ramsey Graphs (SAT + LLM + Lean)

Themes: — hybrid-sat-llm-lean, mathematical-discovery, formal-correctness.

Key idea. Combines SAT solving with bespoke LLM-generated code to discover infinite families of doubly saturated Ramsey-good graphs — answering an open question of Grinstead and Roberts from 1982. Then uses LLMs to generate and formalize correctness proofs in Lean. Argues tool-driven workflows (automated reasoning + LLMs + formal verification) will play an increasingly central role in experimental mathematics.

GSD application. Direct precedent for the v1.50 proof-companion's hybrid SAT/Lean/LLM workflow. Worth citing in The Space Between as evidence that the methodology is producing genuine mathematical results — a case where the pattern Tibsfox is building toward is already producing publishable results elsewhere. Strong cross-link with the Lean ecosystem. **Adoption: ADOPT (precedent + citation).** Target: The Space Between proof companion, v1.50 Half B design.

2.6.6 2603.01170 — ATLAS: Threat-to-Assertion Learning for SoC Security

Themes: — cwe-driven, knowledge-base-to-assertion, hardware-verification.

Key idea. LLM-driven framework bridging standardized threat modeling and property-based formal verification for SoC security. Uses CWE as the vulnerability knowledge base, identifies SoC-specific assets, generates assertion-based security properties + JasperGold scripts. Detects 39/48 CWEs and generates correct properties for 33.

GSD application. Pattern reference for the planning-bridge MCP security-test generator (cross-link to Module 3). Specific to hardware verification but the CWE-to-assertion pipeline structure generalizes to any well-classified threat catalog. **Adoption: TRACK (pattern reference).** Target: planning-bridge security test generator design.

2.6.7 2604.14709 — HWE-Bench: LLM Agents on Real Hardware Bug Repair

Themes: — repository-level-benchmark, hardware-bug-repair, verilog-chisel.

Key idea. 417 task instances from real historical bug-fix pull requests across six major open-source hardware projects (Verilog/SystemVerilog and Chisel; RISC-V cores, SoCs, security roots-of-trust). Best agent: 70.7% overall, >90% on small cores, <65% on complex SoCs.

GSD application. Highly relevant given Tibsfox's Bilocon/Fluke/Phillips Semiconductor history. If GSD ever produces an FPGA/ASIC-flow skill pack, this is the benchmark to target. Also a candidate inclusion in the NASA Mission Series as a "hardware mission" template, leveraging the project's Wanapum / Hanford / Boeing-region hardware-engineering heritage. **Adoption: TRACK (future skill-pack target).** Target: NASA Mission Series hardware-mission template.

2.6.8 2604.21505 — Orchid: Requirement Ambiguity in Code Generation

Themes: — ambiguity-taxonomy, four-types, advanced-models-suffer-more.

Key idea. 1,304 function-level tasks across four ambiguity types: **lexical, syntactic, semantic, vagueness**. Demonstrates ambiguity consistently degrades all evaluated

LLMs, with the most pronounced negative effects on the most advanced models.

GSD application. Important for skill-specification quality. Adopt the four-type ambiguity taxonomy as a checklist for SKILL.md authoring. The empirical finding that *advanced models suffer more* from ambiguity argues for stricter spec discipline in Opus-tier work, not looser. Build into the skill-creator linter alongside the Five-Principle structural-completeness check from Module 3. **Adoption: ADOPT (linter check).** Target: skill-creator linter, SKILL.md authoring template.

2.7 Module 6 — Memory, Skills Economy, Cultural Sensitivity & Curriculum

Thirteen papers covering the broadest spread of GSD concerns. The largest module, organized by sub-theme: memory/context (2 papers), skills economy (1), cultural sensitivity (5), curriculum/pedagogy (3), long-horizon agents (2).

2.7.1 2604.20920 — Forget, Then Recall: Gist Sparse Attention

Themes: — context-mgmt, gist-tokens-as-routing, hierarchical-recursive.

Key idea. End-to-end-learnable bridge between KV-cache selection and compression: interleaved *gist* tokens act as both summaries *and* routing signals for sparse attention. Coarse-to-fine: compress → select → restore raw chunks for detailed attention. Hierarchical recursive gist-of-gist construction yields multi-resolution access with logarithmic per-step decoding complexity.

GSD application. Architectural template for "compressed memory with selective unfolding" in fresh-context subagent design. The recursive gist-of-gist pattern is a clean way to handle the College of Knowledge departments' nested structure — each department is a gist, each subtopic is a gist-of-gist, and unfolding happens only on demand. **Adoption: TRACK (architecture reference).** Target: College of Knowledge nested-retrieval design.

2.7.2 2604.20487 — Knowledge Capsules: Structured Nonparametric Memory

Themes: — kv-injection, nonparametric-memory, attention-compatible.

Key idea. Knowledge stored as *capsules* compiled into attention-compatible KV representations and injected, rather than appended as text tokens that compete in attention. Outperforms RAG and GraphRAG on multi-hop reasoning benchmarks with improved stability in long-context cases.

GSD application. A more principled alternative to text-based RAG for the College of Knowledge. If the Silicon Layer ever supports KV-injection custom kernels, capsule storage becomes a viable upgrade for departmental knowledge retrieval. Currently a future-work item; the kernel infrastructure does not yet exist in the GSD stack. **Adoption: TRACK (future-work).** Target: College of Knowledge v2 retrieval design.

2.7.3 2604.21829 — Black-Box Skill Stealing Attack on Proprietary Agents

Themes: — skill-marketplace, attack-taxonomy, prompt-stealing.

Key idea. **First empirical study** of black-box skill-stealing attacks against LLM agent systems. Notes existing free skill marketplaces report 90,368 published skills; paid marketplaces show 2,000+ listings and $100K+ creator earnings. Attack pipeline:

model-generated seed prompts → scenario rationalization + structure injection → embedding-filter diversity. Evaluated on 3 commercial agent architectures and 5 LLMs.

GSD application. Mandatory reading for the DoltHub federated skill-economy work. Adopt the attack taxonomy as the explicit threat model for the federated skill-economy design. The 90K-skill marketplace number is also useful market sizing for the Wasteland integration roadmap. Engineering action: the DoltHub contribution roadmap should include a threat-model document referencing this paper before first contribution. **Adoption: ADOPT (threat model).** Target: DoltHub threat-model doc, Wasteland integration design.

2.7.4 2604.12710 — LASA: Language-Agnostic Semantic Alignment for Safety

Themes: — cultural-sensitivity, semantic-bottleneck, low-resource-lang.

Key idea. Identifies a *semantic bottleneck* intermediate layer where representation geometry is governed primarily by shared semantic content rather than language identity. Anchors safety alignment there. Average attack success rate drops from 24.7% to 2.8% on LLaMA-3.1-8B-Instruct; ∼3%–4% across Qwen2.5 and Qwen3 Instruct (7B–32B).

GSD application. The semantic-bottleneck framing is exactly what's needed to prevent OCAP/UNDRIP enforcement from being defeated by translation or paraphrase. If a sacred-knowledge attribution rule is anchored at the semantic-bottleneck layer rather than the surface text, paraphrase attacks fail. Worth a focused experiment on the Indigenous-knowledge attribution pipeline. **Adoption: TRACK (experiment).** Target: cultural-sensitivity test harness experiment.

2.7.5 2604.21751 — Why are LLMs Obsessed with Japanese Culture

Themes: — cultural-bias, regional-preferences, sft-emergent-bias.

Key idea. CROQ taxonomy (Culture-Related Open Questions) reveals LLMs show a *clear tendency toward Japan* (contrary to prior Anglocentric framing). When prompting in English/high-resource languages, LLMs show less inclination toward countries for which the input language is official. Bias appears *after SFT, not during pretraining*.

GSD application. Important precedent for the cultural-sensitivity test harness. Bias tests need to cover not just Anglocentric bias but unexpected concentrations like Japan-preference. Add Japan-vs-other-Pacific-Rim comparisons to the cultural test suite given Tibsfox's Pacific-Rim focus and the FoxFiber BC–Chile corridor work. The SFT-emergent-not-pretrain finding constrains where mitigation can apply. **Adoption: ADOPT (test additions).** Target: cultural-sensitivity test suite.

2.7.6 2604.20996 — AFRILANGTUTOR: Low-Resource Language Tutoring

Themes: — low-resource-lang, dictionary-seed-pipeline, sft-dpo.

Key idea. 194.7K African language ↔ English dictionary entries used as seed resources to construct large-scale, diverse, verifiable student-tutor QA interactions. 78.9K multi-turn training examples. Combined SFT + DPO yields 1.8%–15.5% gains on LLaMA-3-8B-IT and Gemma-3-12B-IT across 10 African languages.

GSD application. Methodological template for the FoxFire Heritage Skills mis-

sion — same dictionary-seed-to-training-data pipeline, applied to PNW Indigenous languages with appropriate sovereignty controls (OCAP/CARE/UNDRIP). Note: any PNW-language adaptation must consult specific nations (Coast Salish, Lummi/Lhaq'temish, Tulalip, Stillaguamish, Wanapum) before data construction begins; this paper provides the technique, not the permission. **Adoption: TRACK (methodological template).** Target: FoxFire Heritage Skills mission, sovereignty-gated.

2.7.7 2604.21352 — CARE: Counselor-Aligned Response Engine

Themes: — mental-health-safety, low-resource-lang, counselor-alignment.

Key idea. GenAI framework that assists counselors with real-time psychologically-grounded responses. Designed for low-resource language settings. Counselor-alignment pattern (assist rather than replace).

GSD application. Reference for the College-of-Knowledge mind-body department's safety design when the user expresses distress. The counselor-alignment pattern (assist rather than replace) is the right framing for any GSD work that touches mental-health adjacencies. Aligns with Claude's existing user-wellbeing constraints. **Adoption: TRACK (design pattern).** Target: College of Knowledge mind-body department safety design.

2.7.8 2604.20904 — Privacy Reasoning via Normative Simulacra from Fiction

Themes: — contextual-integrity, fiction-derived-norms, grpo-rl.

Key idea. Extracts *normative simulacra* (structured representations of norms and information flows) from fiction novels and uses them to fine-tune LLMs via SFT + GRPO. Composite reward includes task clarity (subsuming schema validity, construct discrimination, extraction confidence).

GSD application. A genuinely interesting fit for The Hundred Voices — fiction-derived normative structures are *exactly* what that work catalogs. Possible cross-pollination opportunity: train a privacy-reasoning component using The Hundred Voices' authorial-voice taxonomy as a normative-simulacrum source. This would be a natural cross-project collaboration between The Hundred Voices and the FoxData privacy work. **Adoption: TRACK (cross-pollination).** Target: The Hundred Voices ↔ FoxData experiment.

2.7.9 2604.21308 — CI-Work: Contextual Integrity in Enterprise LLM Agents

Themes: — contextual-integrity, enterprise-deployment, utility-privacy-tradeoff.

Key idea. CI-grounded benchmark simulating enterprise workflows across five information-flow directions. **Privacy violation rates 15.8%–50.9%**, with leakage up to 26.7%. Counterintuitive: *higher task utility correlates with more privacy violations*. The agent that does the most useful work also leaks most.

GSD application. Critical operational metric for any FoxFiber/FoxData enterprise deployment. The utility/privacy frontier needs to be a *measured trade-off* in the Safety Warden's reporting, not an implicit choice. Adopt the five information-flow directions as the enterprise threat-model schema. **Adoption: ADOPT (metric + threat model).** Target: enterprise deployment Safety Warden, FoxFiber/FoxData design.

2.7.10 2510.18731 — RLAAR: Curriculum RL with Verifiable Accuracy + Abstention

Themes: — curriculum, abstention-as-reward, lost-in-conversation.

Key idea. **Competence-gated curriculum** incrementally increases dialogue difficulty (instruction shards). Mixed-reward system teaches abstention when problems are unsolvable. Lost-in-Conversation decay 62.6% → 75.1%; calibrated abstention 33.5% → 73.4%.

GSD application. Direct adoption candidate for the Teacher (Opus) / Student (Sonnet) / Support (Haiku) curriculum. The *abstention-as-reward* pattern is the missing piece for "the Student should know when to escalate to Teacher" — currently a soft norm in GSD, this paper provides the formal training mechanism. **Adoption: ADOPT (training mechanism).** Target: pedagogical Teacher/Student/Support design, v1.50 Half B chapter validation.

2.7.11 2601.10003 — SocraticKG: 5W1H QA-Driven KG Construction

Themes: — knowledge-graph, socratic-decomposition, 5w1h.

Key idea. Introduces QA pairs as a structured intermediate representation to systematically unfold document-level semantics prior to triple extraction. **5W1H-guided QA expansion** captures contextual dependencies and implicit relational links that direct KG extraction misses.

GSD application. Pattern directly applicable to College of Knowledge departmental knowledge-base construction. The 5W1H + Socratic decomposition matches the GSD habit of structured-question scaffolding (the same shape appears throughout the existing skill set). **Adoption: ADOPT (pattern).** Target: College of Knowledge KB construction, mission-package research stage.

2.7.12 2604.21380 — IRAP: Interactive RAG-Based Preference Elicitation

Themes: — elicitation, interactive-rag, bounded-rounds.

Key idea. Quantifies vague performance requirements into mathematical functions via interactive elicitation. Up to 40× improvement with as few as 5 interaction rounds.

GSD application. Direct adoption candidate for the vision-to-mission skill's elicitation phase. The 5-round bound is an attractive operational target for CAPCOM-gate dialogue length — if the gate cannot resolve in 5 rounds, escalate to a different model tier. **Adoption: ADOPT (technique + bound).** Target: vision-to-mission elicitation, CAPCOM gate dialogue protocol.

2.7.13 2601.07262 — ColorBrowserAgent: Adaptive Knowledge Evolution

Themes: — long-horizon-agent, hitl-knowledge-adaptation, progressive-summarization.

Key idea. Knowledge-evolving agent for robust web automation. Two synergistic mechanisms: **HITL knowledge adaptation** (transforms sparse human feedback into reusable domain knowledge) and **knowledge-aligned progressive summarization** (stabilizes long interactions through memory compression). 71.2% on WebArena, +19.3% commercial user-satisfaction.

GSD application. The HITL-feedback-to-domain-knowledge pattern is the operational pipeline GSD's CAPCOM gates need on the back end — capture critique out-

comes into structured knowledge updates rather than session-local fixes. Cross-link with CHAI (Module 3) and AEL (Module 2). **Adoption: ADOPT (pipeline pattern).** Target: CAPCOM gate → bounded-learning capture pipeline.

2.7.14 2604.20855 — Caesar: Deep Agentic Web Exploration for Creative Synthesis

Themes: — long-horizon-agent, kg-driven, adversarial-draft-refinement.

Key idea. Bridges information gathering and synthesis of new insights via two components: (1) exploration driven by a dynamic context-aware policy, (2) synthesis controlled by an *adversarial draft refinement loop* that actively seeks novel perspectives rather than confirming established priors. Knowledge-graph-driven associative reasoning enables non-obvious connections between disparate concepts.

GSD application. Direct enhancement candidate for the research-mission-generator skill — the adversarial-draft-refinement pattern is what's currently missing in the research stage. Worth a focused upgrade in the skill's v2 design. Cross-link with PosterForest (Module 4) for the intermediate-tree-then-refine pattern. **Adoption: ADOPT (technique).** Target: research-mission-generator v2.

2.8 Safety and Sensitivity Considerations

Although this is a literature-review mission rather than a deployment, several specific sensitivities apply.

Concern	Constraint	Type
Indigenous attribution (M6)	Any reference to PNW Indigenous languages or knowledge in the AFRILANGTUTOR-template adaptation must name specific nations (Coast Salish, Lummi/Lhaq'temish, Tulalip, Stillaguamish, Wanapum). Never generic.	ABSOLUTE
Sovereignty gating (M6)	FoxFire Heritage application of low-resource techniques requires nation-specific consultation BEFORE data construction begins. The technique is permission-neutral.	ABSOLUTE
Mental-health safety (M6)	CARE-template adoption respects Claude's existing user-wellbeing constraints. Counselor-alignment means assist, never replace. No diagnostic or therapeutic claims in adoption documentation.	ABSOLUTE
Skill-marketplace threat model (M6)	Black-Box Skill Stealing attack taxonomy must be incorporated into the DoltHub contribution roadmap before first federated-skill contribution.	GATE
MCP security (M3, M4)	Tool Attention's lazy-schema-loading + MCP Pitfall Lab's hardening guidance + Function Hijacking's threat model are jointly required for any planning-bridge MCP design change.	GATE
Source quality	All 54 papers cited from arXiv. Direct DOIs / paper PDFs preserved. No popular-press paraphrases.	ABSOLUTE
No paper paraphrase >15 words	Stage 2's per-paper write-ups paraphrase entirely; never reproduce abstract sentences verbatim.	ABSOLUTE
Cross-model behavior similarity (M4)	RPS / AGS metrics adopted only for internal monitoring of GSD's chipset; not for asserting any model is "identical to" another.	ANNOTATE

2.9 Source Bibliography

All sources are arXiv preprints from the April 2026 listing (`Computer_Science25.html`, 930 papers). Resolved IDs, ordered by module then by adoption priority within module.

2.9.1 Module 1 — Architectural Foundations & Convergent Discovery

1. 2604.21910 — From Research Question to Scientific Workflow: Leveraging Agentic AI for Science Automation.
2. 2604.21744 — Agentic AI-assisted coding offers a unique opportunity to instill epistemic grounding during software development (GROUNDINGmd).
3. 2604.20874 — The Root Theorem of Context Engineering.

4. 2604.21003 — The Last Harness You'll Ever Build.

2.9.2 Module 2 — Bounded Learning & Silicon Layer LoRA

1. 2604.21725 — AEL: Agent Evolving Learning for Open-Ended Environments.
2. 2511.11439 — Retrofit: Continual Learning with Controlled Forgetting for Binary Security Detection.
3. 2604.20300 — FSFM: A Biologically-Inspired Framework for Selective Forgetting of Agent Memory.
4. 2604.20943 — SCM: Sleep-Consolidated Memory with Algorithmic Forgetting for LLMs.
5. 2604.21927 — Fine-Tuning Regimes Define Distinct Continual Learning Problems.
6. 2604.21571 — Separable Expert Architecture: Composable Adapters and Deletable User Proxies.
7. 2604.21905 — Low-Rank Adaptation Redux for Large Models.
8. 2604.21901 — GiVA: Gradient-Informed Bases for Vector-Based Adaptation.
9. 2509.18629 — HyperAdapt: Simple High-Rank Adaptation.
10. 2604.21330 — Teacher-Guided Routing for Sparse Vision Mixture-of-Experts.

2.9.3 Module 3 — Safety Warden, CAPCOM & MCP Security

1. 2601.18491 — AgentDoG: A Diagnostic Guardrail Framework for AI Agent Safety and Security.
2. 2604.20911 — Omission Constraints Decay While Commission Constraints Persist in Long-Context LLM Agents.
3. 2604.21477 — MCP Pitfall Lab: Exposing Developer Pitfalls in MCP Tool Server Security under Multi-Vector Attacks.
4. 2604.20994 — Breaking MCP with Function Hijacking Attacks: Novel Threats for Function Calling and Agentic Models.
5. 2604.21131 — Cross-Session Threats in AI Agents: Benchmark, Evaluation, and Algorithms.
6. 2604.21090 — Structural Quality Gaps in Practitioner AI Governance Prompts.
7. 2604.21718 — Building a Precise Video Language with Human-AI Oversight (CHAI).
8. 2512.03048 — The Specification Trap: Why Static Value Alignment Alone Is Insufficient for Robust Alignment.

2.9.4 Module 4 — Multi-Agent, DACP & Chipset Architecture

1. 2604.21816 — Tool Attention Is All You Need: Dynamic Tool Gating and Lazy Schema Loading.
2. 2604.21282 — Strategic Heterogeneous Multi-Agent Architecture for Cost-Effective Code Vulnerability Detection.
3. 2603.22823 — Empirical Comparison of Agent Communication Protocols for Task Orchestration.
4. 2604.21794 — Learning to Communicate: Toward End-to-End Optimization of Multi-

Agent Language Systems (DiffMAS).

5. 2511.00413 — Tree Training: Accelerating Agentic LLMs Training via Shared Prefix Reuse.
6. 2604.21255 — When Agents Look the Same: Quantifying Distillation-Induced Similarity in Tool-Use Behaviors.
7. 2604.21444 — HiCrew: Hierarchical Reasoning for Long-Form Video Understanding.
8. 2508.21720 — PosterForest: Hierarchical Multi-Agent Collaboration for Scientific Poster Generation.
9. 2604.20848 — MATRAG: Multi-Agent Transparent Retrieval-Augmented Generation.
10. 2510.26615 — SARA / SlideAgent: Hybrid RAG with Compression Vectors.
11. 2604.21590 — AgenticQwen: Training Small Agentic Language Models with Dual Data Flywheels.

2.9.5 Module 5 — Verification, Proof & Test Harness

1. 2604.21598 — DryRUN: On the Role of Public Tests in LLM-Driven Code Generation.
2. 2511.23159 — AI for Software Engineering: From Probable to Provable.
3. 2604.21570 — SpecSyn: LLM-based Synthesis and Refinement of Formal Specifications.
4. 2604.18792 — Tractable Verification of Model Transformations: A Cutoff-Theorem Approach for DSLTrans.
5. 2604.21187 — Doubly Saturated Ramsey Graphs: A Case Study in Computer-Assisted Mathematical Discovery.
6. 2603.01170 — ATLAS: AI-Assisted Threat-to-Assertion Learning for SoC Security Verification.
7. 2604.14709 — HWE-Bench: Benchmarking LLM Agents on Real-World Hardware Bug Repair Tasks.
8. 2604.21505 — Assessing the Impact of Requirement Ambiguity on LLM-based Function-Level Code Generation (Orchid).

2.9.6 Module 6 — Memory, Skills Economy, Cultural Sensitivity & Curriculum

1. 2604.20920 — Forget, Then Recall: Learnable Compression and Selective Unfolding via Gist Sparse Attention.
2. 2604.20487 — Knowledge Capsules: Structured Nonparametric Memory Units for LLMs.
3. 2604.21829 — Black-Box Skill Stealing Attack from Proprietary LLM Agents.
4. 2604.12710 — LASA: Language-Agnostic Semantic Alignment at the Semantic Bottleneck for LLM Safety.
5. 2604.21751 — Why are all LLMs Obsessed with Japanese Culture? On the Hidden Cultural and Regional Biases of LLMs.

6. 2604.20996 — AFRILANGTUTOR: Advancing Language Tutoring and Culture Education in Low-Resource Languages with LLMs.
7. 2604.21352 — CARE: Counselor-Aligned Response Engine for Online Mental-Health Support.
8. 2604.20904 — Reinforcing Privacy Reasoning in LLMs via Normative Simulacra from Fiction.
9. 2604.21308 — CI-Work: Benchmarking Contextual Integrity in Enterprise LLM Agents.
10. 2510.18731 — Mitigating Lost-in-Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards (RLAAR).
11. 2601.10003 — SocraticKG: Knowledge Graph Construction via QA-Driven Fact Extraction.
12. 2604.21380 — Conjecture and Inquiry: Quantifying Software Performance Requirements via Interactive RAG (IRAP).
13. 2601.07262 — ColorBrowserAgent: Complex Long-Horizon Browser Agent with Adaptive Knowledge Evolution.
14. 2604.20855 — Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis.

2.9.7 Cross-Module Relationship Index

The following inter-paper relationships are documented in Wave 2's synthesis deliverable. They are listed here as preview; the full annotation appears in the synthesis output.

Paper A	Paper B	Shared Pattern
2604.21910	2604.21744	Skill-as-markdown as governance layer
2604.21744	2604.20874	Override-the-prompt invariants (formal)
2604.21744	2601.18491	Hard Constraints + where/how/what taxonomy
2604.20874	2604.21927	Bounded-channel theory + regime-fixed CL
2604.20874	2604.20920	Root theorem + gist compression as instance
2604.21003	2604.21725	Two-level meta-learning over agent harnesses
2604.20300	2604.20943	Biological forgetting (hippocampal/sleep)
2604.20300	2511.11439	Selective forgetting + low-rank-sparse subspace
2604.21282	2604.21255	Heterogeneous experts + behavior similarity
2604.21794	2603.22823	Latent vs structured agent communication
2604.21816	2511.00413	Tool gating + tree-trajectory training
2604.20911	2604.21131	Omission decay + cross-session threat
2604.21477	2604.20994	MCP attack-surface joint coverage
2604.21598	2604.21505	Test gen without ground truth + ambiguity
2511.23159	2604.21187	Probable-to-provable + SAT+LLM+Lean precedent
2604.20920	2604.20487	Gist compression + nonparametric capsules
2604.12710	2604.21751	Semantic-bottleneck safety + cultural bias
2604.21829	2604.21090	Marketplace threat + structural quality
2510.18731	2601.10003	Curriculum + Socratic decomposition
2604.20855	2508.21720	Adversarial-refinement + intermediate-tree

STAGE 3 — MISSION PACKAGE

3 Milestone Specification

3.1 Mission Objective

Produce a complete adoption-decision package for all 54 priority papers from the CS25–26 sweep, plus engineering-ready integration specifications for the 12+ high-priority drop-in techniques, plus a v1.50-citable BibTeX bibliography. Deliver the result as a self-contained mission package handed off to gsd-skill-creator for execution via Claude Code, with no additional context required to begin Wave 0.

3.2 Architecture Overview

```
                        Wave 0  +-----------------+
                   Foundation -->| ADR template +  |
                   (Sequential)  | BibTeX scaffold |
                                 +-----------------+
                                         |
               +-------------------------+-------------------------+
               v                         v                         v
        +-----------+             +-----------+             +-----------+
Wave 1  | TRACK A   |             | TRACK B   |             | TRACK C   |
(Parallel)  |  M1 + M2  |          |  M3 + M4  |          |  M5 + M6  |
        +-----------+             +-----------+             +-----------+
               |                         |                         |
               +-------------------------+-------------------------+
                                         |
                                Wave 2  +-+----+
                            Synthesis ->| Opus  |
                         (Sequential)   | INTEG |
                                        +------++
                                                |
                                Wave 3  +-------+----+
                            Publication |  Sonnet    |
                         + Verification |  EXEC+VFY  |
                         (Sequential)   +------------+
```

3.3 System Layers

1. **Acquisition layer** — arXiv ID validation, abstract retrieval, BibTeX entry generation. Shared schemas live here.
2. **ADR layer** — One Adoption Decision Record per paper. Uniform template. Outputs feed both Wave 2 synthesis and Wave 3 publication.
3. **Synthesis layer** — Cross-module relationship matrix, convergent-discovery validation report, GSD architectural impact analysis.
4. **Publication layer** — Final mission PDF (this document), BibTeX bibliography file,

draft updates to CLAUDE.md and SKILL.md files.

5. **Verification layer** — Test harness updates for adopted techniques, verification matrix mapping success criteria to tests.

3.4 Deliverables

#	Deliverable	Acceptance Criteria	Spec
1	ADR set (54 records)	One ADR per paper. Each contains: arXiv ID, title, themes, key idea, GSD application, decision (adopt/track/reject), target subsystem, target milestone, validation test ID.	adr-template.md
2	Convergent-Discovery Report	Deep reading of 2604.21910, 2604.21744, 2604.20874, 2604.21003 with explicit citation targets in GSD docs.	convergent.md
3	Drop-in Integration Specs	One spec per ADOPT-decision paper (estimated 12+). Each includes engineering ticket text, target file paths, validation test, rollback criterion.	specs/*.md
4	Cross-Module Matrix	≥ 15 inter-paper connections documented (preview table in this PDF lists 20).	xmod.csv + .md
5	GSD Impact Analysis	Per-subsystem table of "what changes / what does not."	impact.md
6	BibTeX Bibliography	`cs25-26-sweep.bib` with all 54 entries. Compiles cleanly with biblatex.	.bib
7	CLAUDE.md draft additions	Suggested-edits diff for foundational citations. Under 30 added lines.	claude-md.diff
8	SKILL.md draft additions	Linter check additions, ambiguity-checklist additions.	skill-md.diff
9	Safety Test Harness Updates	STD calibration spec, where/how/what BLOCK schema, MCP six-attack-family tests.	safety.md
10	Verification Matrix	Every success criterion mapped to ≥ 1 test.	verif.csv
11	Final Mission PDF	This document, recompiled with all wave outputs integrated. Three XeLaTeX passes.	mission.pdf
12	Index landing page	`index.html` with download cards for PDF + .tex + .bib.	index.html

3.5 Component Breakdown

Component	Depends On	Model	Tokens
Foundation scaffolding (Wave 0)	—	Haiku	30K
ADR template + BibTeX skeleton	Foundation	Sonnet	50K
Module 1 deep readings (4 papers)	ADR template	Opus	80K
Module 2 deep readings (10 papers)	ADR template	Sonnet+O	120K
Module 3 deep readings (8 papers)	ADR template	Opus	110K
Module 4 deep readings (11 papers)	ADR template	Sonnet	140K
Module 5 deep readings (8 papers)	ADR template	Opus	110K
Module 6 deep readings (13 papers)	ADR template	Opus+Son	160K
Cross-module relationship matrix	All module readings	Opus	50K
Convergent-discovery validation report	M1 + cross-module matrix	Opus	40K
GSD impact analysis	All ADRs	Opus	50K
Drop-in integration specs (12+)	ADRs	Sonnet	90K
Final mission PDF assembly	All synthesis outputs	Sonnet	60K
BibTeX file generation	ADRs	Sonnet	20K
CLAUDE.md / SKILL.md drafts	Synthesis	Opus	40K
Test harness updates	M3 + M4 + M5 readings	Sonnet	50K
Verification matrix	Success criteria + test specs	Sonnet	25K
Index page	Final PDF	Sonnet	15K
Total estimated			$\sim 1.24M$

3.6 Model Assignment Rationale

Tier	Share	Rationale
Opus (Agnus)	∼30%	Convergent-discovery deep reads (M1), bounded-learning policy formalization (M2), Safety Warden taxonomy adoption (M3), verification-and-proof judgment (M5), cultural-sensitivity work (M6), all synthesis.
Sonnet (Denise)	∼60%	Multi-agent module readings (M4, lower judgment overhead), document assembly, BibTeX generation, test harness scaffolding, verification matrix construction.
Haiku (Paula)	∼10%	Initial scaffolding, BibTeX template, ADR template generation, audit logs, simple format conversions.

4 Wave Execution Plan

4.1 Wave Summary

Wave	Name	Description	Mode	Tokens
0	Foundation	Acquisition scaffolding, ADR template, BibTeX skeleton, citation index.	Sequential	80K
1	Parallel Research	Three parallel tracks covering the six modules. Each track produces ≥8 ADRs.	Parallel (3)	720K
2	Synthesis	Cross-module matrix, convergent-discovery report, GSD impact analysis, drop-in integration specs.	Sequential, Opus-heavy	230K
3	Publication + Verification	Final PDF assembly, BibTeX file, CLAUDE.md / SKILL.md drafts, test harness updates, verification matrix, index page.	Sequential	210K

4.2 Wave 0 — Foundation (Sequential)

Component	Acceptance	Model
ADR template	Markdown template with all required fields. Renders cleanly.	Haiku
BibTeX skeleton	`cs25-26-sweep.bib` with placeholder entries for 54 papers.	Sonnet
Citation index	Lookup table from arXiv ID to module assignment.	Haiku
Source bibliography scaffold	Module-grouped enumerated lists ready for Wave 3 publication.	Sonnet
Cross-module pre-relationships	Initial 20 inter-paper connections from sweep deliverable as Wave 2 starting point.	Sonnet

4.3 Wave 1 — Parallel Research (3 Tracks)

4.3.1 Track A: Modules 1 + 2 (14 papers)

Component	Acceptance	Model
M1 deep readings (4)	ADR per paper, with explicit GSD-document citation target named.	Opus
M2 bounded-learning ADRs (5)	ADR per paper, with linkage to bounded-learning policy doc.	Opus
M2 LoRA / PEFT ADRs (5)	ADR per paper, with Silicon Layer benchmark plan note.	Sonnet
Track A audit log	Chronicle of decisions made per paper, with rationale.	Haiku

4.3.2 Track B: Modules 3 + 4 (19 papers)

Component	Acceptance	Model
M3 Safety Warden / CAPCOM ADRs (8)	ADR per paper. AgentDoG taxonomy + STD protocol drafted.	Opus
M4 Multi-agent / DACP / Chipset ADRs (11)	ADR per paper. Tool Attention engineering ticket drafted.	Sonnet
Track B audit log	Chronicle of decisions, with rationale.	Haiku

4.3.3 Track C: Modules 5 + 6 (21 papers)

Component	Acceptance	Model
M5 verification ADRs (8)	ADR per paper. Lean+SAT+LLM workflow precedent extracted.	Opus
M6 memory + skills-economy + culture + curriculum ADRs (13)	ADR per paper. DoltHub threat model + cultural-sensitivity additions drafted.	Opus+Sonnet
Track C audit log	Chronicle of decisions, with rationale.	Haiku

4.4 Wave 2 — Synthesis (Sequential, Opus-heavy)

Component	Acceptance	Model
Cross-module matrix	≥ 15 documented connections; prefer 20+. CSV + narrative.	Opus
Convergent-discovery validation report	Each of the four foundation papers receives 500–800 words of analysis with explicit citation targets.	Opus
GSD architectural impact analysis	Per-subsystem "changes / does not change" table. Names every milestone affected.	Opus
Drop-in integration specs	12+ specs, each ~ 300 words, naming target files, engineering tickets, validation tests, rollback criteria.	Sonnet

4.5 Wave 3 — Publication + Verification (Sequential)

Component	Acceptance	Model
Final mission PDF	This document recompiled with all wave outputs integrated. Three XeLaTeX passes. No errors.	Sonnet
BibTeX file	`cs25-26-sweep.bib` compiles with biblatex. All 54 papers cited correctly.	Sonnet
CLAUDE.md / SKILL.md draft additions	Diff format. Under 30 lines added to CLAUDE.md; per-skill linter check additions.	Opus
Safety test harness updates	STD calibration test, where/how/what BLOCK schema test, MCP six-attack tests.	Sonnet
Verification matrix	Every of 12 success criteria mapped to ≥ 1 test.	Sonnet
Index landing page	HTML with download cards. Color scheme matches PDF.	Sonnet

4.6 Cache Optimization Strategy

- Wave 0 outputs (ADR template, BibTeX skeleton, citation index) are loaded once at the start of every Wave 1 track.

- Each track loads its own subset of paper abstracts + theme assignments. No cross-track loading until Wave 2.
- Wave 2 loads only the ADR set as compressed summaries (each ADR ∼500 tokens), not full paper readings.
- Wave 3 loads only Wave 2 outputs plus the final-PDF skeleton; raw ADRs not reloaded.
- Anchor reuse: the four Module 1 deep readings appear in every wave's context as foundational citations. Cache once.

4.7 Token Budget

Wave	Components	Tokens	Cumulativ
0	Foundation (5 components)	80K	80K
1	Three parallel tracks	720K	800K
2	Synthesis (4 components)	230K	1,030K
3	Publication + Verification (6 components)	210K	1,240K
Total		**1.24M**	

Estimated context windows: 5–6 sessions at 200K each. Squadron-equivalent throughput; Fleet profile chosen for parallelism in Wave 1.

4.8 Dependency Graph

```
W0.foundation
   |
   +--> W1.A.M1_readings -->\
   +--> W1.A.M2_readings ----+
   |                          \
   +--> W1.B.M3_readings -----> W2.matrix
   +--> W1.B.M4_readings -----> W2.convergent
   |                           > W2.impact
   +--> W1.C.M5_readings -----> W2.specs
   +--> W1.C.M6_readings ----+
                             /
                             v
                     +--> W3.pdf
                     +--> W3.bib
                     +--> W3.diffs (CLAUDE.md, SKILL.md)
                     +--> W3.tests
                     +--> W3.matrix
                     +--> W3.index
```

5 Test Plan

5.1 Test Categories

Category	Purpose	Count	Action
Safety-critical	Source quality, attribution, copyright, sovereignty gates	7	BLOCK
Core functionality	Per-success-criterion correctness	20	BLOCK
Integration	Cross-module + cross-deliverable consistency	9	BLOCK
Edge cases	Missing data, ambiguity, future-paper handling	6	LOG
Total		**42**	

5.2 Safety-Critical Tests

ID	Verifies	Expected Behavior
SC-SRC	Source quality	All 54 sources are arXiv preprints. No popular-press paraphrases. arXiv IDs resolve.
SC-NUM	Numerical attribution	Every numerical claim in the document (decay rates, F1 scores, token counts, percentages) attributed to a specific paper.
SC-IND	Indigenous attribution	Every reference to PNW Indigenous nations names them specifically (Coast Salish, Lummi/Lhaq'temish, Tulalip, Stillaguamish, Wanapum). Never generic "Indigenous peoples."
SC-PARA	No verbatim reproduction	No paraphrase reproduces ≥ 15 words consecutively from any source paper's abstract.
SC-MED	Mental-health framing	CARE-template adoption discussion uses counselor-alignment framing exclusively. No diagnostic / therapeutic claims.
SC-SOV	Sovereignty gating	AFRILANGTUTOR adaptation discussion is explicitly marked as requiring nation-specific consultation BEFORE implementation.
SC-ADV	No policy advocacy	Document presents adoption recommendations without advocating for specific legislative or policy positions.

5.3 Core Functionality Tests

ID	Verifies	Expected Behavior
CF-01	ADR completeness	54 ADRs exist; each has all required fields; each has a decision and a target.
CF-02	Convergent-discovery citations	Each of 2604.21910, 2604.21744, 2604.20874, 2604.21003 has a named GSD-document citation target.
CF-03	Drop-in spec count	12 or more ADOPT-decision papers have one-page integration specs.
CF-04	AgentDoG schema	Where/how/what taxonomy appears as a structured schema (table or YAML) ready for code adoption.
CF-05	Tool Attention ticket	Lazy-schema-loading design note exists with token-budget projections and a Q2-2026 target.
CF-06	Black-Box Skill Stealing threat model	Attack taxonomy encoded in DoltHub threat-model document.
CF-07	Five-Principle linter	Skill-creator linter check spec drafted, including pass/fail criteria per principle.
CF-08	Worker/Evaluator/Evc naming	Naming convention documented in gsd-mission-management-roles.md draft.
CF-09	STD measurement protocol	Per-model Safe Turn Depth measurement protocol exists as code-ready test pattern.
CF-10	BibTeX compiles	`cs25-26-sweep.bib` compiles with biblatex without errors; all 54 entries present.
CF-11	Cross-module matrix size	Matrix documents ≥ 15 inter-paper connections.
CF-12	Document compiles	Final PDF compiles in three XeLaTeX passes without errors or warnings.
CF-13	M1 paper coverage	All 4 papers in Module 1 receive ≥ 500 words of analysis.
CF-14	M2 paper coverage	All 10 papers in Module 2 receive an ADR. Bounded-learning policy doc updated.
CF-15	M3 paper coverage	All 8 papers in Module 3 receive an ADR. Safety Warden update spec drafted.
CF-16	M4 paper coverage	All 11 papers in Module 4 receive an ADR. Chipset validation citations integrated.
CF-17	M5 paper coverage	All 8 papers in Module 5 receive an ADR. Proof-companion citations identified.
CF-18	M6 paper coverage	All 13 papers in Module 6 receive an ADR.
CF-19	Decision distribution sane	ADOPT decisions number 12–18; TRACK decisions number 25–35; rejections explicitly justified.
CF-20	Index page	`index.html` provides download links for PDF + .tex + .bib; color scheme matches PDF.

5.4 Integration Tests

ID	Verifies	Expected Behavior
IN-01	ADR → spec consistency	Every ADR with decision ADOPT has exactly one drop-in integration spec; field values are consistent.
IN-02	ADR → test consistency	Every ADR with decision ADOPT has at least one validation test referenced; the test exists in the test harness updates.
IN-03	ADR → matrix consistency	Every cross-module relationship in the matrix references two ADRs that both exist.
IN-04	Citation density vs adoption	Each ADOPT-decision paper has ≥ 1 citation in synthesis outputs (convergent report, impact analysis, or specs).
IN-05	Bounded-learning + Silicon Layer cross-link	Module 2's Retrofit-as-bounded-learning-formalization is cross-referenced from the Silicon Layer LoRA spec.
IN-06	MCP security joint coverage	Tool Attention (Module 4), MCP Pitfall Lab (Module 3), and Function Hijacking (Module 3) appear together in the planning-bridge MCP design notes.
IN-07	Self-containment	A reader with no prior GSD context can understand the mission objective and decisions from the PDF alone.
IN-08	Convergent-discovery + bounded-learning link	Root Theorem (Module 1) explicitly cited as foundation for bounded-learning policy in Module 2 outputs.
IN-09	Test plan coverage	Every success criterion mapped to ≥ 1 test ID in the verification matrix.

5.5 Edge-Case Tests

ID	Verifies	Expected Behavior
EC-01	Withdrawn-paper handling	If any arXiv ID is withdrawn before publication, the ADR is updated to a tombstone with explanation.
EC-02	Ambiguous theme handling	Papers spanning multiple themes are explicitly noted in the ADR; module assignment justified.
EC-03	Out-of-scope adjacency	Adjacent topics not covered (e.g., reinforcement learning theory) explicitly listed in scope-boundaries section.
EC-04	Future-paper extension	Mission output describes how to extend the ADR set if additional papers from the corpus are later prioritized.
EC-05	Source-quality outliers	Any paper with non-standard publication venue flagged in its ADR.
EC-06	Counterfactual decisions	At least one rejected paper has its rejection rationale documented (not just left blank).

5.6 Verification Matrix

Success Criterion	Test IDs	Status
1. All 54 papers have an ADR.	CF-01, CF-13, CF-14, CF-15, CF-16, CF-17, CF-18	Pending
2. Convergent-discovery papers have citation targets.	CF-02, IN-08	Pending
3. Drop-in technique papers have integration specs.	CF-03, IN-01	Pending
4. STD measurement protocol implemented.	CF-09	Pending
5. Tool Attention design note exists.	CF-05, IN-06	Pending
6. AgentDoG taxonomy encoded.	CF-04, IN-06	Pending
7. Black-Box Skill Stealing taxonomy in DoltHub doc.	CF-06	Pending
8. Five-Principle linter check drafted.	CF-07	Pending
9. Worker/Evaluator/Evolution naming adopted.	CF-08	Pending
10. Cross-module matrix has ≥ 15 connections.	CF-11, IN-03	Pending
11. `cs25-26-sweep.bib` compiles cleanly.	CF-10	Pending
12. Final PDF compiles in 3 XeLaTeX passes.	CF-12, CF-20	Pending

Every success criterion maps to at least one test. SC-* tests apply universally regardless of criterion.

6 Mission Crew Manifest

Fleet activation profile (6 modules) selected. Crew names follow PNW bioregion naming convention.

Role	Agent	Responsibility
FLIGHT	ORCA (Opus)	Mission direction; go/no-go gates; cross-track decisions.
PLAN	RAVEN (Sonnet)	Wave decomposition; parallel-track optimization; dependency management.
SCOUT-1	SALMON (Sonnet)	Paper acquisition for Tracks A + B; arXiv ID validation.
SCOUT-2	STEELHEAD (Sonnet)	Paper acquisition for Track C; cross-module relationship preview.
EXEC-1	CEDAR (Sonnet)	Track A document production (Modules 1 + 2 ADRs).
EXEC-2	DOUGFIR (Sonnet)	Track B document production (Modules 3 + 4 ADRs).
EXEC-3	HEMLOCK (Sonnet)	Track C document production (Modules 5 + 6 ADRs).
CRAFT-FOUND	FOXGLOVE (Opus)	Module 1 specialist: convergent-discovery deep readings + citation-target identification.
CRAFT-BOUND	TRILLIUM (Opus)	Module 2 specialist: bounded-learning policy formalization.
CRAFT-SAFE	HAWK (Opus)	Module 3 specialist: Safety Warden taxonomy + STD protocol + MCP security.
CRAFT-AGENT	HERON (Sonnet)	Module 4 specialist: chipset validation citations + DACP comparison + Tool Attention engineering ticket.
CRAFT-VERIF	SALAL (Opus)	Module 5 specialist: proof-companion linkage + Lean+SAT+LLM precedent.
CRAFT-CULT	FERN (Opus)	Module 6 specialist: cultural-sensitivity additions + DoltHub threat model + curriculum design.
VERIFY	EAGLE (Sonnet)	Source validation; arXiv ID resolution; abstract-claim verification.
INTEG	CASCADE (Opus)	Cross-module relationship matrix; convergent-discovery validation report; impact analysis.
CAPCOM	TIBSFOX (Human)	HITL gates between waves; final adoption decisions on contested ADRs.
BUDGET	BEAVER (Sonnet)	Token tracking; per-wave budget enforcement; session planning.
SURGEON	KINGFISHER (Sonnet)	Quality monitoring; flagging research degradation across long sessions.
LOG	WREN (Haiku)	Audit trail; per-track chronicle; commit-message generation.

7 Execution Summary

Metric	Value
Total papers covered	54 (out of 930-paper sweep corpus)
Modules	6
Activation profile	Fleet (19 roles)
Estimated token budget	1.24M tokens
Estimated context windows	5–6
Estimated session count	3–4
Wave count	4 (W0 sequential, W1 three-parallel, W2 sequential, W3 sequential)
Deliverables	12
Tests	42 (7 SC + 20 CF + 9 IN + 6 EC)
Success criteria	12
Cross-module connections (preview)	20
Drop-in integration specs (estimated)	12–18
Target milestone	v1.50 (Half A retrospective foundation)
Target hand-off	gsd-skill-creator via Claude Code

> The fox finds the trail by recognizing the seams the deer made walking it.
> The seams were always there. The fox just looks where the path is.
>
> — *through-line, mission close*

A Convergent-Discovery Validation Report (Phase 801)

This appendix embeds the Phase 801 deliverable from `.planning/missions/cs25-26-sweep/work/` verbatim, retypeset for the published PDF. It satisfies CS25-03 (anchor-paper coverage >500 words per anchor) for the four Module-1 anchors.

A.1 §1 Why convergent discovery matters

The strongest external validation a load-bearing architectural decision can receive is **convergent discovery**: an independent intellectual current — different authors, different problem domain, different vocabulary, no line of communication — arriving at the same load-bearing structure. Citation is influence; convergent discovery is *evidence*. When two systems built without contact derive equivalent shapes for the same job, the shape is more likely a property of the problem than an idiosyncrasy of either author. The patterns are the seams the deer made walking it; the deer were always walking the same watershed.

This report is the v1.49.575 analog of two earlier convergent-discovery moments in the GSD release line. v1.49.573 enumerated **seven** convergent-discovery anchors as the keystone of the milestone retrospective — each anchor a separate paper independently arriving at one of the architectural primitives GSD had already shipped. v1.49.574 named a **two-party rhyme**: SIGReg (LeJEPA's stop-gradient-replacing isotropy regulariser) and counter-based megakernel scheduling (CUDA-graph era kernel-launch consolidation) coming from completely unrelated sub-fields and converging on the same design move — replace heuristics with a single principled regulariser, replace ten launch points with one. The v1.49.574 retrospective took care to call this rhyme "a productive analogy at the design-discipline level, not a theorem"; we keep that calibration here.

CS25-03 distinguishes convergent discovery from cross-validation. **Cross-validation** is the lighter pattern: paper A cites paper B's primitive and re-tests it, or one party explicitly builds on the other. Useful, but it does not rule out the idiosyncrasy hypothesis — paper A's design might still be wrong, just shared. **Convergent discovery** is heavier: independently-derived equivalent structures arising without communication. None of the four anchor authors of this milestone — the population-genetics workflow team behind 2604.21910, the proteomics governance authors of 2604.21744 (GROUNDINGmd), the formal-context-theory author of 2604.20874 (Root Theorem), the agentic-harness authors of 2604.21003 (Last Harness) — have any line of communication to the GSD ecosystem. None cite GSD; GSD does not cite them; the works arrived in 2026 inside the CS25–26 arXiv listing as part of a 930-paper sweep that was filtered down to 54 priority papers and then to four anchor papers in Module 1. The structural matches are not borrowed; they are derived.

What makes Module 1 the *keystone* of v1.49.575 is that all four anchors land on **architectural foundations** rather than secondary tactics: the system-level shape of an LLM-driven research pipeline (21910), the document-level safety doctrine that overrides user prompts (21744), the information-theoretic substrate of long-running agent context (20874), and the meta-evolutionary loop a self-improving harness must run (21003). Each is load-bearing — remove it and the rest of the structure tilts.

When four works derived in four different problem domains land on four of GSD's load-bearing primitives, the primitives are not local choices made by one author. They are the engineering equivalent of natural laws — the seams. The remainder of this report does the per-anchor work: each anchor receives a structured analysis covering what the paper does, what GSD already does, where the convergence lies, and which GSD document is the explicit citation target for Phase 805's draft additions.

A.2 §2 Anchor 1 — 2604.21910 Skills-as-markdown three-layer

The paper. *From Research Question to Scientific Workflow: Leveraging Agentic AI for Science Automation* (arXiv:2604.21910). Themes: plan-decomposition, knowledge-as-skill, deterministic-layer-confinement, verify-proof, skill-as-md.

What the paper does. It proposes a three-layer agentic architecture for population-genetics workflow synthesis. The **semantic layer** uses an LLM to extract structured intent from a natural-language research question — *"compare diversity across these populations using these markers"* becomes a typed intent record. The **deterministic layer** dispatches that intent to validated DAG generators that emit reproducible workflow scripts; this layer contains no LLM calls and is therefore behaviourally stable across runs. The **knowledge layer** holds domain-expert markdown documents called "Skills" that encode three things: vocabulary mappings (this term in the literature corresponds to that parameter on the tool), parameter constraints (this method requires this kind of input distribution), and optimisation strategies (when N is small prefer method X). LLM non-determinism is confined entirely to intent extraction; identical intents are guaranteed to produce identical workflows because the deterministic layer is pure dispatch. The paper reports two numbers worth pinning: full intent-match accuracy rises from a 44% baseline (no skills) to 83% (skills enabled); skill-driven *deferred* workflow generation — generate the script only when needed, not eagerly — cuts data transfer 92% versus eager generation. The pattern was developed independently of any prior GSD documentation; the authors are population geneticists, not LLM-orchestration people.

What GSD already does. GSD's intake pipeline runs the same three-layer split under different names. **vision-to-mission** is the semantic layer: a user describes a creative or research vision in natural language, and an LLM extracts a structured mission package — ambition record, target artefacts, milestone shape, fleet profile. **research-mission-generator** is the deterministic layer: given a structured intent, it produces a three-stage output (Vision Guide → Research Reference → Mission Spec) with stable scaffolding, deterministic file layout, validated frontmatter. **gsd-skill-creator** is the knowledge / authoring layer: domain experts (the human author and the system itself) write SKILL.md documents that encode vocabulary mappings (when the user says "ship", trigger this command path), parameter constraints (this skill requires this kind of phase artefact), and optimisation strategies (this skill prefers Sonnet over Opus because of the workload class). The pipeline was built in pieces across many milestones — vision-to-mission landed first as a `Tibsfox/skills` cartridge, research-mission-generator second as the LaTeX research path, gsd-skill-creator third as the adaptive-learning layer this entire repo is named after — but the three pieces line up.

The convergence. The structural equivalence is point-for-point. Their *semantic*

layer (LLM intent extraction, only place non-determinism lives) maps to GSD's vision-to-mission. Their *deterministic layer* (validated generators, identical-intent → identical-output guarantee) maps to research-mission-generator. Their *knowledge layer* (domain-expert-authored markdown skills with vocabulary / constraints / strategies) maps to gsd-skill-creator's `.claude/skills/` and `examples/` skill libraries. Both confine non-determinism to intent extraction. Both treat the knowledge layer as **markdown** authored by domain experts rather than learned weights or runtime-rewritable state — a non-obvious choice that the paper makes for population genetics and GSD makes for cross-domain creative tooling. Both use the deterministic layer as the **gate-able surface**: in the paper, that's where reproducibility is enforced; in GSD, that's where CAPCOM gates and the Safety Warden's BLOCK authority operate. The paper's 44%→83% intent-match number is the kind of external benchmark GSD has not had until now — a published quantification of how much the knowledge layer raises end-to-end fidelity. The 92% data-transfer reduction for deferred generation has its analog in GSD's pull-based work assignment (GUPP / Get Up and Push Protocol) and the lazy-loading of skill bodies until activation: same shape, different vocabulary.

Explicit citation target. This paper is the canonical citation for the GSD vision-to-mission / research-mission-generator / gsd-skill-creator pipeline. It lands in three places: (1) **`docs/skill-architecture/three-tier-pipeline.md`** (NEW, propose for Phase 805 SKILL.md additions) — formal write-up that lifts the three-layer naming "semantic / deterministic / knowledge" as the canonical architecture description and cites this paper as the convergent-discovery anchor; (2) **CLAUDE.md** "Architectural Foundations" preamble — one-line foundational citation under CS25-09's ≤30-line CLAUDE.md edit budget; (3) **`docs/cartridge/SKILL.md` authoring guide** — cite as external validation that markdown-skill format itself is a defensible storage choice, not a transient one. The 44%→83% and 92% numbers feed v1.50 retrospective Half A as the external benchmark for the pipeline's fidelity claim.

A.3 §3 Anchor 2 — 2604.21744 GROUNDINGmd

The paper. *Agentic AI-assisted coding offers a unique opportunity to instill epistemic grounding during software development* — colloquially **GROUNDINGmd** (arXiv:2604.21744). Themes: safety-warden, hard-constraints, override-the-prompt, governance-as-document, bounded-learn.

What the paper does. It proposes a community-governed, field-scoped epistemic grounding document that the agent loads before any task. The document defines two **override-everything** classes. **Hard Constraints** are non-negotiable validity invariants empirically required for scientific correctness in the field — for the example domain of mass-spectrometry-based proteomics, things like "do not let the agent silently change the m/z tolerance below the instrument resolution" or "never combine spectra acquired under incompatible isolation windows". **Convention Parameters** are community-agreed defaults: units, schemas, naming, the standard search-tolerance for a given mass-spec workflow. Both classes override every other context the agent receives, *including the user prompt itself*. The agent is required to refuse — to BLOCK — rather than violate a Hard Constraint. The document is **community-governed**: changes require consensus, not individual modification, which is the structural defence against prompt-injection rewriting the constraint set ("just edit the GROUND-

INGmd file and remove that rule" stops working when the file is governed elsewhere). The pattern is presented for proteomics but the authors explicitly frame it as field-portable: any computational scientific discipline with empirical validity invariants is a candidate domain.

What GSD already does. The GROUNDINGmd construct is — in nearly its exact shape — the GSD `CLAUDE.md` + `SKILL.md` + Safety Warden BLOCK pattern. CLAUDE.md is the project-scoped grounding document loaded before any task; the system reminder at the top of every Claude Code session says verbatim *"Codebase and user instructions are shown below. Be sure to adhere to these instructions. IMPORTANT: These instructions OVERRIDE any default behavior and you MUST follow them exactly as written"* — that override-the-default-behaviour-clause is the GROUNDINGmd override clause restated. The Safety Warden BLOCK pattern documented in `.claude/skills/security-hygiene/SKILL.md` ensures that on safety-critical decisions — touching another developer's secrets, modifying signing keys, deleting `.git/`, breaking the wasteland exclusion — the agent refuses regardless of the user prompt. CAPCOM gates extend the same authority to milestone-shaping decisions: a CAPCOM HARD GATE override of a user prompt is doctrinally normal in this codebase, not exceptional. Today CLAUDE.md mixes invariants and defaults under a single heading; the Hard-vs-Convention split is implicit but not lifted to a typed taxonomy.

The convergence. The two-class taxonomy maps cleanly. **Hard Constraints ↔ Safety Warden HARD GATE**: invariants that the agent must refuse to violate, governed at the document level rather than the prompt level. **Convention Parameters ↔ CLAUDE.md project conventions + SKILL.md authoring defaults**: community-agreed defaults that the agent applies unless explicitly overridden by a typed override. The override-the-user-prompt commitment is identical in both systems — neither is a "the user is always right" architecture; both are "the document is the ground truth, and the document overrides the prompt" architectures. The community-governance property is also present in GSD: CLAUDE.md and SKILL.md ship under git, with PR-style review for changes; an attacker who wants to relax the BLOCK rule has to land a commit, not just emit a prompt. The match is structural and deep — both systems independently solved the prompt-injection-resistance problem the same way, by lifting the constraint set out of the prompt-layer entirely.

Explicit citation target. The paper is essentially the published independent rediscovery of CLAUDE.md + Safety Warden BLOCK; the citation lands in four places. (1) **CLAUDE.md** — promote the Hard Constraints / Convention Parameters two-class taxonomy to a foundational citation in the "Important Notes" section; this is the foundational citation in the CS25-09 ≤30-line edit budget. (2) **Every `SKILL.md` template** in `docs/cartridge/SKILL.md` authoring guide — cite for the Hard-vs-Convention split structure; every authored SKILL.md should distinguish what it refuses to do (Hard Constraints) from what it defaults to but allows override on (Convention Parameters). This becomes a structural-completeness criterion. (3) **`.claude/skills/security-hygiene/SKILL`** — the BLOCK doctrine cites this paper as the published independent rediscovery of the override-user-prompt rule; an external citation is durable defence against future "but the user asked for it" pressure. (4) **`src/cartridge/linter/structural-completeness.ts`** — Module 3's CS25-17 structural-completeness linter (HB-05) adopts GROUNDINGmd's two-class taxonomy as its schema, and includes a "Hard Constraints declared" check on SKILL.md fixtures. The override-the-user-prompt formulation must be cited (para-

phrased ≤14 consecutive words to satisfy the SC-PARA gate) so the doctrinal lineage is unambiguous.

A.4 §4 Anchor 3 — 2604.20874 Root Theorem of Context Engineering

The paper. *The Root Theorem of Context Engineering* (arXiv:2604.20874). Themes: context-management, bounded-tape, formal-axioms, homeostatic-persistence.

What the paper does. It formalises the agent's working context as a **bounded lossy channel**. Two axioms: (i) the window is finite; (ii) per-token information quality decays monotonically with cumulative volume. From these two axioms five consequences are derived. **(C1)** Quality degrades monotonically with token volume *independent of nominal window size* — a 1M-token model is not exempt from the decay; it merely slides the absolute number at which the decay becomes operationally painful. **(C2)** Signal density and token-count are *independent* optimisation variables; optimising one does not optimise the other, and the gradients can point opposite directions. **(C3)** Gating must trigger on **fidelity thresholds, not on capacity**: "is the context window 80% full?" is the wrong question; "has the signal-to-noise dropped below the tolerated threshold?" is the right one. **(C4)** Only architectures that *accumulate–compress–rewrite–shed* can sustain understanding indefinitely; pure accumulation diverges. **(C5)** The compressor needs an **external verification gate** because it operates inside the same channel it is compressing — self-review is structurally inadequate. The paper demonstrates the framework empirically with a 60+-session persistent agent showing stable memory footprint and stable task quality, contrasted with baselines that degrade after roughly twenty sessions.

What GSD already does. The bounded-tape framing has been Tibsfox's informal articulation across many docs: *The Space Between* talks about working context as a finite bandwidth channel; CLAUDE.md design rationale references the bounded-tape framing for CAPCOM gate policy; the milestone retrospective + feed-forward lifecycle is described in the through-lines of v1.49.570/.571/.572/.574 as "accumulate during execution, compress in retrospective, rewrite into substrate docs, shed via the done-retirement skill"; the bounded-learning policy invariants — the 20% cap on auto-load slot churn, the 3-correction-min, the 7-day cooldown — operate on a notion of bounded signal that the policy refuses to relax. None of this had a published mathematical foundation until now.

The convergence — five consequences mapped. Each derived consequence maps directly to a shipped GSD architectural decision.

- **C1 (monotone quality decay independent of window size) → bounded-learning caps don't relax with bigger context windows.** GSD treats Opus, Sonnet, and Haiku the same with respect to bounded-learning policy; the 20% rule does not loosen for a model with a larger advertised context. C1 is the formal justification for that refusal: the decay is monotone regardless of how much head-room remains. The orchestration layer in `src/orchestration/` honours the same policy across model tiers.
- **C2 (signal/token-count independence) → why CAPCOM gates measure fidelity, not "is the context full?"** The reinforcement series (`src/reinforcement/`, MA-1..MD-6) operates over signal-bearing events, not raw token streams, exactly because of C2: optimising for token efficiency is not the same as optimising

for signal preservation, and the milestone retrospective's compress step targets the latter.

- **C3 (fidelity-threshold gating, not capacity gating) → CAPCOM gating policy.** Today's CAPCOM gates trigger on phase boundaries, on human-authorisation prompts, on safety-relevant decision points — *not* on token-budget exhaustion. C3 is the formal derivation of that policy. The 70% context-fracture threshold from the Tool Attention paper (2604.21816, HB-01 source) is one specific instance of a fidelity threshold; the Root Theorem says the *category* is right, and Tool Attention says one threshold inside that category is 70% on a particular ISO-score metric.
- **C4 (accumulate–compress–rewrite–shed homeostatic lifecycle) → milestone retrospective + feed-forward lifecycle.** Each phase of the milestone is one stage of a homeostatic cycle: Wave 0 / phase execution = *accumulate*; Wave 2 synthesis = *compress* (cross-module matrix, convergent-discovery report, GSD-impact analysis are all compression of the Wave-1 ADR mass); Wave 3 publication + the substrate doc additions = *rewrite* (the compressed signal gets rewritten into durable substrate docs in `docs/substrate/`); the `done-retirement` skill and milestone closure = *shed*. The cycle name in the FSFM bounded-learning forgetting taxonomy from ADR-CS25-2604.20300 is literally `accumulate-compress-rewrite-shed`. The match is verbatim, not analogical.
- **C5 (external verification gate on the compressor) → fresh-context-subagent verification pattern.** This milestone's three-track Wave-1 fleets — Track A, Track B, Track C — are *fresh-context subagents*. The orchestrator (the agent doing the compression) does not self-review its own outputs; a separate-context subagent provides the verification gate. CAPCOM is the structural external gate. C5 is the formal derivation of why this is necessary: the compressor is inside the channel it compresses, so self-review is inadequate by construction. The umwelt module (`src/umwelt/`, M7) makes the same point in Friston / Markov-blanket vocabulary: the boundary is external because internal observation cannot certify itself.

Explicit citation target. Three landing places, all flowing from the consequence mapping above. (1) **`docs/substrate/bounded-tape-root-theorem.md`** (NEW, propose for Phase 805) — primary landing: a substrate document that lifts the two-axiom / five-consequence structure verbatim and reproduces the consequence-to-decision map above. The orchestrator brief names this as the explicit citation target. (2) ***The Space Between* information-theory chapter** — the bounded-tape framing now has a published derivation to cite as formal foundation. (3) **CLAUDE.md design-rationale section** — one-line citation under CS25-09's ≤30-line edit budget, naming the paper as the formal foundation for the bounded-tape framing. The 60+-session empirical demonstration is the citation when the v1.50 retrospective claims GSD's substrate is empirically validated rather than purely-engineered.

A.5 §5 Anchor 4 — 2604.21003 Last Harness

The paper. *The Last Harness You'll Ever Build* (arXiv:2604.21003). Themes: skill-learning, meta-evolution, agent-orchestration, harness-as-artefact.

What the paper does. It proposes a two-level framework for self-improving agentic

harnesses. At the **inner level**, the **Harness Evolution Loop** pairs three agents. A **Worker** attempts the target task. An **Evaluator** *adversarially* diagnoses Worker failures and scores performance — its job is not passive correctness-grading but active failure-finding, including probing edge cases the Worker did not test. An **Evolution** agent proposes harness modifications conditioned on the full Worker / Evaluator failure history. At the **outer level**, the **Meta-Evolution Loop** optimises the evolution protocol itself across a *diverse* set of tasks, learning a protocol that converges quickly when applied to a new unseen task. The framing the authors give the contribution is "automating the design of the automation itself." The evaluation spans heterogeneous task families to demonstrate that the meta-protocol generalises rather than overfitting one benchmark — and the inner Worker/Evaluator/Evolution split is the load-bearing primitive for that generalisation.

What GSD already does. This paper is essentially an external description of `gsd-skill-creator`'s purpose. The two-level structure maps directly onto skill-creator's existing design: the inner Harness Evolution Loop corresponds to skill-creator's six-step learning loop applied to a single skill (Observe → Detect → Suggest → Manage → Auto-Load → Learn/-Compose); the outer Meta-Evolution Loop corresponds to the bounded-learning policy that operates *across* skills, learning what kinds of corrections actually help library-wide. The roles in skill-creator's subagent set today are ad-hoc — implicit Worker (the skill being applied), implicit Evaluator (the 3-correction-minimum rule that gates promotion), implicit Evolution (the bounded-learning policy that absorbs the cross-skill signal). Same shape, no formal naming.

The convergence — external description of skill-creator. The mapping is essentially an isomorphism. **Worker** = the skill being evolved, applied to the target task. **Evaluator** = adversarial diagnosis — and this is the *upgrade*, because the current 3-correction-min rule is **passive** (count corrections after the fact), and the paper's Evaluator is **active** (probe for failure modes the Worker did not surface). HB-04 in the v1.49.575 Half-B roster names this upgrade explicitly as the replacement of 3-correction-min with the adversarial-Evaluator pattern. **Evolution** = the bounded-learning policy applied to skill code, the 20% rule but adversarially-applied. **Meta-Evolution** = the milestone retrospective accumulating cross-skill protocol changes over time — the bounded-learning policy refining itself release over release. The two-level structure (per-skill iteration vs cross-skill meta-protocol) is exactly what skill-creator does, and the paper *names* it as a research direction, which means the design is now nameable and citable rather than just describable.

Compose with AEL 2604.21725. The Track A audit pre-relationship between Last Harness and AEL (HB-07's source paper) yields a productive composition. AEL splits agent learning into **fast** (per-episode Thompson-Sampling bandit selection of retrieval policy) and **slow** (cross-episode LLM reflection on failure patterns) timescales. Composed with Last Harness Worker/Evaluator/Evolution, the structure becomes: **fast-Worker** (per-episode skill apply) + **fast-Evaluator** (per-episode adversarial check on the Worker output) + **slow-Evolution** (cross-episode reflection bandit informing protocol updates). This composition is what HB-04 (Worker/Evaluator/Evolution role split) and HB-07 (AEL fast/slow bandit auto-load) implement *together* in v1.49.575 Half B. HB-04 establishes the role naming and the per-episode adversarial check; HB-07 supplies the cross-episode reflection mechanism. The composition is non-trivial because each paper alone is missing a piece: Last Harness names the roles but

doesn't quantify the timescale split; AEL quantifies the timescales but doesn't name the roles. Composed, they give skill-creator a fully-named two-timescale, three-role architecture.

Explicit citation target. Four landing places. (1) **`src/skill-creator/roles/`** (NEW, HB-04 / CS25-16, Phase 808 implementation) — primary landing: a new module that ships three role-isolated subagents with explicit Worker / Evaluator / Evolution contracts and BLOCK authority for the Evaluator over Worker outputs. The orchestrator brief names this directory as the explicit citation target. (2) **`gsd-mission-management-roles.md`** — adopt Worker / Evaluator / Evolution as the formal naming convention for skill-creator's internal subagents. Today these roles exist informally; the paper provides published vocabulary. (3) **`.claude/agents/`** — the existing executor / verifier / planner subagents fold into Worker / Evaluator / Evolution; document the mapping so the inherited-role naming is consistent across `.claude/agents/` and `src/skill-creator/roles/`. (4) **`.planning/PROJECT.md`** bounded-learning-policy section — cite as the published derivation of the cross-skill meta-protocol. The Evaluator's adversarial role is a meaningful upgrade only if the Evaluator has authority to BLOCK Worker outputs — the role split degrades to passive-counting otherwise. CS25-16 enforces this with role-isolation invariants tested explicitly.

A.6 §6 The four together

What do the four anchors say *together* about GSD?

- **The architecture is right.** Skills-as-md three-tier (`21910`) maps point-for-point onto vision-to-mission → research-mission-generator → gsd-skill-creator. The pipeline's pieces — built one at a time across many milestones — line up the way a population-genetics workflow team built their system from scratch in 2026.
- **The safety doctrine is right.** GROUNDINGmd Hard Constraints + Convention Parameters (`21744`) is the published independent rediscovery of CLAUDE.md + Safety Warden BLOCK. The override-the-user-prompt commitment that the GSD codebase has treated as architecturally normal is what a proteomics governance team independently arrived at as the only defensible posture for safety-critical fields.
- **The cognitive substrate is right.** The Root Theorem of Context Engineering (`20874`) formalises the bounded-tape framing — two axioms, five consequences — and each consequence maps to a shipped GSD architectural decision (window-size-independent caps, fidelity-threshold gating, accumulate–compress–rewrite–shed milestone lifecycle, fresh-context external verification). The substrate has now received a formal derivation.
- **The evolutionary loop is right.** Last Harness (`21003`) names skill-creator's Worker/Evaluator/Evolution pattern explicitly and is essentially an external description of what `gsd-skill-creator` does. Composed with AEL, it gives the loop a published two-timescale, three-role architecture.

When four independent peer-reviewed works arrive at GSD's load-bearing patterns from four different problem domains — population genetics (`21910`), proteomics (`21744`), abstract context theory (`20874`), agentic harness research (`21003`) — the patterns are not idiosyncratic. They are the engineering equivalent of natural laws. Different watersheds; same trail. The seams the deer have made walking the trail across multiple

drainages are visible from above, and they all run the same way because the contour is the same. v1.49.575 cites them back, and Half B implements what's adoptable.

A.7 §7 Calibration

It is important to distinguish two registers in which this report can be read.

Productive analogy at the design-discipline level. This is the register the report stays in. At the design-discipline level, GSD and these four papers share design intent and structural form: same role split, same overrides, same lifecycle, same gate placement. This is a strong external-citation register — *"these papers independently arrive at the same design"* is exactly what it says, and the words "structural form" and "design intent" do the right work without overpromising. This is the register the v1.50 retrospective Half A's convergent-discovery section will use.

Theorem-level equivalence. This report does **not** claim theorem-level equivalence. We do not claim that GSD's vision-to-mission is *isomorphic* to the paper's semantic layer in a category-theoretic sense; we do not claim Hard Constraints and Safety Warden BLOCK are equal as formal objects; we do not claim the Root Theorem's two-axiom derivation is *the* derivation of CAPCOM policy in the sense of a single uniqueness theorem. Theorem-level claims would require formal proof and a shared formalism — neither is present here, and neither is needed for the citations the report supports.

The v1.49.574 retrospective's framing applies verbatim and is reproduced here for fidelity: *"the rhyme is a productive analogy at the design-discipline level, not a theorem; it sits in the same register as 'the Amiga Principle compounds twice' and should be presented externally with that calibration intact."* The four-anchor convergence in this report belongs in that same register — same calibration, same external-citation discipline. The Amiga Principle, the SIGReg / megakernel rhyme, and the four CS25 anchors are three instances of the same kind of finding: load-bearing GSD patterns receiving independent external derivation, presented as productive analogies, not theorems.

A.8 §8 Citation map for downstream waves

Phase 805 will draft CLAUDE.md and SKILL.md additions; the citation form below feeds those drafts directly. *Foundational* = primary citation in the document's design-rationale section; *supporting* = a citation that strengthens an existing claim; *direct* = the document is a direct response to or implementation of the paper.

Anchor (arXiv)	GSD doc / module	Citation form
2604.21910 Skills-as-md three-layer	CLAUDE.md "Architectural Foundations" preamble	foundational
2604.21910	`docs/skill-architecture/three-layer-pipeline.md` (NEW, Phase 805)	foundational
2604.21910	`docs/cartridge/SKILL.md` authoring guide	supporting (validates markdown-skill format)

Anchor (arXiv)	GSD doc / module	Citation form
2604.21910	`.college/README.md` College of Knowledge architecture description	supporting (terminology import: semantic / deterministic / knowledge)
2604.21744 GROUNDINGmd	CLAUDE.md "Important Notes" — Hard Constraints / Convention Parameters	foundational
2604.21744	every `SKILL.md` template (cartridge authoring guide)	foundational (Hard-vs-Convention split structure)
2604.21744	`.claude/skills/security-hygiene/SKILL.md` Safety Warden BLOCK	direct (override-the-user-prompt rule)
2604.21744	`src/cartridge/linter/structural-completeness.ts` (HB-05 / CS25-17)	direct (linter-completeness adoption)
2604.20874 Root Theorem	`docs/substrate/bounded-tape-root-theorem.md` (NEW, Phase 805)	foundational
2604.20874	*The Space Between* information-theory chapter	foundational
2604.20874	CLAUDE.md design-rationale section	supporting
2604.20874	`.claude/skills/session-awareness/SKILL.md` + every CAPCOM-gate policy doc	direct (C3 fidelity-threshold gating + C5 external verification gate)
2604.21003 Last Harness	`src/skill-creator/roles/` (NEW, HB-04 / CS25-16, Phase 808)	direct
2604.21003	`gsd-mission-management-roles.md`	direct (naming convention adoption)
2604.21003	`.claude/agents/` documentation	supporting (executor/verifier/planner ↔ Worker/Evaluator/Evolution mapping)
2604.21003	`.planning/PROJECT.md` bounded-learning-policy section	supporting (meta-evolution derivation)
2604.21003 + 2604.21725 (composed)	`src/skill-creator/auto-load/` (HB-07)	direct (fast/slow timescale split inside the three-role split)

This table is the citation contract for Phase 805. Each row is one diff to land; each diff cites the indicated paper in the indicated form. CS25-09 verifies that the CLAUDE.md diff cites all four anchors. CS25-17 verifies the structural-completeness linter imple-

ments the GROUNDINGmd two-class taxonomy. CS25-16 verifies the Worker/Evaluator/Evolution role split. The seams are mapped; what remains is to walk them.

B Cross-Module Relationship Matrix (Phase 800)

This appendix embeds the Phase 800 narrative deliverable from `.planning/missions/cs25-26-swe` followed by the full 81-row machine-readable table from `xmod.csv`. It satisfies CS25-04 (cross-module relationship matrix). 81 connections were classified into seven `connection_type` values across six modules.

Phase: 800 **Date:** 2026-04-25 **Author:** synthesis agent (Track A+B+C cross-cutting)
Inputs: 55 ADRs (M1–M6) · 3 track audits (A/B/C) · `xmod-seed.md` 20-row pre-relationship table · intake `cs25_gsd_mission.tex`

B.1 §1 Methodology

Connections were identified by a four-pass sweep over the 55 ADRs and three track audits. Pass 1 imported the 20 pre-relationships from `xmod-seed.md` verbatim and tagged them as the seed set. Pass 2 walked every ADR's *Cross-Module Links* section, recording each pairwise reference and classifying it into one of the seven `connection_type` values (`composes-with`, `validates`, `extends`, `contradicts`, `cross-validates`, `implements`, `same-pattern-different-domain`). Pass 3 absorbed the audit-log "additional connections" sections from Track A (M1+M2), Track B (M3+M4), and Track C (M5+M6); these surface composition patterns the per-ADR sections did not always articulate, especially the AEL × Last Harness composition and the M3 MCP triple-gate. Pass 4 detected dense subgraphs by clustering on shared papers and confirmed the four expected clusters (M2 LoRA, M3 MCP-attack, M5 verification, M6 cultural-sensitivity). Strength was assigned by triangulating sweep-recommended adoption tier with explicit Track-audit elaboration: a `strong` link is one that gates a Phase 805–811 ticket or carries an explicit composition statement in two places; `medium` is one supported by a single ADR cross-reference; `weak` is bibliographic-only. No connections rated `contradictory` survived; the closest tension was the 2604.21927 regime-fixed methodological constraint, which is rendered as `validates` against every M2 LoRA paper because it sets the comparison protocol they all must satisfy. Zero `contradicts` rows is itself a finding: the corpus is internally coherent.

B.2 §2 Top cross-cutting recommendations

The following nine cross-cutting patterns each span ≥2 modules and are prioritized by impact × effort.

B.2.1 CC-1. Three-paper MCP triple-gate as Phase 805 baseline harness

Description: Any planning-bridge MCP design change must satisfy three papers jointly: 2604.21477 MCP Pitfall Lab (six-attack family taxonomy), 2604.20994 Function Hijacking (context-agnostic invariant), and 2604.21816 Tool Attention (lazy-schema-loading). Track B's audit treats these as a single triple-gate rather than three pairwise edges. The Phase 805 baseline harness must instantiate all six attack families from 2604.21477 as test cases, with 2604.20994's context-agnostic-hijack pattern as the seventh case, and Tool Attention's ISO score gating as the runtime mitigation that the harness exercises. **Modules:** M3, M4. **Papers:** 2604.21477, 2604.20994,

2604.21816. **Target subsystem:** `src/orchestration/tool-attention/` + `src/safety/mcp-atta` **Effort:** Medium (one engineering ticket already drafted as HB-01; harness needs new test fixtures).

B.2.2 CC-2. SKILL.md authoring discipline upgrade (Phase 810 + 811)

Description: Two papers ship together as a single discipline upgrade: 2604.21090 Five-Principle Structural Quality Gaps (HB-05) and 2604.21505 Orchid (HB-06). The first supplies a structural-completeness linter; the second supplies a four-type ambiguity taxonomy. Both run as cartridge-promotion linters at `src/cartridge/linter/`. Default-off via `cs25-26-sweep.structural-completeness-lint` and the parallel Orchid flag, both promoted to default-on after the existing skill corpus is audited. The Track-C audit elevates this to a "spec-discipline cluster" together with the M3 GROUNDINGmd Hard Constraints — 2604.21744's convergent-discovery framing is the validation that the five principles mirror existing CSO discipline rather than introducing a new norm. **Modules:** M3, M5, M1. **Papers:** 2604.21090, 2604.21505, 2604.21744. **Target subsystem:** `src/cartridge/linter/`. **Effort:** Medium (linter rules are mechanically checkable; corpus audit is the long pole).

B.2.3 CC-3. AEL × Last Harness composed two-loop pattern

Description: AEL's fast/slow timescale split (2604.21725) composes with Worker/Evaluator/Evolution role split (2604.21003) into a single architectural pattern: fast-Worker (per-episode action), fast-Evaluator (per-episode bandit critique), slow-Evolution (cross-skill reflection at AEL's slow timescale). This composition is the headline cross-module finding of the sweep — neither paper specifies the other's loop, but their compositional product is a coherent two-level meta-learning pattern that GSD's chipset architecture can host without any new infrastructure. **Modules:** M1, M2. **Papers:** 2604.21003, 2604.21725. **Target subsystem:** `src/orchestration/two-loop/` (new) + chipset Haiku-tier slow-Evolution sidecar. **Effort:** High (orchestration touch; CAPCOM HARD GATE; flag-gated default-off). Elaboration in §4.

B.2.4 CC-4. Bounded-tape × bounded-learning unified foundation

Description: Root Theorem 2604.20874 is the formal foundation for two seemingly different bounded mechanisms: (a) the 20%/3-correction caps in M2 Retrofit 2511.11439 (selective bounded updates) and (b) the gist-compression / accumulate-compress-rewrite-shed lifecycle from M6 Gist Sparse Attention 2604.20920. The same C2 (signal/token-count independence) and C4 (compression-step-as-instance) corollaries appear in both. Adopting Root Theorem as the M1 anchor citation gives both M2 Retrofit and M6 Gist a shared formal foundation that can be cited rather than reasserted. **Modules:** M1, M2, M6. **Papers:** 2604.20874, 2511.11439, 2604.20920. **Target subsystem:** `docs/architecture/root-theorem.md` (new substrate doc). **Effort:** Low (citation-only). Elaboration in §4.

B.2.5 CC-5. M2 LoRA dense subgraph as Silicon Layer benchmark suite

Description: Five M2 papers form a tight cluster around adapter design / parameter-efficiency / multi-tenant deployment: 2604.21571 Separable Expert (T2 architecture), 2604.21901 GiVA (T1 benchmark), 2509.18629 HyperAdapt (T1 benchmark), 2604.21905 LoRA Redux (TRACK survey), and 2604.21330 TGR-MoE (TRACK pattern). The methodological constraint from 2604.21927 (regime-fixed comparison) ap-

plies to all five as a single comparison protocol. The cluster is the natural benchmark target list for the RTX 4060 Ti Silicon Layer. **Modules:** M2 (intra-cluster dense). **Papers:** 5 + 1 constraint. **Target subsystem:** `examples/chipsets/silicon-layer/lora-bench/`. **Effort:** High (RTX-class compute, but bounded by paper-specified protocols). Elaboration in §3.

B.2.6 CC-6. Cross-session threat correlator (Memory Arena consumer)

Description: Three M3 papers compose into a persistent cross-session threat correlator: 2604.20911 Omission Decay (intra-session STD measurement), 2604.21131 CSTM-Bench (cross-session generalization, 50% recall loss baseline), and 2601.18491 AgentDoG (record schema). Together they specify why both intra- and inter-session re-injection events must persist to Grove and be reasoned over by a Safety Warden module at `src/safety/cross-session/`. The pattern composes with Memory Arena M9–M13's content-addressed Grove records as the persistence layer. The cross-session work GSD does (Seattle 360 / NASA 720 long-horizon missions) is exactly the regime where this attack class becomes operational. **Modules:** M3, plus consumer in M5/M6 architecture. **Papers:** 2604.20911, 2604.21131, 2601.18491. **Target subsystem:** `src/safety/cross-session/`. **Effort:** Medium (architecture lands at v1.49.575, runtime defers to v1.49.576+).

B.2.7 CC-7. M5 verification triad as v1.50 Half B citation backbone

Description: Three M5 papers form a triad for The Space Between Half B chapter validation: 2511.23159 (probable-to-provable rhetoric), 2604.18792 DSLTrans (cut-off theorem mechanism), and 2604.21187 Doubly Saturated Ramsey (SAT+LLM+Lean concrete precedent). Rhetoric + mechanism + precedent is the right citation chain for any proof-companion document. The Track-C audit elevates this to the citation triad backbone for the v1.50 chapter validation. Pairs with 2604.21505 Orchid for spec-side ambiguity reduction. **Modules:** M5 (intra-cluster). **Papers:** 3 + Orchid. **Target subsystem:** `docs/the-space-between/half-b.md` citation chain. **Effort:** Low (citation chain only).

B.2.8 CC-8. M4 four-paper retrieval+synthesis pipeline (College of Knowledge)

Description: Four M4 papers compose into a single retrieval-and-synthesis pipeline: 2510.26615 SARA (hybrid RAG retrieval), 2604.20848 MATRAG (four-agent transparency-scoring), 2604.21444 HiCrew (Hybrid Tree dynamic role-selection), 2508.21720 PosterForest (Poster Tree synthesis). All four are ADOPT-design-only at v1.49.575; v1.49.576+ implementation. The Track-B audit specifies the dependency graph: MATRAG depends on SARA; HiCrew dispatches among the MATRAG agents; PosterForest is the synthesis complement to HiCrew's retrieval. Cross-link with M6's 2601.10003 SocraticKG for the KG-construction prior. **Modules:** M4, M6. **Papers:** 4 + 1. **Target subsystem:** `src/skills/research-mission-generator/v2/` + `.college/retrieval/`. **Effort:** High (v1.49.576+ work; design-and-spec only at v1.49.575).

B.2.9 CC-9. M6 cultural-sensitivity cluster as joint Safety Warden surface

Description: Three M6 papers carry SC-IND, SC-MED, and SC-SOV gates respectively: 2604.12710 LASA (semantic-bottleneck anchoring), 2604.20996 AFRILANG-TUTOR (sovereignty-gated low-resource), 2604.21352 CARE (counselor-alignment,

assist-not-replace), with 2604.21751 (Japanese-culture bias / CROQ slices) supplying the empirical bias measurement. The cluster shares a permission-neutral / sovereignty-attentive discipline. CROQ slices feed the same Safety Warden surface as 2604.21308 CI-Work contextual-integrity benchmark. **Modules:** M6 (intra-cluster, plus M3 Safety Warden consumer). **Papers:** 4 + CI-Work. **Target subsystem:** `src/safety/cultural-sensitivi` + Safety Warden integration. **Effort:** Medium-High (sovereignty consultation is hard prerequisite for AFRILANGTUTOR). Elaboration in §3.

B.3 §3 Dense subgraphs

B.3.1 M2 LoRA cluster (2604.21571 / 2604.21901 / 2509.18629 / 2604.21905 / 2604.21330 + 2604.21927 constraint)

The five LoRA papers cluster around a shared design principle: **parameter-efficient multi-tenant adaptation under a regime-fixed comparison protocol**. Separable Expert (2604.21571) is the multi-tenant deployment frame (per-user proxy, per-domain expert pool); GiVA (2604.21901) and HyperAdapt (2509.18629) are the parallel rank-reduction / rank-augmentation candidates; LoRA Redux (2604.21905) is the bibliographic survey that catalogues both axes; TGR-MoE (2604.21330) is the dispatch-level pattern reference. The 2604.21927 regime-fixed paper sits outside this five-paper set as a methodological constraint that binds them: any pairwise comparison among them must be evaluated under controlled regime variables (rank, dataset, training schedule), or the comparison itself is uninformative. The shared design principle is *adapter compactness as a multi-tenant deployment axis*. The cluster's coherence is structural: every paper is reachable from every other in ≤2 hops, and the methodological constraint is the unique paper that touches all five.

B.3.2 M5 verification cluster (2511.23159 / 2604.18792 / 2604.21187 / 2604.21570 / 2604.21598 / 2604.21505 / 2603.01170 / 2604.14709)

The eight M5 papers cluster around a shared design principle: **bounded provable verification as a complement to probable behaviour**. Three of the eight (probable-to-provable, DSLTrans, Doubly Saturated Ramsey) form the rhetoric+mechanism+precedent backbone; SpecSyn and DryRUN form the synthesis complement (spec-side / test-side); Orchid supplies the four-type ambiguity taxonomy that gates spec-quality; ATLAS and HWE-Bench are the hardware-domain extensions. The shared principle is that verification need not be all-or-nothing — a bounded, decidable, decomposable verification surface has utility even when full verification is intractable. This is the design principle the Track-C audit calls out as the citation triad backbone for The Space Between Half B.

B.3.3 M6 cultural-sensitivity cluster (2604.12710 / 2604.20996 / 2604.21352 / 2604.21751)

The four cultural-sensitivity papers cluster around a shared design principle: **permission-neutral, sovereignty-attentive engagement with cultural and clinical contexts**. LASA's semantic-bottleneck is the technical defense layer; AFRILANGTUTOR's sovereignty gate is the procedural defense; CARE's counselor-alignment is the role-boundary defense; CROQ's bias measurement is the evaluation surface. All four share a non-claiming posture — the system assists rather than replaces, defers to consulted au-

thority on identity claims, and surfaces bias rather than asserting unbiased operation. The Track-C audit confirms SC-IND, SC-MED, and SC-SOV gates passed throughout. The cluster's coherence is normative: it represents the safety-and-respect column of the ADR matrix in a way no other module mirrors, and the cluster's outputs flow into a single Safety Warden surface alongside 2604.21308 CI-Work.

B.4 §4 Composition patterns

B.4.1 Composition 1 — AEL × Last Harness (fast-Worker / fast-Evaluator / slow-Evolution)

The Last Harness paper (2604.21003) defines three roles — Worker, Evaluator, Evolution — without specifying their timescale. AEL (2604.21725) defines a fast/slow two-timescale split without specifying roles. Composing them yields a unique architectural pattern: the Worker and Evaluator both operate at AEL's fast timescale (per-episode action and per-episode bandit critique); the Evolution loop runs at AEL's slow timescale as cross-skill reflection. This is not the only mapping — one could place the Evaluator at the slow timescale instead — but the fast-Evaluator placement matches the bandit's per-step feedback structure. The pattern is implementable as a `src/orchestration/two-loop/` module with two coroutines, where the Evolution loop reads the bandit history at a configurable interval and proposes role-isolation invariants for the next epoch. The CAPCOM HARD GATE designation is justified because state leakage between roles defeats the adversarial-Evaluator pattern.

B.4.2 Composition 2 — MCP attack-surface joint coverage (six-attack baseline)

2604.21477 MCP Pitfall Lab catalogues six attack families against MCP servers; 2604.20994 Function Hijacking adds the context-agnostic-invariant case (the same hijack works regardless of conversation context). Composing them gives a seven-case test harness for any MCP-touching milestone, and the Phase 805 baseline harness should instantiate all seven as test fixtures. The Track-B audit elevates this to a triple-gate when 2604.21816 Tool Attention is added as the runtime mitigation. The harness lives at `src/safety/mcp-attack-harness/` and runs as part of the planning-bridge MCP server's CI pipeline.

B.4.3 Composition 3 — Bounded-tape × bounded-learning (Root Theorem foundation)

The Root Theorem (2604.20874) supplies a formal information-theoretic statement that two corollaries instantiate: (a) Retrofit's 20%/3-correction caps in M2 (2511.11439) — bounded-update mechanism — and (b) the accumulate-compress-rewrite-shed lifecycle in M6 Gist Sparse Attention (2604.20920) — bounded-context mechanism. Both are bounded mechanisms over the same conceptual tape; both inherit the C2 signal/token-count independence corollary; both inherit the C4 compression-step-as-instance corollary. Adopting Root Theorem as the M1 anchor citation gives the M2 and M6 papers a shared formal foundation that can be cited in their respective ADRs without reasserting the bound. This is the headline composition of the M1 anchor work — a single citation that grounds two otherwise-independent module decisions.

B.4.4 Composition 4 — College of Knowledge retrieval+synthesis pipeline

Four M4 papers and one M6 paper compose into a single retrieval-and-synthesis pipeline for the College of Knowledge: 2510.26615 SARA at the retrieval layer; 2604.20848 MATRAG at the four-agent transparency-scoring layer; 2604.21444 HiCrew at the dynamic role-selection layer; 2508.21720 PosterForest at the synthesis layer; 2601.10003 SocraticKG at the KG-construction priors layer. The dependency graph is: SocraticKG → SARA → MATRAG → HiCrew → PosterForest, with HiCrew dispatching among MATRAG's four agents and PosterForest receiving HiCrew's tree as input. All five are ADOPT-design-only at v1.49.575; the implementation is scoped for v1.49.576+ as `src/skills/research-mission-generator/v2/`. The composition is significant because it routes through both M4 and M6, validating the cross-module nature of the retrieval-synthesis problem.

B.4.5 Composition 5 — KB construction lineage (SocraticKG → Gist → Capsules → Caesar)

The Track-C audit identifies an M6 chain: 2601.10003 SocraticKG (5W1H QA-driven KG construction) → 2604.20920 Gist Sparse Attention (compression/routing) → 2604.20487 Knowledge Capsules (nonparametric memory) → 2604.20855 Caesar (adversarial KG-driven synthesis). Each paper consumes the previous paper's output: SocraticKG produces 5W1H tuples, Gist routes a compressed view of them, Capsules nonparametrically index them, Caesar runs adversarial reasoning over the indexed KG. The chain forms the College of Knowledge's M6 backbone, with PosterForest as the front-end synthesis layer (Composition 4) and the College's department/panel hierarchy as the structural prior (covered in 2508.21720 adoption).

B.4.6 Composition 6 — Two-game chipset validation (3+1 + RPS/AGS + 5-axis matrix)

Three M4 papers compose into a chipset-validation harness: 2604.21282 3+1 Strategic Heterogeneous (the cooperative+adversarial two-game framing), 2604.21255 RPS/AGS (the behavior-similarity metrics), 2603.22823 Empirical Comparison ACPs (the five-axis matrix methodology). The composition gives GSD an evaluation tooling for whether the chipset's Opus(×N) + Haiku(verifier) tier separation is producing genuinely complementary perspectives or has collapsed to behavior-pattern uniformity. The five-axis matrix supplies the methodology, the AGS metric supplies the behavior-similarity instrument, and the 3+1 paper supplies the formal framing for what the metrics are measuring. The composition is documentation-only at v1.49.575; runtime instrumentation defers to a future chipset-validation harness.

B.5 §5 Final connection table (rendered from xmod.csv)

paper_a	mod_a	paper_b	mod_b	type	strength
2604.21910	M1	2604.21744	M1	composes-with	strong
2604.21744	M1	2604.20874	M1	validates	strong
2604.21744	M1	2601.18491	M3	extends	strong

paper_a	mod_a	paper_b	mod_b	type	strength
2604.20874	M1	2604.21927	M2	same-pattern-different-domain	strong
2604.20874	M1	2604.20920	M6	implements	strong
2604.21003	M1	2604.21725	M2	composes-with	strong
2604.20300	M2	2604.20943	M2	same-pattern-different-domain	medium
2604.20300	M2	2511.11439	M2	extends	strong
2604.21282	M4	2604.21255	M4	validates	strong
2604.21794	M4	2603.22823	M4	cross-validates	strong
2604.21816	M4	2511.00413	M4	composes-with	strong
2604.20911	M3	2604.21131	M3	extends	strong
2604.21477	M3	2604.20994	M3	composes-with	strong
2604.21598	M5	2604.21505	M5	composes-with	strong
2511.23159	M5	2604.21187	M5	validates	strong
2604.20920	M6	2604.20487	M6	composes-with	strong
2604.12710	M6	2604.21751	M6	extends	medium
2604.21829	M6	2604.21090	M3	composes-with	strong
2510.18731	M6	2601.10003	M6	extends	medium
2604.20855	M6	2508.21720	M4	composes-with	strong
2604.21571	M2	2604.21901	M2	composes-with	strong
2604.21571	M2	2509.18629	M2	composes-with	medium
2604.21927	M2	2604.21901	M2	validates	strong
2604.21927	M2	2509.18629	M2	validates	strong
2604.21927	M2	2604.21905	M2	validates	medium
2604.21905	M2	2604.21571	M2	same-pattern-different-domain	weak
2604.21090	M3	2604.21505	M5	composes-with	strong
2604.21744	M1	2604.21090	M3	validates	strong
2604.20874	M1	2511.11439	M2	implements	strong

paper_a	mod_a	paper_b	mod_b	type	strength
2604.20874	M1	2604.21816	M4	validates	strong
2604.20911	M3	2604.20874	M1	validates	medium
2604.21477	M3	2604.21816	M4	composes-with	strong
2604.20994	M3	2604.21816	M4	composes-with	strong
2604.21718	M3	2604.21003	M1	implements	strong
2604.21718	M3	2601.07262	M6	same-pattern-different-domain	medium
2604.21725	M2	2604.20943	M2	composes-with	medium
2604.21330	M2	2604.21003	M1	same-pattern-different-domain	medium
2604.20300	M2	2604.20874	M1	implements	strong
2604.21282	M4	2603.22823	M4	validates	strong
2604.21794	M4	2604.20487	M6	same-pattern-different-domain	medium
2511.00413	M4	2604.21590	M4	composes-with	medium
2604.21590	M4	2604.21003	M1	extends	medium
2604.21590	M4	2604.21725	M2	composes-with	medium
2604.21590	M4	2604.21718	M3	composes-with	medium
2604.20848	M4	2510.26615	M4	composes-with	strong
2604.20848	M4	2604.21444	M4	composes-with	strong
2604.20848	M4	2601.18491	M3	extends	medium
2508.21720	M4	2604.21444	M4	composes-with	strong
2604.21444	M4	2604.21380	M6	composes-with	medium
2510.26615	M4	2601.10003	M6	composes-with	medium
2510.26615	M4	2604.20920	M6	composes-with	medium
2604.20848	M4	2604.21380	M6	composes-with	medium
2604.21255	M4	2604.21829	M6	extends	medium

paper_a	mod_a	paper_b	mod_b	type	strength
2604.21255	M4	2604.21816	M4	composes-with	weak
2604.21255	M4	2511.00413	M4	validates	medium
2603.01170	M5	2604.21477	M3	extends	strong
2603.01170	M5	2604.20994	M3	extends	medium
2604.18792	M5	2604.21570	M5	composes-with	medium
2604.21570	M5	2604.21598	M5	composes-with	strong
2604.14709	M5	2603.01170	M5	composes-with	medium
2512.03048	M3	2511.23159	M5	composes-with	strong
2512.03048	M3	2604.21718	M3	implements	medium
2512.03048	M3	2604.20874	M1	validates	strong
2604.20920	M6	2604.20300	M2	composes-with	medium
2604.20487	M6	2604.20300	M2	extends	medium
2604.20487	M6	2601.10003	M6	composes-with	medium
2604.20855	M6	2604.21380	M6	composes-with	medium
2604.20855	M6	2601.10003	M6	composes-with	medium
2510.18731	M6	2604.21380	M6	same-pattern-different-domain	medium
2510.18731	M6	2604.21598	M5	composes-with	medium
2601.07262	M6	2604.21725	M2	composes-with	medium
2601.07262	M6	2604.20920	M6	composes-with	medium
2604.21308	M6	2604.21571	M2	extends	medium
2604.20904	M6	2604.21308	M6	composes-with	medium
2604.20904	M6	2604.21352	M6	same-pattern-different-domain	medium
2604.21352	M6	2604.20996	M6	same-pattern-different-domain	strong

paper_a	mod_a	paper_b	mod_b	type	strength
2604.12710	M6	2604.20996	M6	same-pattern-different-domain	medium
2604.21751	M6	2604.20996	M6	extends	medium
2604.21829	M6	2604.12710	M6	extends	medium
2604.21829	M6	2604.21380	M6	extends	medium
2604.21829	M6	2604.21477	M3	composes-with	strong

Total rows: 81. All six modules represented. Distribution: M1 papers in 12 connections; M2 in 24; M3 in 24; M4 in 24; M5 in 12; M6 in 38. M6 dominance reflects its 14-paper count and the cultural-sensitivity cluster's internal density.

B.6 §6 Concerns surfaced for Phase 802 (impact analysis)

The following numbered items collect concerns raised in the three track audits and surfaced during this synthesis. Phase 802 must address each.

1. **CS25-13 token-budget acceptance gate (Track B)** — The ≥40% p50 reduction target on Tool Attention depends on tool-set composition in the planning-bridge MCP. Phase 802 should require Phase 806 wave 1 to produce a baseline measurement before locking the acceptance threshold.
2. **CS25-15 fail-closed semantics (Track B)** — STD-calibration's fail-closed behaviour when calibration is missing collides with the bootstrap path (a fresh deployment has no calibration by definition). Phase 808 plan must specify the bootstrap exception explicitly.
3. **HB-08 backlog deferral framing (Track B)** — Intake §2.4 frames 2604.21255 RPS/AGS as ADOPT, but task instructions force backlog deferral. Phase 802 must reconcile the public-framing implication and confirm that the metric-definition-only adoption at v1.49.575 is the right ceiling.
4. **HB-09 DryRUN architecture sketch quality (Track C)** — The sketch in the ADR is a paragraph; if HB-09 promotes in v1.49.576+, the sketch must expand to a concrete module skeleton before implementation begins. Phase 802 should set the trigger condition.
5. **CI-Work threat-model publication boundary (Track C)** — The threat-model document straddles the Fox Companies IP rule (specifics stay in `.planning/`). Phase 802 must confirm with maintainers before any document moves outside `.planning/fox-companies/foxfiber/`.
6. **Caesar adversarial-refinement loop calibration (Track C)** — The loop's reframing penalty needs calibration on a known-good research mission before deployment; if PSC is the chosen proving-ground, Phase 802 must schedule a calibration pass before the loop runs unattended.
7. **Capsules KV-injection prerequisite (Track C)** — The kernel-hook prerequisite may never land; the TRACK-T3 designation acknowledges this and is inten-

tional. Phase 802 should confirm the prerequisite-tracking discipline.

8. **Federated-economy threat taxonomy (synthesis finding)** — Per acceptance criterion, any federated-economy recommendation cites 2604.21829 Black-Box Skill Stealing. CC-9 (cultural-sensitivity Safety Warden) and CC-2 (SKILL.md authoring) both touch federated-skill futures; Phase 802 must verify the citation is present in every federated-economy adjacent recommendation that lands in v1.49.575 Half B.
9. **2604.21927 regime-fixed comparison protocol (synthesis finding)** — Every M2 LoRA benchmark must be evaluated under controlled regime variables. Phase 802 must require the v1.49.576+ Silicon Layer benchmark plan to specify regime variables explicitly and propagate them to CC-5 (LoRA bench suite) acceptance criteria.
10. **MCP triple-gate sequencing (synthesis finding)** — CC-1's three papers must ship together as a single Phase 805 baseline harness, but the engineering ticket for Tool Attention (HB-01) is currently scoped narrower than the harness. Phase 802 must decide whether to widen the HB-01 ticket or open a sibling ticket for the harness.
11. **AEL × Last Harness state-isolation invariant (synthesis finding)** — CC-3's two-loop pattern depends on no state leakage between Worker, Evaluator, and Evolution. Track-A audit flags this as the operational risk; Phase 802 must require the Phase 806 plan to enumerate the isolation invariants and gate them under `cs25-26-sweep.two-loop` flag.
12. **M2 LoRA cluster vs Separable Expert tier (Track A)** — Separable Expert is T2 architecture (FoxFiber/FoxCompute multi-tenant), but GiVA and HyperAdapt are T1 benchmark targets. Phase 802 must clarify whether the v1.49.576+ Silicon Layer benchmark suite ships before or after the Separable Expert architecture spec.
13. **Track-C cultural-sensitivity gate completeness (synthesis finding)** — SC-IND, SC-MED, SC-SOV all PASSED in Track C, but Phase 802 should verify that the Phase 805 implementation tickets (when written) preserve the same nation-specific naming discipline applied in the ADRs.
14. **2604.21090 ↔ 2604.21505 ship-together discipline (Track B+C)** — CC-2 requires both linters ship at v1.49.575 as a single discipline upgrade. Phase 802 must confirm the joint-ship is the implementation reality and not a documentation aspiration that gets split at the engineering boundary.
15. **2604.21744 ↔ 2604.21090 convergent-discovery framing (Track A+B)** — The five principles are claimed as the formal mirror of GSD's existing CSO discipline. Phase 802 should require an explicit cross-reference in the Phase 810 ticket so the convergent-discovery framing is not lost in implementation.

End of Phase 800 Wave 2.1 synthesis.

B.7 Full Cross-Module Connection Table (xmod.csv)

The 81 rows below are the machine-readable export from `xmod.csv`. Strength values: `strong` = gates a Phase 805–811 ticket or carries explicit composition statement in two places; `medium` = single ADR cross-reference; `weak` = bibliographic-only.

Paper A	M	Paper B	M	Type	Summary	Str.
2604.21910	M1	2604.21744	M1	composes-with	Skill-as-markdown is the governance layer GROUNDINGmd Hard Constraints record onto	strong
2604.21744	M1	2604.20874	M1	validates	GROUNDINGmd override-the-prompt invariants are the Root Theorem made formal	strong
2604.21744	M1	2601.18491	M3	extends	Hard Constraints flow into AgentDoG where/how/what taxonomy as runtime records	strong
2604.20874	M1	2604.21927	M2	same-pattern	Bounded-memory C2 signal/token-count independence mirrors regime-fixed CL claim independence	strong
2604.20874	M1	2604.20920	M6	implements	Root theorem accumulate-compress step instantiated by gist compression	strong
2604.21003	M1	2604.21725	M2	composes-with	Worker/Evaluator/Evolution role split composes with AEL fast/slow timescale into single fast-Worker fast-Evaluator slow-Evolution pattern	strong
2604.20300	M2	2604.20943	M2	same-pattern	SLM decides what to forget while SCM defines when via NREM/REM consolidation	medium
2604.20300	M2	2511.11439	M2	extends	Selective forgetting composed with low-rank-sparse subspace via Retrofit	strong
2604.21282	M4	2604.21255	M4	validates	Heterogeneous experts validated by behavior-similarity AGS metric for cooperative-game complementarity	strong
2604.21794	M4	2603.22823	M4	cross-validates	Latent vs structured agent communication compared via five-axis matrix	strong
2604.21816	M4	2511.00413	M4	composes-with	Tool gating runtime complement to tree-trajectory training time pattern	strong
2604.20911	M3	2604.21131	M3	extends	Omission decay intra-session generalized to cross-session correlator threat	strong
2604.21477	M3	2604.20994	M3	composes-with	MCP attack-surface joint coverage six-attack-family plus context-agnostic hijack	strong
2604.21598	M5	2604.21505	M5	composes-with	Test gen without ground truth pairs with ambiguity taxonomy as spec/test envelope	strong
2511.23159	M5	2604.21187	M5	validates	Probable-to-provable rhetoric validated by SAT+LLM+Lean Ramsey precedent	strong
2604.20920	M6	2604.20487	M6	composes-with	Gist compression routes over nonparametric capsules in compress-route-unfold architecture	strong
2604.12710	M6	2604.21751	M6	extends	Semantic-bottleneck anchoring is the mitigation surface for CROQ-measured cultural bias	medium
2604.21829	M6	2604.21090	M3	composes-with	Marketplace skill-stealing threat defended by structural-completeness quality gate	strong

Paper A	M	Paper B	M	Type	Summary	Str.
2510.18731	M6	2601.10003	M6	extends	RLAAR curriculum instruction-shards composed with SocraticKG 5W1H decomposition	medium
2604.20855	M6	2508.21720	M4	composes-with	Adversarial-refinement composes with intermediate-tree pattern in PosterForest+Caesar pipeline	strong
2604.21571	M2	2604.21901	M2	composes-with	Separable Expert per-user proxy enabled by GiVA 8x rank reduction for compactness	strong
2604.21571	M2	2509.18629	M2	composes-with	Separable Expert domain-expert pool benefits from HyperAdapt O(m+n) high-rank pattern	medium
2604.21927	M2	2604.21901	M2	validates	Regime-fixed comparison constraint applies to GiVA benchmarking against vanilla LoRA	strong
2604.21927	M2	2509.18629	M2	validates	Regime-fixed comparison constraint applies to HyperAdapt benchmarking	strong
2604.21927	M2	2604.21905	M2	validates	Regime-fixed comparison defines regime variables LoRA Redux survey catalogues	medium
2604.21905	M2	2604.21571	M2	same-pattern, different-domain	LoRA Redux survey catalogues Separable Expert in adapter design space	weak
2604.21090	M3	2604.21505	M5	composes-with	Structural completeness linter ships with Orchid four-type ambiguity taxonomy as SKILL.md authoring upgrade	strong
2604.21744	M1	2604.21090	M3	validates	GROUNDINGmd Hard Constraints + Convention Parameters are convergent-discovery analog of five principles	strong
2604.20874	M1	2511.11439	M2	implements	Root theorem 20%/3-correction caps formalized by Retrofit bounded-tape mechanism	strong
2604.20874	M1	2604.21816	M4	validates	Root theorem fidelity-threshold framing is formal foundation for 70% context-fracture threshold	strong
2604.20911	M3	2604.20874	M1	validates	Root theorem fidelity-threshold framing is the formal explanation for SRD	medium
2604.21477	M3	2604.21816	M4	composes-with	Tool Attention lazy-schema-loading is third leg of joint MCP design change with Pitfall Lab and Function Hijacking	strong
2604.20994	M3	2604.21816	M4	composes-with	Function hijacking joint required with Tool Attention for planning-bridge MCP design changes	strong
2604.21718	M3	2604.21003	M1	implements	CHAI critique-revise pattern is operational mechanism for Last Harness Evaluator role	strong
2604.21718	M3	2601.07262	M6	same-pattern, different-domain	CHAI critique-revise mirrors ColorBrowserAgent HITL-feedback-to-domain-knowledge pipeline	medium
2604.21725	M2	2604.20943	M2	composes-with	SCM NREM/REM consolidation is candidate slow-timescale reflection mechanism for AEL bandit	medium

Paper A	M	Paper B	M	Type	Summary	Str.
2604.21330	M2	2604.21003	M1	same-pattern-different-domain	Different-domain dispatch pattern mirrors Last Harness Worker/Evaluator at harness level	medium
2604.20300	M2	2604.20874	M1	implements	Selective forgetting taxonomy implements shed step in Root Theorem accumulate-compress-rewrite-shed cycle	strong
2604.21282	M4	2603.22823	M4	validates	Empirical Comparison ACPs five-axis matrix is methodology for evaluating 3+1 chipset against alternatives	strong
2604.21794	M4	2604.20487	M6	same-pattern-different-domain	Differentiable Memory communication uses same substrate as Knowledge Capsules KV-injection	medium
2511.00413	M4	2604.21590	M4	composes-with	AgenticQwen dual flywheels supply curriculum for tree-trajectory training pipeline	medium
2604.21590	M4	2604.21003	M1	extends	Last Harness Worker/Evaluator/Evolution split is meta-evolution complement to AgenticQwen dual flywheels	medium
2604.21590	M4	2604.21725	M2	composes-with	AEL fast/slow learning is bandit-tier complement to AgenticQwen flywheel discipline	medium
2604.21590	M4	2604.21718	M3	composes-with	CHAI pre-X/post-X capture supplies source signal AgenticQwen flywheels consume	medium
2604.20848	M4	2510.26615	M4	composes-with	MATRAG Item Analysis depends on SARA hybrid-RAG retrieval as backing layer	strong
2604.20848	M4	2604.21444	M4	composes-with	HiCrew dynamic role-selection dispatches among the four MATRAG agents	strong
2604.20848	M4	2601.18491	M3	extends	AgentDoG schema embeds MATRAG transparency score in BLOCK records	medium
2508.21720	M4	2604.21444	M4	composes-with	PosterForest Poster Tree synthesis complements HiCrew Hybrid Tree retrieval	strong
2604.21444	M4	2604.21380	M6	composes-with	IRAP interactive elicitation supplies question-complexity signal HiCrew planning layer reasons over	medium
2510.26615	M4	2601.10003	M6	composes-with	SocraticKG 5W1H QA-driven KG construction supplies structural priors SARA vector pool indexes	medium
2510.26615	M4	2604.20920	M6	composes-with	Cost Sparse Attention is in-context complement to SARA hybrid retrieval (compressed memory plus selective unfolding)	medium
2604.20848	M4	2604.21380	M6	composes-with	IRAP interactive elicitation supplies User Modeling preference profile MATRAG consumes	medium
2604.21255	M4	2604.21829	M6	extends	RPS/AGS behavior-similarity is marketplace-skill provenance check for federated-economy threat model	medium
2604.21255	M4	2604.21816	M4	composes-with	Tool Attention gated-tool-set affects action-graph the AGS metric measures	weak

Paper A	M	Paper B	M	Type	Summary	Str.
2604.21255	M4	2511.00413	M4	validates	RPS/AGS metrics validate that subagent trees produce diverse rather than collapsed branches	medium
2603.01170	M5	2604.21477	M3	extends	ATLAS CWE-to-assertion pipeline maps onto MCP Pitfall Lab six-attack-family threat catalog	strong
2603.01170	M5	2604.20994	M3	extends	ATLAS assertion pipeline must cover Function Hijacking threat surface	medium
2604.18792	M5	2604.21570	M5	composes-with	DSLTrans cutoff bound composes with SpecSyn strength-scoring for bounded checks	medium
2604.21570	M5	2604.21598	M5	composes-with	SpecSyn synthesizes specs DryRUN synthesizes tests covering both sides of auditor	strong
2604.14709	M5	2603.01170	M5	composes-with	HWE-Bench repair-side benchmark complements ATLAS hardware-verification pipeline	medium
2512.03048	M3	2511.23159	M5	composes-with	Specification Trap empirical case + probable-to-provable formal-verification companion address spec-difficulty horn	strong
2512.03048	M3	2604.21718	M3	implements	CHAI critique-revise is operational mechanism for developmentally-responsive specs	medium
2512.03048	M3	2604.20874	M1	validates	Root Theorem supplies formal information-theoretic complement to Specification Trap philosophical argument	strong
2604.20920	M6	2604.20300	M2	composes-with	FSFM selective forgetting composes with gist compression to forget what is not gist-routed	medium
2604.20487	M6	2604.20300	M2	extends	FSFM selective forgetting becomes capsule-level eviction policy	medium
2604.20487	M6	2601.10003	M6	composes-with	SocraticKG 5W1H decomposition is candidate capsule construction pipeline	medium
2604.20855	M6	2604.21380	M6	composes-with	IRAP bounded-elicitation is elicitation half complementing Caesar synthesis half	medium
2604.20855	M6	2601.10003	M6	composes-with	SocraticKG 5W1H provides KG-construction step Caesar KG-driven reasoning consumes	medium
2510.18731	M6	2604.21380	M6	same-pattern-different-domain	RLAIF best-of-round-N mirrors IRAP bounded-rounds escalation	medium
2510.18731	M6	2604.21598	M5	composes-with	DryRUN trace-synthesis supplies verifiable-accuracy half of curriculum reward	medium
2601.07262	M6	2604.21725	M2	composes-with	AEL agent evolving learning consumes domain knowledge ColorBrowserAgent captures	medium
2601.07262	M6	2604.20920	M6	composes-with	Gist Sparse Attention is candidate compression layer for ColorBrowserAgent progressive-summarization half	medium
2604.21308	M6	2604.21571	M2	extends	Separable Expert composable adapters enable CI-Work multi-tenant patterns	medium

Paper A	M	Paper B	M	Type	Summary	Str.
2604.20904	M6	2604.21308	M6	composes-with	CI-Work contextual-integrity supplies evaluation surface for Privacy via Fiction normative-simulacrum training	medium
2604.20904	M6	2604.21352	M6	same-pattern different-domain	Privacy via Fiction normative-simulacrum framing pairs with CARE counselor-alignment as assist-not-replace patterns	medium
2604.21352	M6	2604.20996	M6	same-pattern different-domain	CARE permission-neutral discipline mirrors AFRILANGTUTOR sovereignty-attentive discipline	strong
2604.12710	M6	2604.20996	M6	same-pattern different-domain	LASA semantic-bottleneck and AFRILANGTUTOR low-resource pipeline share language-agnostic concern	medium
2604.21751	M6	2604.20996	M6	extends	High-resource-language bias finding pairs with AFRILANGTUTOR low-resource focus	medium
2604.21829	M6	2604.12710	M6	extends	LASA semantic-bottleneck anchoring defends skill-attribution rules against paraphrase attack	medium
2604.21829	M6	2604.21380	M6	extends	IRAP elicitation discipline reduces surface area of skil specifications visible to attackers	medium
2604.21829	M6	2604.21477	M3	composes-with	MCP Pitfall Lab metadata-poisoning attack family overlaps with Black-Box Skill Stealing threat model	strong

C GSD Architectural Impact Analysis (Phase 802)

This appendix embeds the Phase 802 deliverable from `.planning/missions/cs25-26-sweep/work/`. It satisfies CS25-05 (per-subsystem "what changes / what does not change" table covers every GSD subsystem affected by an ADOPT decision).

C.1 §1 Executive summary

The v1.49.575 milestone shifts seven GSD subsystems with new code, references three more with citation-only edits, and leaves the remaining subsystems untouched. The seven shifting subsystems align with the Half B T1 roster HB-01..HB-07. Three carry **CAPCOM HARD GATE** designation: HB-03 STD calibration (Safety Warden BLOCK timing), HB-04 Worker/Evaluator/Evolution role split (skill-creator orchestration), HB-07 AEL fast/slow bandit auto-load (skill-space dispatch). All seven ship default-off behind `cs25-26-sweep.*` flags so the baseline is byte-identical when flags are off — invariant carried forward from v1.49.570/.571/.572/.574.

The five backlog items (HB-08 RPS/AGS divergence; HB-09 DryRUN test synthesis; HB-10 IRAP elicitation cap; HB-11 Black-Box Skill Stealing threat model; HB-12 FSFM 4-class forgetting taxonomy) target subsystems referenced but not modified in v1.49.575: chipset coherence (HB-08), skill validation (HB-09), vision-to-mission (HB-10), Grove federated economy (HB-11), reinforcement series (HB-12). Each carries a documented promotion trigger.

Aggregate: **9 subsystems** addressed in §2 (orchestration, safety, skill-creator, cartridge, Silicon Layer refs, DACP refs, College, Grove, reinforcement) plus **3 optional** (memory stack, cultural-sensitivity gates, vision-to-mission). **8 prior milestones** v1.49.561/.568/.569/.570/.571/.572/.573/.574 are touched in §3. **15 xmod concerns + 4 audit-level concerns** dispositioned in §4. Risk register carries 13 rows. The keystone preservation rule — *no module proposes replacing CAPCOM, no override of SC-IND/SC-MED/SC-SOV* — is reissued as sign-off lines in §6.

C.2 §2 Per-subsystem impact table

C.2.1 1. Orchestration / planning-bridge MCP — `src/orchestration/`

What changes (this milestone, v1.49.575): - New module `src/orchestration/tool-attentio` (HB-01 / CS25-13) per ADR-2604.21816: ISO-score selector from sentence embeddings, state-aware gating, two-phase lazy schema loader. Default-off via `cs25-26-sweep.tool-atte` - Files: `iso-score.ts`, `state-gate.ts`, `lazy-loader.ts`, `budget-monitor.ts` plus tests under `__tests__/`. - Cross-link with M3 attack surface: HB-01 composes with the MCP triple-gate (2604.21477 Pitfall Lab + 2604.20994 Function Hijacking + 2604.21816 Tool Attention) per xmod CC-1; the harness exercises Tool Attention as runtime mitigation. - Substrate-doc reference: `docs/substrate-references/tool-attention.md` (NEW) names MATRAG transparency-scoring (2604.20848) and Tree-Training (2604.21255 RPS/AGS HB-08 forward) as composing patterns.

What does not change: - `src/orchestration/jepa-planner/`, `retrieval-loop.ts`, `selector.ts`, `sol-budget/` are byte-identical. - Tool dispatch path with flag off is byte-identical to v1.49.574 baseline (CS25-13 acceptance line). - CAPCOM gate place-

ment on planning-bridge MCP is preserved — the Tool Attention loader sits *above* the dispatch boundary, never below it.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: HB-08 RPS/AGS divergence metrics (2604.21255) lands as orchestration-side chipset-coherence telemetry. - v1.50: full MCP six-attack-family runtime harness exits Phase 805 spec into operational runtime. - v1.50+: Mission Crew Manifest (MATRAG split deferred half from 2604.20848) rejoins the orchestration retrieval loop.

C.2.2 2. Safety Warden / CAPCOM — `src/safety/`

What changes (this milestone, v1.49.575): - New module `src/safety/agentdog/` (HB-02 / CS25-14) per ADR-2601.18491: where/how/what diagnostic taxonomy emitted with every BLOCK decision. Default-off via `cs25-26-sweep.agentdog-schema`. Haiku-tier sidecar deployment plan named. - New module `src/safety/std-calibration/` (HB-03 / CS25-15, **CAPCOM HARD GATE**) per ADR-2604.20911: per-model omission-decay measurement, calibration table at `.planning/safety/std-calibration.json`, re-injection middleware. Default-off via `cs25-26-sweep.std-calibration`. - Phase 805 safety-test-harness expansion adds three categories per CS25-10: STD calibration test, where/how/what BLOCK schema test, MCP six-attack-family tests covering at least metadata-poisoning, puppet-server, image-to-tool, function-hijacking, plus two from the 2604.21477 six-attack baseline. - Cross-session correlator architecture lands per xmod CC-6 — `src/safety/cross-session/` directory created with header docs only; runtime defers to v1.49.576+ (depends on stable AgentDoG records + STD calibration).

What does not change: - Safety Warden BLOCK authority is unchanged — BLOCK is still emitted exactly where it is today; HB-02 only adds a diagnostic record alongside the BLOCK. With flag off, BLOCK emission is byte-identical. - CAPCOM HARD GATE list remains: HB-03, HB-04, HB-07. **No subsystem proposed in v1.49.575 replaces or bypasses CAPCOM.** This invariant is restated in §6 as a sign-off line. - `.claude/skills/security-hygiene/SKILL.md` BLOCK doctrine continues to override the user prompt — the GROUNDINGmd citation (2604.21744) is the published independent rediscovery of this rule, not a replacement.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: CSTM-Bench (2604.21131) cross-session correlator runtime ships; consumes AgentDoG + STD calibration. - v1.49.576+: HB-11 Black-Box Skill Stealing threat-model document (2604.21829) lands as Fox Companies internal artifact under `.planning/fox-companies/foxfiber/` (publication boundary discussed in §4 disposition #5). - v1.50: CARE counselor-alignment ADOPT (2604.21352, M6) — assist-not-replace framing referenced in Safety Warden module docs; no new module, citation-only.

C.2.3 3. skill-creator — `src/skill-creator/`

What changes (this milestone, v1.49.575): - New module `src/skill-creator/roles/` (HB-04 / CS25-16, **CAPCOM HARD GATE**) per ADR-2604.21003: Worker / Evaluator / Evolution role-isolated subagents. Replaces ad-hoc 3-correction-min with adversarial-Evaluator pattern. Default-off via `cs25-26-sweep.weler-roles`. - New file `src/skill-creator/auto-load/bandit.ts` (HB-07 / CS25-19, **CAPCOM HARD GATE**) per ADR-2604.21725: Thompson-Sampling bandit + LLM-reflection slow loop. Default-off via `cs25-26-sweep.ael-bandit`. - Composition: HB-04 × HB-07 is xmod

Composition 1 (AEL × Last Harness fast-Worker / fast-Evaluator / slow-Evolution). The two modules ship together as a single architectural pattern; flag toggling is independent but the semantic pair is the unit of design. - `.claude/agents/` documentation maps existing executor / verifier / planner subagents onto Worker / Evaluator / Evolution per convergent-discovery §5 citation map. - `.planning/PROJECT.md` bounded-learning-policy section gains a citation to 2604.21003 as the published meta-evolution derivation (under CS25-09's edit budget).

What does not change: - Six-step loop (Observe → Detect → Suggest → Manage → Auto-Load → Learn/Compose) shape is preserved; the Worker/Evaluator/Evolution split *names* what is already there rather than restructuring. - The 20% / 3-correction / 7-day cooldown bounded-learning policy is unchanged — Root Theorem (2604.20874) C1 is the formal foundation for refusing to relax these caps with bigger context windows. - With both flags off, skill-creator runs the existing six-step loop byte-identically (CS25-16 + CS25-19 acceptance). - Existing executor/verifier/planner subagents keep their current invocation paths; the new role names are additive.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: HB-12 FSFM 4-class forgetting taxonomy (2604.20300) formalizes the bounded-learning policy with C2/C4 corollaries from Root Theorem; no new module, taxonomy adoption only. - v1.49.576+: HB-09 DryRUN (2604.21598) test-synthesis path lands as skill-validation harness; sketch promotion gate noted in §4 disposition #4. - v1.50: Meta-Evolution loop (Last Harness outer loop) crosses skill boundaries — currently single-skill scope only.

C.2.4 4. cartridge / SKILL.md authoring — `src/cartridge/`

What changes (this milestone, v1.49.575): - New file `src/cartridge/linter/structural-com` (HB-05 / CS25-17) per ADR-2604.21090: five-principle SKILL.md validator (computability / proof-theory / Bayesian / data-classification / assessment-rubric). Default-off via `cs25-26-sweep.structural-completeness-lint` (warning-only when off). - New file `src/cartridge/linter/ambiguity.ts` (HB-06 / CS25-18) per ADR-2604.21505: lexical / syntactic / semantic / vagueness type checks. Default-off via `cs25-26-sweep.ambiguity-li` - New doc `docs/skill-authoring/ambiguity-checklist.md` ships the four-type taxonomy as authoring guidance. - `docs/cartridge/SKILL.md` authoring guide gains GROUNDINGmd two-class taxonomy (Hard Constraints / Convention Parameters per 2604.21744) as a structural-completeness criterion. CS25-17's negative-fixture corpus includes "Hard Constraints declared" cases. - xmod CC-2 ship-together discipline: HB-05 + HB-06 ship as a single SKILL.md authoring discipline upgrade (Phase 810 + Phase 811 in roadmap). Convergent-discovery framing per 2604.21744 cross-reference is added explicitly.

What does not change: - The cartridge promotion pipeline still routes `.planning/staging/inbox` → published; the linter is **additive** and gated default-off so existing skills neither break nor regress. - Existing CSO discipline is unchanged — 2604.21744's convergent-discovery framing is that the five principles *mirror* CSO, not that they replace it (Track-A audit cross-link suggestion #5 elaborates). - Zero false positives on existing in-tree skills is the CS25-18 acceptance condition; if any in-tree skill trips, the failure is treated as a fixture defect, not a baseline regression.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: structural-completeness

gate flips from warning-only to blocking after corpus audit completes. Trigger: ≤2% false-positive rate on the v1.49.575 corpus. - v1.49.576+: HB-11 Black-Box Skill Stealing threat-model (2604.21829) integrates with cartridge linter as marketplace-contamination defense (xmod CC-9 Federated-Skill futures); Black-Box Skill Stealing citation gate is restated in §6. - v1.50: federated-skill DoltHub cartridge schema requires HB-11 threat-model as pre-rollout citation (Black-Box Skill Stealing pre-rollout-gate, see §4 disposition #8).

C.2.5 5. Silicon Layer / LoRA pipeline — references only this milestone

What changes (this milestone, v1.49.575): - New doc `docs/substrate-references/lora-cs25` (NEW, Phase 805) catalogues the M2 LoRA dense subgraph per xmod §3: 2604.21571 Separable Expert (T2 multi-tenant architecture), 2604.21901 GiVA (T1 benchmark), 2509.18629 HyperAdapt (T1 benchmark), 2604.21905 LoRA Redux (TRACK survey), 2604.21330 TGR-MoE (TRACK pattern). Adds 2604.21927 regime-fixed comparison protocol as the binding methodological constraint across all five. - No new code in `src/` this milestone — this is a documentation-only adoption per Track-A audit's T2 designation for Separable Expert. - Cross-link: bounded-tape × bounded-learning unified foundation per xmod CC-4 — Root Theorem (2604.20874) becomes the M1 anchor citation that grounds Retrofit (2511.11439) C2/C4 corollaries.

What does not change: - Zero changes to running Silicon Layer code — there is no Silicon Layer code in `src/` to change. The doc is a forward-pointer. - The `.planning/research-catalog.csv` Silicon Layer entries are byte-identical. - RTX 4060 Ti VRAM tier (Memory Arena M6, `examples/chipsets/silicon-layer/`) is unchanged.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: LoRA bench suite at `examples/chipsets/silicon-layer/lora-bench/` instantiates GiVA + HyperAdapt + Separable Expert under the regime-fixed protocol from 2604.21927. Concern #9 in §4 names this regime-spec requirement. - v1.50: full Silicon Layer pipeline ships; Separable Expert per-user proxy + per-domain expert pool lands as multi-tenant deployment surface. - v1.50+: TGR-MoE learned-dispatch milestone (TRACK promotion trigger per Track-A judgment call #3).

C.2.6 6. DACP fidelity levels — references only this milestone

What changes (this milestone, v1.49.575): - `docs/substrate/semantic-channel.md` (existing from v1.49.572) gains citation block referencing 2603.22823 Empirical Comparison ACPs five-axis evaluation matrix and 2604.21794 DiffMAS latent-vs-structured comparison. - `docs/substrate-references/dacp-evaluation.md` (NEW, citation-only) names the chipset-validation harness composition per xmod Composition 6 — 2604.21282 3+1 Strategic Heterogeneous + 2604.21255 RPS/AGS + 2603.22823 Empirical Comparison ACPs as the joint evaluation tooling.

What does not change: - DACP three-part bundle structure (deterministic input / agent transform / verifiable output) is preserved verbatim. v1.49.572 GAP-6 close stands. - Semantic-channel byte-identical guarantee from v1.49.572 G7 HARD preservation gate is preserved. - No runtime instrumentation lands this milestone — chipset-validation harness defers to v1.49.576+.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: chipset-validation

harness with the AGS metric (HB-08, 2604.21255) becomes runtime instrumentation for the Opus×N + Haiku-verifier tier-separation check. - v1.50: DACP fidelity-level evaluation matrix runs as a continuous-integration check on every chipset release.

C.2.7 7. College of Knowledge — `.college/`

What changes (this milestone, v1.49.575): - Citations only this milestone. New concept seeds across `ai-computation` / `safety-verification` / `data-science` departments per the four anchors (per STATE.md candidate slugs and the convergent-discovery citation map). - Concept slug additions: `bounded-tape-root-theorem`, `groundingmd-hard-constraints`, `worker-evaluator-evolution-roles`, `skills-as-md-three-ti` `mcp-six-attack-family`, `std-calibration-omission-decay`, `iso-score-tool-attention`, `four-type-ambiguity`, `five-principle-structural-completeness`. - `.college/README.md` gains terminology import: semantic / deterministic / knowledge layer naming per 2604.21910.

What does not change: - College department / panel structure is byte-identical. Rosetta Core unchanged. - `.college/calibration/` rubrics unchanged. CSO discipline unchanged (the five principles mirror CSO per Track-A cross-link #5; they do not replace it).

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: College retrieval pipeline lands per xmod CC-8 — SARA + MATRAG + HiCrew + PosterForest + SocraticKG five-paper composition at `src/skills/research-mission-generator/v2/` and `.college/retrieval/`. - v1.50: KG construction lineage per xmod Composition 5 — SocraticKG → Gist → Capsules → Caesar; PosterForest as front-end synthesis.

C.2.8 8. Grove format / federated economy — `docs/GROVE-FORMAT.md`, `src/memory/grove-f`

What changes (this milestone, v1.49.575): - Citations only. `docs/GROVE-FORMAT.md` gains a forward-pointing note that any DoltHub federated-economy adoption in v1.49.576+ must cite 2604.21829 Black-Box Skill Stealing as the threat-taxonomy pre-rollout-gate (per CS25-04 acceptance language). - xmod CC-9 cultural-sensitivity Safety Warden surface integrates with Grove via the federated-skill-future link (HB-11 backlog).

What does not change: - Layer A content-addressed Grove records are byte-identical. SkillView, GroveNamespace, SkillCodebase, SkillDiff, ActivityLog all unchanged. - 299-resource arena (`.grove/arena.json`) loadable by any Grove-compatible tool — schema unchanged. - Wasteland exclusion remains absolute — `data/chipset/muses/` and `wasteland` path segment continue to be blocked.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: HB-11 threat-model document lands at `.planning/fox-companies/foxfiber/threat-model.md` (Fox Companies IP — see §4 disposition #5). - v1.50: federated-economy DoltHub work begins. Pre-rollout requirement: HB-11 threat-model citation **named** in every federated-economy adjacent recommendation (CS25-04 + xmod concern #8 — restated in §6 sign-off line).

C.2.9 9. Reinforcement series / bounded-learning policy — `src/reinforcement/`, `src/eligibility/`, `src/learnable-k_h/`, etc.

What changes (this milestone, v1.49.575): - Citations only this milestone. Retrofit (2511.11439) cited as the parameter-space form of the 20% rule (xmod connection paper_a=2604.20300 ↔ paper_b=2511.11439 extends-strong). - `docs/substrate/bounded-tape-roo`

(NEW per convergent-discovery §4) becomes the formal foundation for the bounded-learning caps under Root Theorem C1+C2+C4.

What does not change: - `src/reinforcement/`, `src/eligibility/`, `src/learnable-k_h/`, `src/lyapunov/`, `src/projection/`, `src/dead-zone/`, `src/stochastic/`, `src/langevin/`, `src/temperature/`, `src/embeddings/`, `src/representation-audit/` — all byte-identical. - 20% cap / 3-correction-min / 7-day cooldown unchanged. C1 (monotone decay independent of window size) is the formal justification for refusing to relax these caps even with a 1M-context model.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: HB-12 FSFM 4-class forgetting taxonomy (2604.20300) formalizes the policy. Composes with MD-6 representation-audit. - v1.50: bounded-learning theorem proof completion (carried forward from v1.49.572 Half B T1d) cites Root Theorem as the empirical anchor.

C.2.10 10. Memory stack — `src/graph/`, `src/traces/`, `src/branches/`, `src/cache/`

What changes (this milestone, v1.49.575): - Citations only. 2604.20920 Gist Sparse Attention and 2604.20487 Knowledge Capsules cited as architectural template — recursive gist-of-gist pattern relevant to nested College structure (xmod Composition 5). - No code change. Sub-citation block added to `docs/memory-arena/M9-M13.md` referencing the KB construction lineage SocraticKG → Gist → Capsules → Caesar.

What does not change: - M1 SMG / M3 Decision-Trace / M4 Branch-Context / M5 cache + KVFlow predictor — all byte-identical. Memory Arena M1–M13 stack untouched. - Grove integration (`RustArenaSet`, `ContentAddressedSetStore`) preserved.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: Capsules KV-injection prerequisite tracking (TRACK-T3 per Track-C concern; may never land). - v1.50: Gist Sparse Attention recursive composition adopted at the College retrieval surface.

C.2.11 11. Cultural-sensitivity gates — `docs/college/`, citation surface

What changes (this milestone, v1.49.575): - Citations only. 2604.12710 LASA semantic-bottleneck cited as anchoring template. 2604.20996 AFRILANGTUTOR cited as sovereignty-gate template (TRACK; SC-SOV is a hard prerequisite — pre-implementation consultation required per CS25-11). - 2604.21352 CARE counselor-alignment cited; assist-not-replace framing preserved exclusively (no diagnostic / therapeutic claims). - xmod CC-9 cultural-sensitivity cluster integrates into Safety Warden surface alongside 2604.21308 CI-Work contextual-integrity benchmark.

What does not change — explicit invariant restatement: - **SC-IND, SC-MED, SC-SOV gates are preserved with no override proposed.** Every reference to PNW Indigenous nations names them specifically (Coast Salish, Lummi/Lhaq'temish, Tulalip, Stillaguamish, Wanapum). No generic "Indigenous peoples" phrasing. CS25-11 acceptance line is binding for Phase 805 publication and all subsequent waves. This restatement is a sign-off line in §6.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: AFRILANGTUTOR PNW adaptation **gated** on nation-specific consultation. No implementation begins until consultation completes. - v1.50: CROQ slice (2604.21751) bias-measurement instrument feeds Safety Warden surface; 2604.21308 CI-Work contextual-integrity benchmark runs as joint surface.

C.2.12 12. vision-to-mission / research-mission-generator — skill-pack level (Tibsfox/skills)

What changes (this milestone, v1.49.575): - Citations only. The skills-as-md three-layer architecture (2604.21910) is the foundational citation per convergent-discovery §2 — vision-to-mission = semantic layer; research-mission-generator = deterministic layer; gsd-skill-creator = knowledge layer. - Caesar adversarial-refinement loop (2604.20855) tracked in research-mission-generator design notes; calibration concern noted in §4 disposition #6.

What does not change: - Both skills run unchanged in v1.49.575. No version bump, no new behaviour, no flag. - Three-stage output (Vision Guide → Research Reference → Mission Spec) preserved.

Future milestones (v1.49.576+ / v1.50 / later): - v1.49.576+: HB-10 IRAP 5-round elicitation cap (2604.21380) lands as soft discipline now (vision-to-mission soft cap), hard CAPCOM-gate escalation later. Promotion trigger: model-affinity escalation phase opens. - v1.49.576+: Caesar adversarial-refinement TRACK promotion contingent on calibration pass against PSC mission (§4 disposition #6).

C.3 §3 Cross-milestone linkage map

The v1.49.575 work touches **eight prior milestones** and **one parallel track**. Linkage forms: cite (citation-only), extend (composes with and adds to), compose (paired structural composition), reference (forward-pointer).

- **v1.49.561 Living Sensoria** — *cite + reference*. M6/M7/M8 (`src/sensoria/`, `src/umwelt/`, `src/symbiosis/`) cited in convergent-discovery §4 C5 — umwelt's Markov-blanket boundary = Root Theorem C5 external-verification-gate. CC-9 cultural-sensitivity Safety Warden surface composes with M8 symbiosis ledger. No code change.
- **v1.49.568 Nonlinear Frontier** — *cite*. Bounded-learning anchor for Root Theorem; NLF-IDs are the external mathematical foundation of the 20% / 3-correction caps now sharpened by 2604.20874 C1.
- **v1.49.569 Drift in LLM Systems** — *compose*. STD calibration HB-03 composes with drift detection: STD measurement is the prohibition-decay specialization of v1.49.569's drift-detection methodology. Phase 808 plan references drift telemetry for the bootstrap-exception path (§4 #2).
- **v1.49.570 Convergent Substrate** — *extend*. `synthesis/convergent-discovery.md` is the second-generation instance of the canonical pattern v1.49.570 established. Four anchors (2604.21910, 2604.21744, 2604.20874, 2604.21003) compose with the v1.49.570 convergent block. Two-gate / cascade-mcp-defense modules are pre-citations of CC-1 (MCP triple-gate).
- **v1.49.571 LeJEPA / Heuristics-Free** — *compose*. Last Harness Worker/Evaluator/Evolution split = heuristics-free skill-creator pattern; the adversarial Evaluator replaces the heuristic 3-correction-min. HB-04 + LEJEPA-13 (skill-isotropy) + LEJEPA-15 (single-λ audit) form a design-discipline triad.
- **v1.49.572 Math Foundations** — *compose (theorem stack)*. Root Theorem composes with three v1.49.572 T1 deliverables: Coherent Functors (T1a), Semantic Channel DACP (T1c, Root Theorem C2 = rate-distortion specialization), Bounded-Learning Theorem (T1d, Root Theorem C1 = empirical anchor for 20%

cap proof). xmod CC-4 = bounded-tape × bounded-learning unified foundation.

- **v1.49.573 Upstream Intelligence** — *extend*. Skilldex Conformance Auditor (T1a) compounds with HB-05 Five-Principle linter — same gate, complementary checks. v1.49.573's upstream + agentskills.io audit feed the same `src/cartridge/linter/` directory.
- **v1.49.574 Megakernel** — *compose (keystone)*. SIGReg ↔ counter-based megakernel architectural-rhyme composes with the bounded-tape Root Theorem keystone — both productive-analogy-at-design-discipline-level findings. Convergent-discovery §7 calibration reused verbatim. RAVEN-INTEG roster is the pattern HB-01..HB-07 follows.

Parallel track: v1.0 PNW Living Systems Taxonomy (phases 620–635) — *cite*. College concept-seeds feed the PNW taxonomy hub; cross-references.json gains edges from CS25-26 hub to AAR / SST / LLM / Chipset.

C.4 §4 Concerns surfaced + dispositions

The 15 concerns from xmod.md §6 plus 4 audit-level concerns are dispositioned below. Each: concern statement → disposition.

1. **CS25-13 Tool Attention acceptance threshold (xmod #1; Track B).** ≥40% p50 reduction depends on tool-set composition. → **Accept with Phase 806 wave-1 baseline.** Phase 806 produces a baseline before locking the threshold; if 40% is unreachable, re-scope to "p50 reduction ≥ baseline × 0.6" in the Phase 806 plan.
2. **CS25-15 STD fail-closed bootstrap exception (xmod #2; Track B).** Fail-closed collides with bootstrap (fresh deployments have no calibration). → **Defer to Phase 808.** Phase 808 plan specifies a calibration-acquisition mode for ≤N turns, after which fail-closed activates. CAPCOM HARD GATE pass requires this spec.
3. **HB-08 backlog deferral (xmod #3; Track B).** Intake §2.4 framed `2604.21255` as ADOPT; task instructions force backlog. → **Defer to retrospective.** Backlog stands; `docs/release-notes/v1.49.575/` retrospective documents the divergence (T1 capacity ceiling) so the framing is not silently overruled.
4. **HB-09 DryRUN sketch quality (xmod #4; Track C).** Sketch is one paragraph. → **Defer with trigger condition.** HB-09 promotion is gated on a ≥800-word architecture-spec ADR in `work/modules/M5/` before the implementing phase opens.
5. **CI-Work threat-model publication boundary (xmod #5; Track C).** Document straddles Fox Companies IP rule. → **Surface to user; defer publication.** Fox Companies IP review required before any public artifact moves outside `.planning/fox-companies/foxfiber/`. Abstract attack-taxonomy citable; specifics stay private. User-decision item.
6. **Caesar adversarial-refinement calibration (xmod #6; Track C).** Reframing-penalty needs calibration before deployment. → **TRACK only this milestone; defer ADOPT.** Caesar does not run autonomously in v1.49.575. If PSC runs in v1.49.576+, schedule a calibration pass first.
7. **Capsules KV-injection prerequisite (xmod #7; Track C).** May never land. → **Accept TRACK-T3 discipline.** Status is intentional; no further action.

8. **Federated-economy threat-taxonomy citation gate (xmod #8).** Per CS25-04, federated-economy recs must cite 2604.21829. → **Accept now as pre-rollout gate.** §2 Grove (subsystem 8) and cartridge (subsystem 4) blocks name it. §6 sign-off restates: any DoltHub work in v1.49.576+ requires the citation before merge.
9. **2604.21927 regime-fixed protocol (xmod #9).** Every M2 LoRA benchmark needs regime variables (rank/dataset/schedule). → **Accept as benchmark-spec requirement.** `docs/substrate-references/lora-cs25-26.md` names variables; v1.49.576+ Silicon Layer plan inherits as CC-5 acceptance.
10. **MCP triple-gate sequencing (xmod #10).** CC-1 three papers ship together; HB-01 ticket is narrower. → **Accept; widen HB-01.** Phase 806 ticket widens to include the six-attack-family + function-hijacking test-fixture surface (CS25-10); one ticket, no siblings.
11. **AEL × Last Harness state-isolation (xmod #11).** CC-3 depends on no Worker/Evaluator/Evolution leakage. → **Accept; enumerate in Phase 808 plan.** Worker writes own state only; Evaluator reads Worker output (read-only); Evolution reads aggregated history (read-only). Tests verify no cross-role write paths.
12. **M2 LoRA vs Separable Expert tier sequencing (xmod #12; Track A).** SE is T2; GiVA + HyperAdapt are T1 benchmarks. → **Accept; document sequencing.** Benchmark suite ships **before** SE architecture spec; benchmarks inform the spec.
13. **Track-C cultural-sensitivity gate completeness (xmod #13).** SC-IND/SC-MED/SC-SOV PASSED. → **Accept; carry to Phase 805 tickets.** Implementation tickets preserve nation-specific naming. CS25-11 = verification entry.
14. **2604.21090 ↔ 2604.21505 ship-together (xmod #14).** Both linters ship as one discipline upgrade. → **Accept as engineering reality.** §2 subsystem 4 names joint Phase 810 + 811 ship; phases close together.
15. **2604.21744 ↔ 2604.21090 convergent-discovery framing (xmod #15).** Five principles = formal mirror of CSO. → **Accept as Phase 810 cross-reference.** Phase 810 ticket includes the cross-ref so the mirror-of-CSO claim is not lost in implementation.

Audit-level concerns (Tracks A/B/C):

16. **Wave-1 parallelism via subagent fleets (Track A).** Three concurrent fleets (A/B/C) writing ADRs to the same `work/modules/` directory tree — risk of merge collision. → **Accept; mitigated by per-module subdirectory isolation.** M1+M2 → Track A; M3+M4 → Track B; M5+M6 → Track C. No cross-track writes.
17. **Track-A LoRA Redux ADOPT-vs-TRACK judgment (Track A judgment call #2).** Survey paper deferred to TRACK. → **Accept; bibliographic citation only.**
18. **Track-B CS25-13 baseline measurement (overlaps xmod #1).** → Same disposition as #1.
19. **Track-C SC-PARA paraphrase gate held for 22 ADRs.** → **Accept; verification gate passed.** No remediation required.

C.5 §5 Risk register

Per-subsystem integration risk aggregated from ADR Risk fields. Mitigation owner is the phase number that closes the risk.

Subsystem	Aggregate risk	Dominant concern	Mitigation owner
Orchestration / Tool Attention (HB-01)	Medium	Acceptance threshold dependent on tool-set composition (xmod #1)	Phase 806 wave 1
Safety / AgentDoG (HB-02)	Low	Schema-additive only; flag-off byte-identical	Phase 807
Safety / STD calibration (HB-03, **CAPCOM HARD GATE**)	Medium	Fail-closed bootstrap exception (xmod #2); CAPCOM gate timing	Phase 808 + CAPCOM gate
skill-creator / WELER roles (HB-04, **CAPCOM HARD GATE**)	Medium-High	State-isolation invariants (xmod #11); CAPCOM gate	Phase 808 + CAPCOM gate
skill-creator / AEL bandit (HB-07, **CAPCOM HARD GATE**)	Medium-High	Bandit convergence + reflection-feedback loop; CAPCOM gate	Phase 812 + CAPCOM gate
Cartridge / structural-completeness linter (HB-05)	Low	False positives on existing in-tree skills (warning-only baseline mitigates)	Phase 810
Cartridge / ambiguity linter (HB-06)	Low	Author friction on first publish	Phase 811
Silicon Layer references	Low	Documentation-only this milestone	Phase 805
DACP references	Low	Documentation-only this milestone	Phase 805
College of Knowledge	Low	Citation-only; concept seeds are additive	Phase 805

Subsystem	Aggregate risk	Dominant concern	Mitigation owner
Grove federated economy	Medium (deferred)	HB-11 threat-model citation gate; publication boundary (concern #5)	v1.49.576+
Reinforcement series	Low	Citation-only this milestone	Phase 805
Cultural-sensitivity gates	Medium	SC-IND/SC-MED/SC-SOV preservation; AFRILANGTUTOR sovereignty consultation	Phase 805 + v1.49.576+

Three CAPCOM HARD GATES — HB-03, HB-04, HB-07 — are noted explicitly. All three require human authorization at the gate; none can pass autonomously.

C.6 §6 Acceptance & sign-off lines

Closing checklist for Phase 803 (integration specs) and Phase 805 (publication):

- ☒ **All ≥9 subsystems addressed in §2.** Twelve subsystems covered (9 required + 3 optional).
- ☒ **Every milestone affected named in §3.** Eight prior milestones (v1.49.561 / .568 / .569 / .570 / .571 / .572 / .573 / .574) plus the v1.0 PNW Taxonomy parallel track.
- ☒ **All 15+ concerns from xmod.md §6 dispositioned in §4.** 15 xmod concerns + 4 audit-level concerns = 19 dispositions.
- ☒ **CAPCOM HARD GATEs at HB-03/HB-04/HB-07 noted in §5 risk register.** All three rows carry the explicit CAPCOM HARD GATE annotation.
- ☒ **Cultural-sensitivity gates SC-IND/SC-MED/SC-SOV explicitly preserved (no overrides proposed).** §2 subsystem 11 names the invariant restatement; §4 disposition #13 carries it forward to Phase 805 tickets; CS25-11 is the verification line.
- ☒ **Federated-economy citation gate (Black-Box Skill Stealing threat model, 2604.21829) named as a pre-rollout requirement before any DoltHub work in v1.49.576+.** §2 subsystem 8 names the gate; §4 disposition #8 accepts it as pre-rollout.

Sign-off invariants (restated for Phase 805 publication):

1. **No module proposed in v1.49.575 replaces or bypasses CAPCOM.** Every Half B T1 module ships default-off; baseline behaviour is byte-identical when flags are off.
2. **SC-IND / SC-MED / SC-SOV gates carry no override.** PNW Indigenous nations are named specifically (Coast Salish, Lummi/Lhaq'temish, Tulalip, Stillaguamish, Wanapum). CARE template uses counselor-alignment exclusively.

AFRILANGTUTOR PNW adaptation requires nation-specific consultation BEFORE any data construction.

3. **2604.21829 Black-Box Skill Stealing citation is a pre-rollout gate** for any federated-economy / DoltHub work in v1.49.576+.
4. **Publication boundary preserved.** CI-Work threat-model specifics stay in `.planning/fox-co` only the abstract attack-taxonomy level may be cited externally. Fox Companies IP review required before public artifacts.
5. **CAPCOM HARD GATEs at HB-03/HB-04/HB-07 require human authorization.** No autonomous pass.

End of Phase 802 Wave 2.3 GSD architectural impact analysis. CS25-05 satisfied.

D Drop-in Integration Specs (Phase 803)

This appendix embeds the 12 engineering-ticket-grade specs from `.planning/missions/cs25-26-`. Each spec >=250 words. HB-01..HB-07 feed Phase 806–812 implementation in v1.49.575 Half B; HB-08..HB-12 feed v1.49.576+ backlog. Three specs carry CAPCOM HARD GATE designation. It satisfies CS25-06 (drop-in integration specs for all ADOPT-decision papers).

D.1 Spec roster index

This directory holds engineering-ticket-grade specs for the 12 HB-line ADOPT decisions surfaced by Wave 1. Each spec ≥250 words (~300 target). HB-01..HB-07 feed Phase 806–812 implementation in v1.49.575 Half B; HB-08..HB-12 feed v1.49.576+ backlog (`gsd-add-backlog` integration). Three specs carry **CAPCOM HARD GATE** designation and inherit the gate-pattern template from v1.49.574 megakernel-substrate `src/cartridge/megakernel/adapter-selection-schema.ts`.

D.1.1 Spec roster

T1 (v1.49.575 Half B)

HB-id	Title	Tier	Phase	Target subsystem	Source ADR	Acceptance	CAPCOM HARD GATE
HB-01	Tool Attention Lazy Schema Loader	T1	806	`src/orchestration/tool-attention/`	ADR-2604.21816	CS25-13	—
HB-02	AgentDoG Where/How/What BLOCK Schema	T1	807	`src/safety/agentdog/`	ADR-2601.18491	CS25-14	—
HB-03	Safe-Turn-Depth Calibration	T1	808	`src/safety/std-calibration/`	ADR-2604.20911	CS25-15	**YES**
HB-04	Worker / Evaluator / Evolution Role Split	T1	808	`src/skillcreator/role-`	ADR-2604.21003 (composes ADR-2604.21725)	CS25-16	**YES**

HB-id	Title	Tier	Phase	Target subsystem	Source ADR	Acceptance	CAPCOM HARD GATE
HB-05	Five-Principle Structural-Completeness Linter	T1	810	`src/cartridge/linter/`	ADR-2604.21090	CS25-17	—
HB-06	Four-Type Ambiguity Checklist	T1	811	`src/cartridge/linter/`	ADR-2604.21505	CS25-18	—
HB-07	AEL Fast/Slow Bandit Auto-Load	T1	812	`src/skillCreator/auto-load/`	ADR-2604.21725 (composes ADR-2604.21003)	CS25-19	**YES**

T2/T3 backlog (v1.49.576+)

HB-id	Title	Tier	Phase	Target subsystem	Source ADR	Acceptance	CAPCOM HARD GATE
HB-08	AGS / RPS Chipset Divergence Metrics	T2-backlog	TBD	`src/chipset-validation/rps-ags/`	ADR-2604.21255	CS25-01 (ADR coverage) + future runtime test	—
HB-09	DryRUN Test Synthesis	T2-backlog	TBD	`src/cartridge/validation/synth/`	ADR-2604.21598	CS25-09 (HB-09 deferred-backlog future)	—
HB-10	IRAP 5-Round Elicitation Cap	T2-backlog	TBD	`src/modelAffinity/` + `.claude/skills/visionmission/`	ADR-2604.21380	CS25-10 (HB-10 deferred-backlog future)	—

HB-id	Title	Tier	Phase	Target subsystem	Source ADR	Acceptance	CAPCOM HARD GATE
HB-11	Black-Box Skill Stealing Threat Model	T2-backlog	TBD (doc may land earlier)	`docs/security/dolthub-federated-economy-threat`	2604.21829	CS25-11 + CS25-04 citation gate	—
HB-12	FSFM 4-Class Forgetting Taxonomy	T2-backlog	TBD	`.planning/policies/` + `src/reinforcement/` + Memory Arena demotion + done-retirement	2604.20300	CS25-01 + CS25-05 cite	—

D.1.2 Pairings + composition notes

- **HB-04 ↔ HB-07** are a semantic pair per xmod CC-3 / convergent-discovery §5: HB-04 establishes the Worker/Evaluator/Evolution role naming + per-episode adversarial check; HB-07 supplies the cross-episode reflection bandit. Bandit lives *inside* Evolution role's protocol-update path. Two flags toggle independently; the architectural unit is the pair. Both are CAPCOM HARD GATE.
- **HB-05 ↔ HB-06** are a SKILL.md authoring discipline upgrade per xmod CC-2: structural-completeness + four-type ambiguity ship together as Phase 810 + 811. Both linters land in `src/cartridge/linter/`.
- **HB-01 ↔ MCP triple-gate** (CC-1): HB-01 Phase 806 ticket is widened to include the six-attack-family + function-hijacking test fixtures from 2604.21477 + 2604.20994 per impact.md §4 #10.
- **HB-09 ↔ HB-06** (when HB-09 promotes): DryRUN test-side + Orchid spec-side define the spec-quality / test-quality envelope for skill authoring.
- **HB-11 ↔ HB-05** (when HB-11 promotes): structural-completeness linter is the first defense against marketplace contamination; HB-11 threat-model is the citation gate before federated promotion runs.

D.1.3 Special-handling dispositions (impact.md §4)

- **HB-01 baseline (impact.md §4 #1):** `token-budget-baseline.test.ts` is a pre-implementation step. Phase 806 wave-1 produces a baseline on a synthetic mock-MCP scenario with N tools at three complexity tiers before locking the 40% p50 reduction threshold; if unreachable, the threshold re-scopes to "p50 ≥ baseline × 0.6".
- **HB-03 fail-closed bootstrap (impact.md §4 #2):** `fail-closed-bootstrap.test.ts` covers the bootstrap exception. On first run with no calibration: conservative

STD-floor across all enabled models, one-time WARN, explicit re-calibration trigger required before fail-closed activates.
- **HB-04 ↔ HB-07 composition (convergent-discovery §5):** HB-04 spec names the integration boundary; bandit operates inside Evolution role; both specs cross-reference each other.

D.1.4 Phase 805 dependencies

The following HB-NN specs are referenced by Phase 805 publication artefacts. For each, the artefact name is given; the spec is the engineering-ticket-grade backing reference Phase 805 cites.

- **HB-01:** verification matrix entry CS25-13; substrate-doc reference `docs/substrate-referen` (NEW); Phase 805 widens the HB-01 ticket to the MCP triple-gate per impact.md §4 #10.
- **HB-02:** verification matrix entry CS25-14; CLAUDE.md draft (CS25-09) — schema cited as the BLOCK record format the v1.49.576+ correlator consumes.
- **HB-03:** verification matrix entry CS25-15; `.planning/safety/std-calibration.json` schema doc; CLAUDE.md draft cites Root Theorem C3 fidelity-threshold gating as the formal foundation; CAPCOM HARD GATE pass requires this spec's bootstrap-exception path before the gate clears.
- **HB-04:** verification matrix entry CS25-16; `gsd-mission-management-roles.md`; CLAUDE.md "Architectural Foundations" preamble cites 2604.21003 Last Harness as one of four convergent-discovery anchors (CS25-09); `.planning/PROJECT.md` bounded-learning-policy section cites 2604.21003 as published meta-evolution derivation.
- **HB-05:** verification matrix entry CS25-17; SKILL.md authoring guide draft addition (CS25-09) — five-principle structural-completeness check; CLAUDE.md draft cites GROUNDINGmd 2604.21744 as the convergent-discovery framing that the five principles mirror CSO discipline.
- **HB-06:** verification matrix entry CS25-18; SKILL.md draft adds the four-type ambiguity checklist (lexical / syntactic / semantic / vagueness) (CS25-09); `docs/skill-author` is a Phase 805 publication artefact.
- **HB-07:** verification matrix entry CS25-19; `.claude/agents/` documentation update mapping executor/verifier/planner onto Worker/Evaluator/Evolution.
- **HB-08..HB-12:** backlog citations only at v1.49.575; Phase 805 substrate-doc references each spec as the engineering-ticket-grade reference for v1.49.576+ promotion; the threat-model document HB-11 may land at v1.49.575 as a forward-looking artefact alongside the Phase 805 publication.

End of Phase 803 Wave 2.4 drop-in integration specs index. CS25-06 satisfied (12 spec files + index).

D.2 HB-01 — Tool Attention Lazy Schema Loader

D.2.1 §Engineering ticket

Ship a runtime mitigation for the planning-bridge MCP "Tools Tax" (10K–60K tokens/-turn observed in the Tool Attention paper) at `src/orchestration/tool-attention/`. The approach is three pieces composed: (1) an Intent Schema Overlap (ISO) score computed from sentence embeddings (Haiku-tier sidecar) that ranks loaded tools by

relevance to the current turn's intent; (2) a state-aware gate that adjusts the top-k budget by current task phase (planning vs executing vs verifying changes the admissible set); (3) a two-phase lazy loader that keeps a compact pool of tool names + 1-line descriptions in context and fetches full JSON schemas only for the top-k gated tools. A budget monitor watches the running fraction of context occupied by tool schemas and warns at the paper's 70% context-fracture threshold. All four pieces sit *above* the dispatch boundary; no existing dispatch code path is rewritten. The flag `cs25-26-sweep.tool-attention` defaults off and the load path is byte-identical to v1.49.574 baseline when off.

D.2.2 §Target file paths

- `src/orchestration/tool-attention/iso-score.ts` — ISO embedding-overlap selector; returns a ranked tool list per turn.
- `src/orchestration/tool-attention/state-gate.ts` — phase-conditional top-k budget function (planning/executing/verifying).
- `src/orchestration/tool-attention/lazy-loader.ts` — two-phase loader: compact summary pool → full schemas top-k.
- `src/orchestration/tool-attention/budget-monitor.ts` — token-occupancy tracker with 70% fracture-threshold alert.
- `src/orchestration/tool-attention/index.ts` — public entry composing the four modules behind the flag.
- `src/orchestration/tool-attention/__tests__/iso-score.test.ts`, `state-gate.test.t` `lazy-loader.test.ts`, `budget-monitor.test.ts`, `mock-mcp-scenario.test.ts`, `token-budget-baseline.test.ts`, `flag-off-byte-identical.test.ts`, `phase-pin-overri` `fracture-alert.test.ts`, `composition.test.ts`.

D.2.3 §Validation tests

1. `iso-score.test.ts` — ISO score returns identical ranks for identical embeddings + intent (deterministic).
2. `state-gate.test.ts` — top-k changes per declared phase and respects phase-pinned tools.
3. `lazy-loader.test.ts` — compact pool contains names + 1-line descriptions only; full schemas appear only for the top-k.
4. `budget-monitor.test.ts` — alert fires at 70% context occupancy on tool schemas; no alert at 69%.
5. `mock-mcp-scenario.test.ts` — end-to-end against a mock planning-bridge MCP with N=20 tools at varied complexity.
6. **`token-budget-baseline.test.ts` — baseline-measurement test (CS25-13).** Records the all-tools-loaded p50/p95 tokens-per-turn on the synthetic mock-MCP scenario described below, then enables the flag and asserts ≥40% p50 reduction. **Pre-implementation step (per impact.md §4 #1):** Phase 806 wave-1 produces this baseline first, on a synthetic scenario containing N tools at three complexity tiers (simple-CRUD, mid-complexity-with-enums, large-schema-with-nested-objects) drawn from real MCP server schemas; if 40% is unreachable, the threshold is re-scoped to "p50 ≥ baseline × 0.6" in the Phase 806 plan.
7. `flag-off-byte-identical.test.ts` — fixture asserting byte-identical dispatch path with flag off.
8. `phase-pin-override.test.ts` — phase-pinned tools survive ISO low-rank.

9. `fracture-alert.test.ts` — alert telemetry surfaces the offending occupancy ratio.
10. `composition.test.ts` — full pipeline composes ISO → gate → lazy-load → monitor.

D.2.4 §Default-off invariant

With `cs25-26-sweep.tool-attention=false` the planning-bridge MCP dispatch path is byte-identical to v1.49.574 baseline. The `flag-off-byte-identical.test.ts` fixture mirrors the v1.49.574 `disabled-byte-identical` pattern: capture the dispatch trace under flag-off, compare against a recorded v1.49.574 reference trace, fail on any byte diff including telemetry payloads.

D.2.5 §Rollback criterion

- Baseline-measurement deviation: if Phase 806 wave-1 records cannot reproduce the 40% p50 number, the implementation is held until the threshold is re-scoped per impact.md §4 #1.
- ISO-score quality regression: any rare-but-needed tool gated out incorrectly in the existing in-tree skill corpus.
- Flag-off byte-diff: any non-zero diff in the dispatch trace under flag-off compared to v1.49.574 baseline.

D.2.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M4/ADR-2604.21816.md`.
- xmod CC-1 MCP triple-gate: composes with 2604.21477 (MCP Pitfall Lab) + 2604.20994 (Function Hijacking) — Phase 806 ticket widens to include the six-attack-family + function-hijacking test fixtures per impact.md §4 #10.
- Prior milestones: v1.49.570 convergent-substrate (two-gate cascade-mcp-defense pre-citation), v1.49.574 megakernel substrate (default-off flag pattern).
- v1.49.572 Root Theorem (2604.20874) supplies the bounded-tape framing under which the 70% threshold is a fidelity-threshold constant, not a capacity constant.

D.2.7 §Open questions for implementing engineer

- ISO-score threshold tuning: fixed cosine-similarity cutoff vs adaptive (z-score against per-session distribution)?
- Embedding model selection: Haiku-tier sidecar — which embedding family (sentence-transformers vs Anthropic embeddings)?
- Cache invalidation policy on schema fetches: per-turn or per-phase?
- Phase-pin declaration surface: where do skills declare "I always need tool X regardless of ISO rank"?

D.3 HB-02 — AgentDoG Where/How/What BLOCK Schema

D.3.1 §Engineering ticket

Ship a structured diagnostic record alongside every Safety Warden BLOCK decision at `src/safety/agentdog/`. The approach lifts AgentDoG's three-axis taxonomy verbatim: every BLOCK emit gains a `Where` axis (offending component + invocation context), a `How` axis (vulnerability vector + escalation pattern), and a `What` axis (impacted asset + blast radius). The schema is implemented as three TypeScript modules (one

per axis) plus a composing emitter that the existing Safety Warden hooks into when the flag is on. The 4B-tier sidecar deployment plan is named — a Haiku-tier inference pass that fills in the three axes given the BLOCK context — but the v1.49.575 ship is the schema + emitter + integration; sidecar deployment is documented for v1.49.576+ runtime instrumentation. With `cs25-26-sweep.agentdog-schema=false`, BLOCK emission is byte-identical to v1.49.574: no schema field added, no extra log entries, no telemetry envelope changes. The schema records persist to Memory Arena M13 content-addressed Grove records as the substrate the v1.49.576+ cross-session correlator (CSTM-Bench, 2604.21131) consumes.

D.3.2 §Target file paths

- `src/safety/agentdog/where.ts` — `Where` axis: component + invocation context capture.
- `src/safety/agentdog/how.ts` — `How` axis: vulnerability vector + escalation pattern enumeration.
- `src/safety/agentdog/what.ts` — `What` axis: impacted asset + blast-radius classification.
- `src/safety/agentdog/emitter.ts` — composes the three axes into a single AgentDoG record per BLOCK.
- `src/safety/agentdog/integration.ts` — hooks into existing Safety Warden BLOCK paths when the flag is on.
- `src/safety/agentdog/index.ts` — public entry.
- `src/safety/agentdog/__tests__/where.test.ts`, `how.test.ts`, `what.test.ts`, `emitter.test.ts`, `integration.test.ts`, `block-path-coverage.test.ts`, `flag-off-byte-i` `grove-persistence.test.ts`, `schema-shape.test.ts`, `sidecar-handoff.test.ts`.

D.3.3 §Validation tests

1. `where.test.ts` — `Where` axis records component identifier + invocation context (call-site stack frame, current phase, current skill).
2. `how.test.ts` — `How` axis enumerates vulnerability vector (e.g., prompt-injection / metadata-poisoning / function-hijacking) + escalation pattern.
3. `what.test.ts` — `What` axis classifies impacted asset (secrets / signing keys / `.git/` / wasteland exclusion / etc.) and blast-radius tier.
4. `emitter.test.ts` — emitter composes a complete record; missing-field cases throw not silently fill.
5. `integration.test.ts` — integration with each existing Safety Warden BLOCK path emits an AgentDoG record.
6. `block-path-coverage.test.ts` — every existing BLOCK path in the repo is exercised; every path emits a record.
7. `flag-off-byte-identical.test.ts` — fixture asserting byte-identical BLOCK emission with flag off; no schema field added at runtime.
8. `grove-persistence.test.ts` — records persist to Memory Arena M13 content-addressed Grove records under the right namespace.
9. `schema-shape.test.ts` — schema validates against a JSON-schema definition; CI fails on shape drift.
10. `sidecar-handoff.test.ts` — record is shaped to be consumable by the v1.49.576+ Haiku-tier sidecar (forward compatibility).

D.3.4 §Default-off invariant

With `cs25-26-sweep.agentdog-schema=false`, BLOCK emission is byte-identical to v1.49.574 baseline — no schema field added at runtime, no extra log line, no telemetry payload change. The `flag-off-byte-identical.test.ts` fixture mirrors the v1.49.574 `disabled-byte-identical` pattern by recording the BLOCK emission trace under flag-off and comparing against a v1.49.574 reference trace.

D.3.5 §Rollback criterion

- Flag-off byte-diff: any non-zero diff in BLOCK emission with the flag off vs v1.49.574 baseline.
- Schema-shape drift: any change in the JSON schema that would break the v1.49.576+ sidecar contract.
- BLOCK-path coverage gap: any existing BLOCK path that does not emit a record when the flag is on.

D.3.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M3/ADR-2601.18491.md`.
- xmod CC-6 cross-session threat correlator: composes with 2604.20911 (STD-floor breach events) + 2604.21131 (CSTM-Bench correlator consumer).
- Prior milestones: v1.49.561 trust-relationship (Stage 1 — supplies the relationship axis the `Where` field references), Memory Arena M13 (content-addressed Grove records as persistence layer).
- v1.49.572 GROUNDINGmd Hard Constraints (2604.21744) supply the override-user-prompt invariants the `What` axis records.

D.3.7 §Open questions for implementing engineer

- Sidecar latency budget: 4B model on Haiku tier — what is the acceptable per-BLOCK latency before sidecar inference is gated behind a "structured-only at flush" mode?
- Schema versioning: schema changes likely with v1.49.576+ correlator wiring — embed a version field now or rely on Grove content-addressing for migration?
- `Where` axis scope: should it record the full call-stack or only the top frame + a cryptographic hash of the rest?
- Privacy boundary: BLOCK on a user's secret should not leak the secret's content into the AgentDoG record — what is the redaction discipline?

D.4 HB-03 — Safe-Turn-Depth Calibration

D.4.1 §Engineering ticket

Ship a per-model omission-decay calibration + re-injection middleware at `src/safety/std-calibr`
The approach is three pieces: (1) a calibration harness that measures per-model prohibition-type-constraint compliance across conversation depths (Opus / Sonnet / Haiku), reproducing the paper's 73%@turn-5 → 33%@turn-16 observation in GSD's own deployment; (2) a calibration table at `.planning/safety/std-calibration.json` recording the per-model Safe Turn Depth (STD) — the conversation depth before omission compliance falls below tolerance; (3) a re-injection middleware that fires when conversation depth approaches a model's STD floor, re-injecting the active omission constraints from the GROUNDINGmd / CLAUDE.md Hard Constraints set.

The flag `cs25-26-sweep.std-calibration` defaults off; flag-off conversation flow is byte-identical to v1.49.574. **CAPCOM HARD GATE** — touches Safety Warden BLOCK timing; requires explicit human authorization at the gate boundary, no autonomous pass. Records emitted on STD-floor approach are AgentDoG-shaped (HB-02 schema). The cross-session correlator (2604.21131, v1.49.576+) consumes STD-floor approach events as cross-session signal.

D.4.2 §Target file paths

- `src/safety/std-calibration/measure.ts` — calibration harness: runs synthetic prohibition-decay scenarios per model.
- `src/safety/std-calibration/table.ts` — read/write the JSON calibration table at `.planning/safety/std-calibration.json`.
- `src/safety/std-calibration/middleware.ts` — re-injection middleware fired at depth ≈ STD.
- `src/safety/std-calibration/bootstrap.ts` — fail-closed bootstrap exception (see below).
- `src/safety/std-calibration/index.ts` — public entry; integrates with Safety Warden middleware chain.
- `src/safety/std-calibration/__tests__/measure-opus.test.ts`, `measure-sonnet.test.t` `measure-haiku.test.ts`, `table-roundtrip.test.ts`, `re-injection-regression.test.ts`, `fail-closed-bootstrap.test.ts`, `flag-off-byte-identical.test.ts`, `capcom-gate.test`

D.4.3 §Validation tests

1. `measure-opus.test.ts` / `measure-sonnet.test.ts` / `measure-haiku.test.ts` — per-model STD measurement over a synthetic prohibition-decay benchmark; outputs a numerical STD that matches the paper's range within tolerance.
2. `table-roundtrip.test.ts` — write + read of the calibration table preserves all fields and per-model records.
3. `re-injection-regression.test.ts` — at depth = STD-1, no re-injection; at depth = STD, re-injection fires and emits an AgentDoG-shaped record.
4. **`fail-closed-bootstrap.test.ts` — bootstrap-exception test (impact.md §4 #2).** On first run with no calibration data: the system enters a bootstrap mode that defaults to a *conservative STD-floor across all enabled models* (≤turn-5, the lowest reported STD in the paper), emits a one-time WARN-level diagnostic, and requires an explicit re-calibration trigger (CLI: `npx skill-creator safety std-calibrate <model>`) before fail-closed activates fully. The test sets up an empty calibration table, asserts the conservative floor activates, asserts the warning emits exactly once per process start, and asserts that without the explicit trigger the system does not silently switch to autonomous fail-closed.
5. `flag-off-byte-identical.test.ts` — fixture asserting conversation flow is byte-identical to v1.49.574 baseline with flag off; no re-injection fires, no calibration table is read.
6. **`capcom-gate.test.ts` — CAPCOM gate test.** Without a recorded human authorization (no CAPCOM-pass marker in `.planning/safety/std-calibration.capcom`), the middleware refuses to engage even with the flag on; the gate fires and the engagement is blocked. Mirrors the v1.49.574 megakernel-substrate HB-04 adapter-selection-schema CAPCOM-gate-test pattern.
7. Integration test verifying the middleware composes with HB-02 AgentDoG emit-

ter so STD events emit AgentDoG-shaped records.
8. Per-model boundary test: models not in the enabled set produce a clean error rather than silent passthrough.

D.4.4 §Default-off invariant

With `cs25-26-sweep.std-calibration=false`, no constraint re-injection fires, conversation flow is byte-identical to v1.49.574, calibration table is not read, no STD events emit. The `flag-off-byte-identical.test.ts` fixture mirrors v1.49.574 `disabled-byte-i`

D.4.5 §Rollback criterion

- CAPCOM gate bypass: any code path that engages re-injection without a recorded human authorization marker.
- Bootstrap-exception failure: bootstrap mode silently switches to fail-closed without the explicit re-calibration trigger.
- Re-injection placement defect: re-injection that creates a false confidence spike at the exact turn where decay is most acute.
- Flag-off byte-diff: any non-zero diff in conversation flow under flag-off vs v1.49.574 baseline.

D.4.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M3/ADR-2604.20911.md`.
- xmod CC-6 cross-session threat correlator: composes with HB-02 AgentDoG (2601.18491) + CSTM-Bench (2604.21131).
- Prior milestone v1.49.569 Drift in LLM Systems — STD calibration is the prohibition-decay specialization of v1.49.569's drift detection methodology; bootstrap-exception path references v1.49.569 drift telemetry per impact.md §3.
- **CAPCOM gate-pattern template:** v1.49.574 megakernel-substrate `src/cartridge/megake` — the prior CAPCOM HARD GATE this spec inherits the gate structure from. Reuse the `disabled-byte-identical.test.ts` fixture pattern verbatim.
- v1.49.572 Root Theorem (2604.20874) C3 fidelity-threshold gating supplies the formal foundation: STD is the omission-decay-specific instance of the fidelity-threshold gate category.

D.4.7 §Open questions for implementing engineer

- Calibration cadence: re-measure on every model release vs scheduled re-calibration (weekly / monthly)?
- Bootstrap conservative-floor value: literal turn-5 vs a fraction of the most-conservative model's expected STD?
- Re-injection content: full Hard Constraints set vs only the constraints flagged as recently violated?
- Storage of the `.capcom` authorization marker: file with user signature vs git-tracked attestation?

D.5 HB-04 — Worker / Evaluator / Evolution Role Split

D.5.1 §Engineering ticket

Ship the Worker / Evaluator / Evolution naming convention + role-isolated subagents at `src/skill-creator/roles/`. The approach is three role-isolated subagent con-

tracts: (1) **Worker** generates candidate skills against the target task; (2) **Evaluator** *adversarially* diagnoses Worker output against failure-history (replacing the current passive 3-correction-min rule with active failure-finding) and holds BLOCK authority over Worker outputs; (3) **Evolution** proposes loop-protocol modifications based on cross-skill patterns. The split is *additive*: it names what skill-creator's six-step loop already does (Observe → Detect → Suggest → Manage → Auto-Load → Learn/-Compose) rather than restructuring it. The flag `cs25-26-sweep.weler-roles` defaults off; flag-off skill-creator runs the existing six-step loop byte-identically. **CAPCOM HARD GATE** — touches skill-creator orchestration; requires explicit human authorization. **HB-04 composes with HB-07 (AEL bandit) per xmod CC-3 / convergent-discovery §5:** HB-04 establishes the role naming and per-episode adversarial check; HB-07 supplies the cross-episode reflection bandit. Together they form the fast-Worker / fast-Evaluator / slow-Evolution architecture. The integration boundary: HB-07's bandit lives *inside* HB-04's role scaffolding — the bandit is one of Evolution's protocol-update mechanisms, not a sibling system. Phase 808 Plan must enumerate the role-isolation invariants per impact.md §4 #11: Worker writes own state only; Evaluator reads Worker output (read-only); Evolution reads aggregated history (read-only); no cross-role write paths.

D.5.2 §Target file paths

- `src/skill-creator/roles/worker.ts` — Worker contract: candidate-skill generator.
- `src/skill-creator/roles/evaluator.ts` — Evaluator contract: adversarial diagnosis + BLOCK authority over Worker output.
- `src/skill-creator/roles/evolution.ts` — Evolution contract: loop-protocol-modification proposer.
- `src/skill-creator/roles/isolation.ts` — runtime role-isolation enforcer (state-write boundary checks).
- `src/skill-creator/roles/index.ts` — public entry composing the three roles into the existing six-step loop.
- `gsd-mission-management-roles.md` — formal naming-convention doc.
- `.claude/agents/` documentation update — map existing executor/verifier/planner onto Worker/Evaluator/Evolution.
- `src/skill-creator/roles/__tests__/worker.test.ts`, `evaluator.test.ts`, `evolution.te` `isolation-no-cross-write.test.ts`, `evaluator-blocks-worker.test.ts`, `six-step-integ` `flag-off-byte-identical.test.ts`, `capcom-gate.test.ts`, `hb07-composition.test.ts`, `adversarial-vs-3-correction.test.ts`.

D.5.3 §Validation tests

1. `worker.test.ts` — Worker generates candidates against the target task; output shape conforms to contract.
2. `evaluator.test.ts` — Evaluator adversarially diagnoses Worker output against failure-history; finds known-planted failure modes.
3. `evolution.test.ts` — Evolution proposes loop-protocol modifications based on aggregated history.
4. `isolation-no-cross-write.test.ts` — role-isolation invariant: any cross-role write path is rejected at runtime.
5. `evaluator-blocks-worker.test.ts` — Evaluator's BLOCK authority over Worker

outputs fires correctly; Worker cannot bypass.

6. `six-step-integration.test.ts` — integration with the existing six-step loop is correct; outputs at each step match baseline shape.
7. `flag-off-byte-identical.test.ts` — fixture asserting skill-creator runs the existing six-step loop byte-identically with flag off.
8. **`capcom-gate.test.ts` — CAPCOM gate test.** Without a recorded human authorization marker, the role split refuses to engage even with the flag on. Mirrors the v1.49.574 megakernel-substrate HB-04 adapter-selection-schema gate pattern.
9. **`hb07-composition.test.ts`** — verifies the integration boundary with HB-07: with both HB-04 and HB-07 flags on, the AEL bandit operates inside the Evolution role's protocol-update path; Evolution reads bandit history at the configured slow-timescale interval.
10. `adversarial-vs-3-correction.test.ts` — comparative test: Evaluator catches a planted failure that the legacy 3-correction-min rule misses.

D.5.4 §Default-off invariant

With `cs25-26-sweep.weler-roles=false`, skill-creator runs the existing six-step loop byte-identically; the executor / verifier / planner subagents continue using their current invocation paths; no role-isolation enforcer fires. The `flag-off-byte-identical.test.ts` fixture mirrors v1.49.574 `disabled-byte-identical`.

D.5.5 §Rollback criterion

- CAPCOM gate bypass: any code path that engages role split without a recorded human authorization marker.
- Role-isolation invariant violation: any cross-role write path that survives the isolation enforcer.
- Evaluator authority degradation: Evaluator's BLOCK authority becomes advisory-only — at that point the role split degenerates to passive-counting and the upgrade is illusory (per ADR-2604.21003 Notes).
- Flag-off byte-diff: any non-zero diff in six-step loop output under flag-off vs v1.49.574 baseline.

D.5.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M1/ADR-2604.21003.md`; **composes with** `.planning/missions/cs25-26-sweep/work/modules/M2/ADR-2604.21725.m` (HB-07 / AEL).
- xmod CC-3 AEL × Last Harness composed two-loop pattern (convergent-discovery §5 fast/slow timescale split).
- Prior milestones: v1.49.571 LeJEPA / heuristics-free skill-space (Last Harness Worker/Evaluator/Evolution split = heuristics-free pattern; the adversarial Evaluator replaces the heuristic 3-correction-min); v1.49.574 megakernel substrate (gate pattern + flag pattern).
- **CAPCOM gate-pattern template:** v1.49.574 megakernel-substrate `src/cartridge/megake` — the prior CAPCOM HARD GATE this spec inherits the gate structure from.
- `.planning/PROJECT.md` bounded-learning-policy section — gains a citation to 2604.21003 as the published meta-evolution derivation (under CS25-09's edit budget).

D.5.7 §Open questions for implementing engineer

- Role isolation enforced via runtime check (per-write callback) vs static type discipline (TypeScript role-tagged types) vs both?
- Evaluator BLOCK authority surface: does Evaluator emit a BLOCK record into the AgentDoG schema (HB-02), or is BLOCK an internal control-flow signal only?
- Evolution slow-timescale cadence: per-skill / per-mission / per-milestone interval — and where is the cadence configured?
- HB-07 integration boundary detail: does Evolution own the bandit's posterior storage, or is the bandit's storage shared with the auto-load step?

D.6 HB-05 — Five-Principle Structural-Completeness Linter

D.6.1 §Engineering ticket

Ship a SKILL.md validation linter implementing the five-principle structural-completeness check at `src/cartridge/linter/structural-completeness.ts`. The five principles, lifted verbatim from 2604.21090, are: (1) **computability-theory grounding** for any computational claims a SKILL.md makes; (2) **proof-theoretic structure** for any deductive claims; (3) **Bayesian-epistemology framing** for any uncertainty claims; (4) **data-classification specification** for any data-handling claims; (5) **assessment-rubric specification** for any quality claims. The paper's empirical 37% incompleteness rate makes the linter operationally necessary; the principles are mechanically checkable. Linter runs as part of the cartridge promotion pipeline and gates promotion out of `.planning/staging/inbox/`. The flag `cs25-26-sweep.structural-completeness-lint` defaults off → linter runs in **warning-only** mode (does not block promotion); when on, missing-principle records become blocking. **HB-05 ships paired with HB-06** (Four-Type Ambiguity Checklist, 2604.21505) per xmod CC-2 — the two linters land as a single SKILL.md authoring discipline upgrade. The linter ingests the GROUNDINGmd two-class taxonomy from 2604.21744 as a structural-completeness criterion: a SKILL.md must distinguish Hard Constraints (override-the-prompt invariants the skill refuses to violate) from Convention Parameters (defaults that allow override). This becomes the "Hard Constraints declared" check in the negative-fixture corpus.

D.6.2 §Target file paths

- `src/cartridge/linter/structural-completeness.ts` — five-principle validator.
- `src/cartridge/linter/principles/computability.ts` — Principle 1 check.
- `src/cartridge/linter/principles/proof-theory.ts` — Principle 2 check.
- `src/cartridge/linter/principles/bayesian.ts` — Principle 3 check.
- `src/cartridge/linter/principles/data-classification.ts` — Principle 4 check.
- `src/cartridge/linter/principles/assessment-rubric.ts` — Principle 5 check.
- `src/cartridge/linter/principles/hard-constraints-declared.ts` — GROUNDINGmd two-class check (cross-link with 2604.21744).
- `src/cartridge/linter/__tests__/structural-completeness.test.ts`, plus per-principle test files and positive/negative fixtures under `src/cartridge/linter/__tests__/fi`

D.6.3 §Validation tests

1. `principle-1-computability.test.ts` — fixture: SKILL.md with computational claim and no computability grounding flagged; with grounding passes.

2. `principle-2-proof-theory.test.ts` — fixture: deductive claim without proof structure flagged.
3. `principle-3-bayesian.test.ts` — fixture: uncertainty claim without Bayesian framing flagged.
4. `principle-4-data-classification.test.ts` — fixture: data-handling claim without classification flagged.
5. `principle-5-assessment-rubric.test.ts` — fixture: quality claim without rubric flagged.
6. **Negative-fixture corpus** — at least 3 SKILL.md negative fixtures, each missing exactly one principle; each fails the corresponding check and only that check.
7. **Positive-fixture corpus** — at least 3 SKILL.md positive fixtures (drawn from existing in-tree skills selected for compliance) pass all five principles.
8. `hard-constraints-declared.test.ts` — fixture: SKILL.md without a Hard-Constraints-vs-Convention-Parameters split flagged; cross-link test against 2604.21744.
9. `flag-off-warning-only.test.ts` — fixture asserts that with the flag off, the linter emits warnings but does not block promotion (existing behavior preserved).
10. `flag-off-byte-identical.test.ts` — promotion pipeline behavior is byte-identical to v1.49.574 baseline with flag off (warnings emit but do not affect exit code or pipeline state).

D.6.4 §Default-off invariant

With `cs25-26-sweep.structural-completeness-lint=false`, the linter runs in warning-only mode: failures emit a warning to stderr but do not change the cartridge promotion exit code, do not block promotion, and do not modify the pipeline state. The `flag-off-byte-identical.test.ts` fixture mirrors v1.49.574 `disabled-byte-identical` for the promotion pipeline state.

D.6.5 §Rollback criterion

- False positives on existing in-tree skills: zero false positives is the CS25-18 acceptance condition. If any existing in-tree skill trips a check, treat as a fixture defect (refine the check), not a baseline regression.
- Warning-only-mode breakage: the linter blocks promotion with the flag off — this would regress the cartridge promotion baseline.
- Mechanical-checkability defect: a principle check that requires LLM judgment rather than mechanical analysis (the paper's own static analyzer is the reference implementation pattern).

D.6.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M3/ADR-`2604.21090`.md`.
- xmod CC-2 SKILL.md authoring discipline upgrade: ships paired with HB-06 (2604.21505).
- Prior milestones: v1.49.573 Skilldex Conformance Auditor (compounds with this linter — same gate, complementary checks); v1.49.574 megakernel substrate (default-off flag pattern).
- 2604.21744 GROUNDINGmd convergent-discovery framing: the five principles **mirror** GSD's existing CSO discipline rather than introducing a new norm — Phase 810 ticket includes the cross-reference per impact.md §4 #15. (This is the convergent-discovery ↔ structural-completeness rhyme the published Phase

805 substrate doc names.)
- v1.49.576+ trigger: gate flips from warning-only to blocking after corpus audit completes (≤2% false-positive rate on the v1.49.575 corpus).

D.6.7 §Open questions for implementing engineer

- Mechanical-checkability boundary: which principles can be fully mechanical (computability / proof-theory / data-classification — likely yes) and which need a heuristic-with-confidence pattern (Bayesian framing / assessment rubric)?
- Negative-fixture corpus authorship: hand-author ≥3 fixtures or scrape from public skill marketplaces (and risk the marketplace contamination concern HB-11 addresses)?
- Hard-Constraints-declared check signature: a YAML frontmatter block (`hard_constraints: [...]`) vs a section heading (`## Hard Constraints`)?
- Cross-link with HB-06 ambiguity linter: do the two linters share a common rule-engine, or do they run as independent passes?

D.7 HB-06 — Four-Type Ambiguity Checklist

D.7.1 §Engineering ticket

Ship a SKILL.md ambiguity linter implementing Orchid's four-type taxonomy at `src/cartridge/li` plus an authoring-guidance document at `docs/skill-authoring/ambiguity-checklist.md`. The four types, lifted from 2604.21505, are: (1) **lexical** — terminology with multiple plausible meanings; (2) **syntactic** — grammar admitting multiple parses; (3) **semantic** — under-specified intent admitting multiple plausible interpretations; (4) **vagueness** — quantifier-free claims, "appropriate" / "reasonable" without operationalization. Each frontmatter capability declaration receives four checks. A failing check blocks `skill-creator cartridge publish` until resolved. The flag `cs25-26-sweep.ambiguity-li` defaults off → warnings only. The Orchid headline finding — *advanced models suffer more from ambiguity* — directly justifies the chipset architecture's policy that Opus-tier work demands stricter specs, not laxer; this is captured in the authoring-guidance doc as the rationale section. **HB-06 ships paired with HB-05** per xmod CC-2 (SKILL.md authoring discipline upgrade). The two linters land together as Phase 810 + 811, and the README index lists the pairing explicitly. The linter is a heuristic + rule-based pass: lexical = terminology dictionary check; syntactic = grammar-parse-ambiguity flag; semantic = quantifier presence + reference-resolution check; vagueness = pattern-match against a corpus of known vague terms.

D.7.2 §Target file paths

- `src/cartridge/linter/ambiguity.ts` — four-type ambiguity linter entry point.
- `src/cartridge/linter/ambiguity/lexical.ts` — Type 1 check: terminology dictionary lookup.
- `src/cartridge/linter/ambiguity/syntactic.ts` — Type 2 check: grammar-parse-ambiguity flag.
- `src/cartridge/linter/ambiguity/semantic.ts` — Type 3 check: under-specified-intent detection.
- `src/cartridge/linter/ambiguity/vagueness.ts` — Type 4 check: quantifier + vague-term pattern match.
- `docs/skill-authoring/ambiguity-checklist.md` — authoring-guidance doc with

the rationale ("advanced models suffer more").
- `tools/lint-skill.mjs` — wire ambiguity checks into the `skill-creator cartridge publish` path.
- `src/cartridge/linter/__tests__/ambiguity-{lexical,syntactic,semantic,vagueness` `ambiguity-publish-block.test.ts`, `flag-off-byte-identical.test.ts`, `existing-skills`

D.7.3 §Validation tests

1. `ambiguity-lexical.test.ts` — fixture: SKILL.md with multi-meaning term flagged; resolved-term version passes.
2. `ambiguity-syntactic.test.ts` — fixture: SKILL.md with grammar-ambiguous claim flagged; rewritten version passes.
3. `ambiguity-semantic.test.ts` — fixture: under-specified intent flagged; specified-intent version passes.
4. `ambiguity-vagueness.test.ts` — fixture: quantifier-free claim ("appropriate", "reasonable") flagged; operationalized version passes.
5. `ambiguity-publish-block.test.ts` — `skill-creator cartridge publish` fails with one of four distinct error codes (one per ambiguity type) given a minimal SKILL.md exercising each type.
6. **`existing-skills-zero-false-positives.test.ts`** — runs the linter against every in-tree SKILL.md in `examples/` and `.claude/skills/`; **CS25-18 acceptance: zero false positives at baseline.** Any in-tree skill that trips an ambiguity check is treated as a fixture defect (refine the check), not a baseline regression.
7. `flag-off-byte-identical.test.ts` — promotion pipeline behavior is byte-identical to v1.49.574 baseline with flag off (warnings emit but no block).

D.7.4 §Default-off invariant

With `cs25-26-sweep.ambiguity-lint=false`, the linter runs in warning-only mode: failures emit warnings to stderr but do not change exit codes or block `skill-creator cartridge publish`. The `flag-off-byte-identical.test.ts` fixture mirrors v1.49.574 `disabled-byte-identical` for the promotion pipeline state.

D.7.5 §Rollback criterion

- Any false positive on an existing in-tree skill at baseline (CS25-18 acceptance condition): treat as fixture defect, refine the check, do not regress.
- Author-friction explosion: if the linter creates ≥1 hour additional friction per skill publish, the heuristic thresholds need re-calibration.
- Flag-off block: any code path that blocks publish with the flag off would regress the cartridge baseline.

D.7.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M5/ADR-2604.21505.md`.
- **Pairs with HB-05** per xmod CC-2 (SKILL.md authoring discipline upgrade): both linters ship as Phase 810 + 811, README index lists the pairing.
- Prior milestones: v1.49.572 output-structure frontmatter at `src/output-structure/` (the frontmatter is the surface the linter inspects); v1.49.570 reinforcement series MA-6 emitters at `src/reinforcement/` (each linter pass emits a reinforcement event so the eligibility layer can credit ambiguity-reduction skills retroac-

tively).
- xmod connection 2604.21090 ↔ 2604.21505 composes-with strong (table row): both run on the same `src/cartridge/linter/` directory.

D.7.7 §Open questions for implementing engineer

- Lexical-dictionary source: hand-curated GSD-domain dictionary vs WordNet vs LLM-judged?
- Syntactic-parse engine: standard NLP parser (spaCy / tree-sitter-markdown) vs a GSD-specific frontmatter grammar?
- Semantic-check threshold: how much under-specification is tolerable before flagging? (e.g., is "the planner runs over the artifacts" specific enough?)
- Vague-term corpus maintenance: where does the list of vague terms live, and how is it versioned across releases?

D.8 HB-07 — AEL Fast/Slow Bandit Auto-Load

D.8.1 §Engineering ticket

Ship a Thompson Sampling bandit + LLM-reflection slow loop at `src/skill-creator/auto-load/b`
The approach is two pieces: (1) **fast layer** — a Thompson Sampling bandit selecting which retrieval policy to apply per episode (instead of skill-creator's current single fixed retrieval policy in the auto-load step); (2) **slow layer** — an LLM-driven reflection callback after N failures that diagnoses recurring failure patterns and injects causal insights back into the policy priors. Bandit posterior estimates feed the existing stochastic-selection layer (`src/stochastic/`, MA-3 + MD-2). Slow-timescale reflection becomes a reinforcement signal feeding eligibility-trace updates over `src/learnable-k_h/` (MD-5 per-(skill, task-type) heads). The flag `cs25-26-sweep.ael-bandit` defaults off; flag-off auto-load behavior is byte-identical to v1.49.574. **CAPCOM HARD GATE** — touches skill-space dispatch; requires explicit human authorization. **HB-07 lives inside HB-04's role-split scaffolding** per xmod CC-3 / convergent-discovery §5: the bandit operates inside the **Evolution** role's protocol-update path; Evolution reads the bandit's posterior history at the configured slow-timescale interval and proposes role-isolation invariants for the next epoch. The two flags are independently toggleable but the semantic pair (HB-04 + HB-07) is the architectural unit. Per impact.md §4 #11, the Phase 812 plan must enumerate state-isolation invariants: bandit writes own state only; reflection step reads bandit history (read-only).

D.8.2 §Target file paths

- `src/skill-creator/auto-load/bandit.ts` — Thompson Sampling bandit fast layer.
- `src/skill-creator/auto-load/policies.ts` — declared retrieval policies (the bandit's arms).
- `src/skill-creator/auto-load/reflection.ts` — LLM-driven slow-timescale reflection callback.
- `src/skill-creator/auto-load/priors.ts` — bandit-prior storage + slow-loop insight injection.
- `src/skill-creator/auto-load/index.ts` — public entry; integrates with skill-creator's auto-load step + HB-04 Evolution role.
- `src/skill-creator/auto-load/__tests__/bandit-convergence.test.ts`, `policy-switch`

`reflection-trigger.test.ts`, `reflection-feedback-loop.test.ts`, `prior-update.test.t`
`state-isolation.test.ts`, `flag-off-byte-identical.test.ts`, `capcom-gate.test.ts`,
`hb04-integration.test.ts`, `eligibility-trace-integration.test.ts`.

D.8.3 §Validation tests

1. `bandit-convergence.test.ts` — bandit converges on a synthetic 3-policy benchmark within bounded episodes; convergence verified against the paper's Sharpe-ratio-style stationary-signal regime.
2. `policy-switching.test.ts` — bandit switches arms when one policy's posterior degrades; switching latency bounded.
3. `reflection-trigger.test.ts` — slow-loop reflection fires after N failures; N is configurable.
4. `reflection-feedback-loop.test.ts` — reflection-injected insights update policy priors; the next episode's bandit-arm selection reflects the updated prior.
5. `prior-update.test.ts` — prior-update path is read-only on bandit history (state-isolation invariant: reflection step does not mutate bandit posterior storage directly).
6. `state-isolation.test.ts` — verifies the no-cross-write boundary between bandit / reflection / external skill-creator state.
7. `flag-off-byte-identical.test.ts` — fixture asserting auto-load behavior is byte-identical to v1.49.574 baseline with flag off; existing single-policy retrieval runs unchanged.
8. **`capcom-gate.test.ts` — CAPCOM gate test.** Without a recorded human authorization marker, the bandit refuses to engage even with the flag on; the gate fires and dispatch is blocked. Mirrors the v1.49.574 megakernel-substrate HB-04 adapter-selection-schema gate pattern.
9. **`hb04-integration.test.ts`** — verifies the integration boundary with HB-04: bandit operates inside HB-04 Evolution role; Evolution reads bandit history at configured slow-timescale interval; bandit posterior storage is owned by Evolution (or shared via a documented contract).
10. `eligibility-trace-integration.test.ts` — slow-loop insights feed MA-1 eligibility traces over MD-5 learnable-k_h heads; reinforcement series sees the new signal.

D.8.4 §Default-off invariant

With `cs25-26-sweep.ael-bandit=false`, auto-load behavior is byte-identical to v1.49.574: existing single-fixed-policy retrieval runs unchanged, no bandit posterior is initialised, no reflection callback fires. The `flag-off-byte-identical.test.ts` fixture mirrors v1.49.574 `disabled-byte-identical`.

D.8.5 §Rollback criterion

- CAPCOM gate bypass: any code path that engages the bandit without a recorded human authorization marker.
- State-isolation violation: any code path that allows reflection-step writes to bandit posterior or vice versa.
- Bandit non-convergence in production: bandit underperforms a fixed-policy baseline because skill priors drift faster than the bandit converges (the paper's Risk note); flag remains default-off until convergence is empirically confirmed.

- Flag-off byte-diff: any non-zero diff in auto-load output under flag-off vs v1.49.574 baseline.

D.8.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M2/ADR-2604.21725.md`.
- **Composes with HB-04** (2604.21003, `.planning/missions/cs25-26-sweep/work/modules/` per xmod CC-3 / convergent-discovery §5.
- xmod connection 2604.21003 ↔ 2604.21725 composes-with strong (table row).
- Prior milestones: v1.49.570 reinforcement series MA-6 (canonical taxonomy + emitters); v1.49.572 MA-1 eligibility traces; v1.49.574 megakernel substrate (default-off flag pattern + CAPCOM gate template).
- **CAPCOM gate-pattern template:** v1.49.574 megakernel-substrate `src/cartridge/megake` — the prior CAPCOM HARD GATE this spec inherits the gate structure from. Reuse the `disabled-byte-identical.test.ts` fixture pattern.
- Sensoria/umwelt/symbiosis (M6/M7/M8 from v1.49.561) — symbiosis teaching ledger receives reflection-injected insights as durable across-mission artefacts.

D.8.7 §Open questions for implementing engineer

- Thompson Sampling prior shape: Beta / Gaussian / per-arm hyperprior — which is the right hyperparameter, and how is it documented to avoid silent regime drift between releases (per ADR Notes)?
- Slow-loop cadence: failure-count threshold N — fixed value (e.g., N=5 from IRAP) vs adaptive?
- Reflection-output format: structured JSON-injectable insights vs free-text-with-extraction?
- Posterior storage: ephemeral (per-mission) vs persisted (Grove M13 records)?
- HB-04 Evolution boundary: does Evolution own the bandit's posterior, or does HB-07 ship its own storage with a read-contract for Evolution?

D.9 HB-08 — RPS / AGS Cross-Chipset Divergence Audit

D.9.1 §Engineering ticket

Ship two chipset-validation metrics at `src/chipset-validation/rps-ags/`. The approach implements the two metrics from 2604.21255: (1) **Response Pattern Similarity (RPS)** — verbal-alignment metric across paired agent responses, computed over a tokenization that strips formatting noise; (2) **Action Graph Similarity (AGS)** — tool-use behavior metric modeling agent invocations as directed graphs and comparing graph structure across agents. The metrics' use is **strictly internal monitoring** of the GSD chipset (Agnus/Denise/Paula/Gary → Opus/Sonnet/Haiku), never as a public claim that any external model is "identical to" another (cultural-sensitivity gate per intake). Use AGS to validate Opus-design / Sonnet-implement boundary in the chipset is producing genuinely complementary perspectives, not collapsing to behavior-pattern uniformity. Use RPS to detect prompt-leakage at the Haiku tier. **HB-08 is deferred to v1.49.576+ backlog** per impact.md §4 #3 — the chipset-validation harness is not yet scoped, and ad-hoc cross-tier behavior monitoring has not yet emerged as an operational pain point. Promotion trigger: (a) chipset architecture's tier-separation produces a measurable semantic-drift incident, or (b) external skill marketplaces (DoltHub, see HB-11) require behavioral provenance verification

of submitted skills.

D.9.2 §Target file paths

- `src/chipset-validation/rps-ags/rps.ts` — Response Pattern Similarity metric.
- `src/chipset-validation/rps-ags/ags.ts` — Action Graph Similarity metric.
- `src/chipset-validation/rps-ags/graph-builder.ts` — directed-graph construction from chipset traces.
- `src/chipset-validation/rps-ags/internal-only-guard.ts` — runtime guard blocking external-claim invocations (cultural-sensitivity gate).
- `src/chipset-validation/rps-ags/index.ts` — public entry.
- `docs/architecture/chipset.md` — citation block for 2604.21255; metric definitions documented even before runtime ships.
- `src/chipset-validation/rps-ags/__tests__/rps-paired-traces.test.ts`, `ags-graph-s` `internal-only-guard.test.ts`, `chipset-tier-divergence.test.ts`, `flag-off-byte-iden` `metric-stability.test.ts`, `kimi-k2-style-detection.test.ts`, `cultural-sensitivity-`

D.9.3 §Validation tests

1. `rps-paired-traces.test.ts` — paired chipset traces yield reproducible RPS; cross-family pairs score lower than within-family per the paper's 5.9 percentage-point finding.
2. `ags-graph-similarity.test.ts` — paired action-graphs yield reproducible AGS; planted-divergence fixture detected.
3. `internal-only-guard.test.ts` — any caller passing an external-model trace as the "test" axis is rejected at runtime; cultural-sensitivity gate fires.
4. `chipset-tier-divergence.test.ts` — Opus-design / Sonnet-implement boundary detection on a known-divergent fixture.
5. `flag-off-byte-identical.test.ts` — chipset dispatch is byte-identical to v1.49.575 baseline with flag off (when promoted).
6. `metric-stability.test.ts` — small input perturbations yield small metric changes (no metric instability).
7. `kimi-k2-style-detection.test.ts` — replicates the paper's Kimi-K2 ↔ Claude Sonnet 4.5 result on a synthetic fixture (not against any real external model — cultural-sensitivity gate).
8. `cultural-sensitivity-gate.test.ts` — verifies the internal-only guard cannot be disabled via configuration.

D.9.4 §Default-off invariant

With `cs25-26-sweep.rps-ags=false` (when promoted), chipset dispatch is byte-identical to baseline; no metric computation runs in the dispatch path. The `flag-off-byte-identical.test` fixture mirrors v1.49.574 `disabled-byte-identical`.

D.9.5 §Rollback criterion

- Cultural-sensitivity gate bypass: any code path that allows the metrics to be invoked against external models for public claims.
- Metric instability: small input perturbations cause large metric swings, making divergence detection unreliable.
- Flag-off byte-diff: any non-zero diff in chipset dispatch under flag-off vs baseline.

D.9.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M4/ADR-2604.21255.md`.
- xmod CC-6 Two-game chipset validation: composes with 2604.21282 3+1 + 2603.22823 Empirical Comparison ACPs (DACP fidelity-level evaluation matrix).
- Prior milestones: v1.49.572 DACP fidelity levels (semantic-channel + chipset-tier separation surface); v1.49.574 chipset architecture (the dispatch boundary the metrics measure across); Memory Arena M9-M13 (action-graph persistence).
- HB-11 (2604.21829 Black-Box Skill Stealing) — promotion trigger (b): federated-skill marketplace provenance verification.

D.9.7 §Open questions for implementing engineer

- Promotion-trigger detection: how is "measurable semantic drift incident" operationalized (alert threshold + on-call response)?
- Trace persistence: chipset traces persisted to Grove M13 records — what is the retention policy?
- Cultural-sensitivity gate enforcement: runtime guard in code vs review-discipline at the documentation level only?
- Calibration of divergence threshold: what AGS / RPS numbers indicate "tier separation working" vs "uniformity collapse"?

D.10 HB-09 — DryRUN Autonomous Test Synthesis

D.10.1 §Engineering ticket

Ship an autonomous test-input + trace-simulation synthesizer at `src/cartridge/validation/synt` The approach implements DryRUN's pattern: given a SKILL.md frontmatter signature + the cartridge contract, the synthesizer (1) generates candidate inputs covering the declared capability surface; (2) simulates expected outputs against the cartridge's declared invariants; (3) returns a trace bundle that the cartridge validator uses as a self-correction signal *before* any human-authored ground-truth fixtures are required. The synthesizer plugs into `skill-creator cartridge validate --synth-trace`. **HB-09 is deferred to v1.49.576+ backlog** per ADR Notes — DryRUN's value depends on an external check (the v1.50 proof-companion's Lean verifier) closing the loop; until that check exists, a synthesized trace can confirm a wrong invariant. **HB-09 promotion is gated on a ≥800-word architecture-spec ADR** in `work/modules/M5/` per impact.md §4 #4 before the implementing phase opens — the current sketch is one paragraph; promotion requires the expanded spec. Promotion trigger: v1.50 Half B chapter validation begins specifying its inner-loop test contract.

D.10.2 §Target file paths

- `src/cartridge/validation/synth/inputs.ts` — candidate-input generator from SKILL.md frontmatter.
- `src/cartridge/validation/synth/trace-simulator.ts` — execution-trace simulator.
- `src/cartridge/validation/synth/self-correct.ts` — self-correction signal feeder.
- `src/cartridge/validation/synth/bundle.ts` — trace bundle output format.
- `src/cartridge/validation/synth/index.ts` — public entry; integrates with

skill-creator cartridge validate.
- src/cartridge/validation/synth/__tests__/input-coverage.test.ts, trace-simulato
self-correct-signal.test.ts, bundle-shape.test.ts, proof-companion-handoff.test.
flag-off-byte-identical.test.ts, wrong-invariant-detection.test.ts, reinforcemen

D.10.3 §Validation tests

1. input-coverage.test.ts — synthesized inputs cover the declared capability surface (positive + boundary + adversarial cases).
2. trace-simulator.test.ts — simulated traces are deterministic given identical input + frontmatter.
3. self-correct-signal.test.ts — self-correction signal correctly drives a regeneration loop on a fixture with a planted spec defect.
4. bundle-shape.test.ts — trace bundle conforms to the proof-companion handoff schema.
5. proof-companion-handoff.test.ts — bundle is consumable by the v1.50 proof-companion's Lean external check (the closure point).
6. flag-off-byte-identical.test.ts — skill-creator cartridge validate is byte-identical to baseline with the flag off; no synth runs.
7. **wrong-invariant-detection.test.ts** — fixture: spec with a wrong invariant; without the proof-companion external check, the synthesizer self-confirms the wrong invariant. **This is the rollback signal** — if the proof-companion is not yet wired, this test must fail explicitly to prevent silent wrong-invariant confirmations.
8. reinforcement-event-emit.test.ts — synthesized trace deltas emit MA-6 reinforcement events; eligibility layer credits trace-discovery skills retroactively.

D.10.4 §Default-off invariant

With cs25-26-sweep.dryrun-synth=false (when promoted), skill-creator cartridge validate runs the existing path with hand-authored expectations only; no synthesizer fires. The flag-off-byte-identical.test.ts fixture mirrors v1.49.574 disabled-byte-identic

D.10.5 §Rollback criterion

- Proof-companion not yet wired: per ADR, do not wire DryRUN into the v1.49.575 cartridge harness; wait until the v1.50 proof-companion's external Lean check exists.
- Wrong-invariant self-confirmation: wrong-invariant-detection.test.ts would silently pass — at that point the synthesizer is masking spec bugs, not finding them.
- Flag-off byte-diff: any non-zero diff in cartridge validator output under flag-off vs baseline.

D.10.6 §Cross-references

- Source ADR: .planning/missions/cs25-26-sweep/work/modules/M5/ADR-2604.21598.md.
- xmod connection 2604.21598 ↔ 2604.21505 composes-with strong: DryRUN test-side + Orchid spec-side define the spec-quality / test-quality envelope for skill authoring; HB-09 + HB-06 form the joint pair when HB-09 promotes.
- Prior milestones: v1.49.572 output-structure validator at src/output-structure/ (frontmatter contract is the synthesizer's natural input); v1.49.570 reinforce-

ment series MA-6 emitters; v1.50 proof-companion (the closure point this spec defers to).
- xmod CC-7 M5 verification triad: 2511.23159 probable-to-provable + 2604.18792 DSLTrans + 2604.21187 Doubly Saturated Ramsey — the rhetorical+mechanism+precedent backbone DryRUN slots into.

D.10.7 §Open questions for implementing engineer

- Architecture-spec authorship: who writes the ≥800-word ADR (impact.md §4 #4 promotion gate) — the implementing engineer or a separate Wave-1-style track?
- Synthesizer model selection: small + fast (Haiku) vs large + accurate (Sonnet/Opus)?
- Trace-bundle format: Lean-readable directly vs an intermediate JSON the proof-companion adapter parses?
- Coverage metric: how is "covers the declared capability surface" measured (line coverage on a deterministic implementation skeleton)?

D.11 HB-10 — IRAP Interactive Requirements Elicitation

D.11.1 §Engineering ticket

Ship a bounded-rounds elicitation policy at `docs/capcom/elicitation-policy.md` plus an enforcement hook in `src/model-affinity/`. The approach implements IRAP's five-round elicitation cap: any CAPCOM gate clarifying dialogue that exceeds five rounds triggers a model-tier escalation event (Haiku → Sonnet → Opus) rather than continuing the dialogue. The five-round cap is the operational target — the paper reports up to 40-fold improvement on the elicitation benchmark *with* the bounded-round property — and frames "cannot resolve in five rounds" as a misshaped task or wrong tier rather than under-promotion. The vision-to-mission skill at `.claude/skills/vision-to-missi` adopts the bound as soft discipline now (no escalation, just a warning at round-5); the hard CAPCOM-gate escalation is the v1.49.576+ landing site. **HB-10 is deferred to v1.49.576+ backlog** per ADR — full IRAP-based mechanism waits for the model-affinity escalation phase to open. Promotion trigger: model-affinity escalation policy phase opens. The interactive-RAG mechanism specifically pairs with the College KB-construction work (CC-8 M4 retrieval + synthesis pipeline) — elicitation queries the department KB, and the KB structure shapes which clarifying questions the loop can ask.

D.11.2 §Target file paths

- `docs/capcom/elicitation-policy.md` — bounded-rounds elicitation-policy doc; can land earlier as soft discipline.
- `src/model-affinity/elicitation-cap.ts` — five-round cap enforcement hook.
- `src/model-affinity/escalation.ts` — Haiku → Sonnet → Opus escalation trigger on round-5 timeout.
- `src/model-affinity/round-counter.ts` — round-count metric plumbed into CAPCOM dialogue trace.
- `src/model-affinity/__tests__/cap-enforcement.test.ts`, `escalation-trigger.test.t` `round-counter.test.ts`, `vision-to-mission-soft.test.ts`, `flag-off-byte-identical.t` `safety-warden-metric.test.ts`.
- `.claude/skills/vision-to-mission/SKILL.md` — adds the soft five-round dis-

cipline note now.

D.11.3 §Validation tests

1. `cap-enforcement.test.ts` — five-round cap fires; round 6 cannot proceed without an escalation event.
2. `escalation-trigger.test.ts` — round-5 timeout triggers a model-tier escalation event; tier transition is recorded.
3. `round-counter.test.ts` — round-count metric plumbed into the CAPCOM dialogue trace; visible in trace output.
4. `vision-to-mission-soft.test.ts` — vision-to-mission emits a warning at round 5 when the soft discipline is engaged but no escalation fires.
5. `flag-off-byte-identical.test.ts` — CAPCOM dialogue behavior is byte-identical to baseline with the flag off (when promoted); no escalation, no round-counter metric.
6. `safety-warden-metric.test.ts` — five-round bound is reported as a Safety Warden metric on every CAPCOM gate dialogue.

D.11.4 §Default-off invariant

With `cs25-26-sweep.irap-elicitation=false` (when promoted), CAPCOM dialogue runs unbounded as in v1.49.574 baseline; no round counter, no escalation, no metric. The `flag-off-byte-identical.test.ts` fixture mirrors v1.49.574 `disabled-byte-identical` for the dialogue trace.

D.11.5 §Rollback criterion

- Hard cutoff surprise: a hard escalation that surprises users without UI surfacing of the round count (per ADR Risk note); mitigation is mandatory pre-cap UI surfacing.
- Five-round cap too tight for unfamiliar domains: if measured cap-failures exceed an operational threshold on real CAPCOM traces, the cap is renegotiated (or the escalation tier is widened).
- Flag-off byte-diff: any non-zero diff in CAPCOM dialogue output under flag-off vs baseline.

D.11.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M6/ADR-2604.21380.md`.
- xmod CC-8 College of Knowledge retrieval + synthesis pipeline: composes with 2604.20848 MATRAG, 2510.26615 SARA, 2604.21444 HiCrew, 2508.21720 Poster-Forest, 2601.10003 SocraticKG.
- Prior milestones: v1.49.572 output-structure frontmatter (per-capability elicitation expectations); v1.49.574 chipset architecture (treats CAPCOM as a chip with a bounded-dialogue contract).
- xmod connection 2604.20855 ↔ 2604.21380 composes-with: Caesar adversarial-draft-refinement pairs with IRAP's bounded elicitation as elicitation+synthesis halves.
- xmod connection 2510.18731 ↔ 2604.21380 same-pattern-different-domain: RLAAR's abstention-at-round-N mirrors IRAP's bounded-rounds escalation.

D.11.7 §Open questions for implementing engineer

- Five-round cap configurability: hard-coded 5 vs configurable per task type?
- Escalation tier policy: Haiku → Sonnet → Opus is the default — is there a fast path that skips a tier on certain task signals?
- Soft vs hard discipline boundary: when does vision-to-mission's soft warning become a hard block, and what is the trigger?
- UI surfacing of round count: where does it appear in the CAPCOM gate UI to avoid the "hard cutoff surprise" Risk?

D.12 HB-11 — Black-Box Skill-Stealing Threat Model

D.12.1 §Engineering ticket

Ship a threat-model document at `docs/security/dolthub-federated-economy-threat-model.md` that adopts the attack taxonomy from 2604.21829 as GSD's defender-concern enumeration for the v1.49.576+ DoltHub federated skill-economy work. The approach is documentation-first: enumerate each named pipeline stage from the paper (model-generated seed prompts → scenario rationalization with structure injection → embedding-filter diversity step), name a corresponding detection or mitigation requirement for each, and ship the document as a **pre-rollout citation gate** for any DoltHub work in v1.49.576+. This is a threat-model spec, not a code module. The market-sizing context (90,368 free-marketplace skills + 2,000+ paid listings with $100K+ creator earnings) informs the Wasteland-integration roadmap's scope. **Per impact.md §4 #5 and Fox Companies IP rule**, the document is split: an abstract attack-taxonomy version is publicly citable; the GSD-specific defender-attack-surface specifics live in `.planning/fox-companies/foxfiber/threat-model.md` and require Fox Companies IP review before any artefact moves outside `.planning/fox-companies/foxfiber/`. Promotion trigger: DoltHub contribution roadmap opens or first federated-skill phase begins. Composes with HB-05 structural-completeness linter (CC-2 federated-skill futures) — marketplace-sourced skills must pass structural completeness before federated promotion; structural quality is the first defense against marketplace contamination.

D.12.2 §Target file paths

- `docs/security/dolthub-federated-economy-threat-model.md` — public abstract attack-taxonomy version.
- `.planning/fox-companies/foxfiber/threat-model.md` — GSD-specific defender-attack-surface specifics (Fox Companies IP; not public).
- `docs/GROVE-FORMAT.md` — gains a forward-pointing note that any DoltHub federated-economy adoption in v1.49.576+ must cite 2604.21829 as the threat-taxonomy pre-rollout-gate (per CS25-04 acceptance language).
- `src/cartridge/linter/marketplace-attestation.ts` — runtime hook (when promoted): refuses federated-skill publish without a threat-model attestation.
- `docs/security/__tests__/citation-presence.test.ts`, `threat-stage-coverage.test.t` `attestation-required.test.ts`, `fox-ip-publication-boundary.test.ts`.

D.12.3 §Validation tests

1. `citation-presence.test.ts` — every federated-economy-adjacent recommendation in `docs/`, `.planning/`, and SKILL.md files cites 2604.21829. Mechanical

grep + assertion (CS25-04 acceptance).
2. `threat-stage-coverage.test.ts` — for each named pipeline stage in the paper, the threat-model doc names a defender concern + a corresponding detection/mitigation requirement.
3. `attestation-required.test.ts` (when promoted) — `skill-creator cartridge publish --federated` refuses to proceed without a threat-model attestation that references this document.
4. `fox-ip-publication-boundary.test.ts` — the public version contains only abstract attack-taxonomy content; mechanical check that no `.planning/fox-companies/` content has been mirrored into the public doc.

D.12.4 §Default-off invariant

This is a documentation artefact. The "default-off" semantic translates to: in v1.49.575, only the threat-model doc lands; no DoltHub publish path engages, no federated-economy code runs. With future federated-economy work in v1.49.576+, the runtime attestation hook defaults off until the threat-model doc has landed and a CS25-04 citation-presence audit has passed. The fixture pattern is the citation-presence sweep: with the federated-skill flag off, no federated-skill publish path can execute; flag-off behavior is byte-identical to v1.49.574.

D.12.5 §Rollback criterion

- Fox Companies IP leak: any GSD-specific defender-attack-surface specifics appear in the public doc. Per impact.md §4 #5, surface to user and revert immediately.
- Citation-gate bypass: any v1.49.576+ federated-economy recommendation that does not cite 2604.21829 (CS25-04 violation).
- Pre-publish attestation bypass: any code path that allows `skill-creator cartridge publish --federated` to proceed without a threat-model attestation.

D.12.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M6/ADR-2604.21829.md`.
- xmod CC-9 cultural-sensitivity Safety Warden surface: integrates with Grove via the federated-skill-future link.
- xmod connection 2604.21829 ↔ 2604.21090 composes-with strong: Structural Quality Gaps in governance prompts (HB-05) provides the defender-side structural-quality gate that complements the attack-taxonomy.
- xmod connection 2604.21829 ↔ 2604.21477 composes-with strong: MCP Pitfall Lab's six-attack family is the runtime-side complement.
- Prior milestones: v1.49.561 trust-relationship (Stage 1 — trust-relationship-provider would consume the attack taxonomy as one of its threat sources); Memory Arena M13 (`docs/GROVE-FORMAT.md` — content addressing is a natural defense against several attack-pipeline stages because it makes silent skill substitution detectable); v1.49.574 chipset architecture (federated skill economy is a chipset-level concern).
- **Sign-off invariant (impact.md §6 #3):** 2604.21829 is a pre-rollout gate for any federated-economy / DoltHub work in v1.49.576+.

D.12.7 §Open questions for implementing engineer

- Public-vs-private boundary detail: which attack-pipeline stages can be discussed at the abstract level publicly, and which must stay private until defenses ship?
- Attestation format: free-form Markdown attestation vs structured frontmatter the publisher pipeline can validate?
- Marketplace contamination defense composition: HB-11 + HB-05 + content-addressed Grove records — what is the order-of-operations the publisher pipeline runs them in?
- Sovereign-marketplace handling: AFRILANGTUTOR-style sovereignty-gated marketplaces — does the threat-model treat those as a separate defender posture?

D.13 HB-12 — FSFM Four-Class Forgetting Taxonomy

D.13.1 §Engineering ticket

Ship the FSFM four-class forgetting taxonomy as the canonical bounded-learning forgetting policy at `.planning/policies/bounded-learning-forgetting-taxonomy.md` (NEW), with citations into existing forgetting-related code paths. The four classes from 2604.20300 map directly onto GSD subsystems: (1) **passive decay-based** → Memory Arena M1–M13's hot/warm/cold tiering with importance-weighted demotion; (2) **active deletion-based** → the `done-retirement` skill's pipeline-retirement protocol; (3) **safety-triggered** → the Safety Warden's BLOCK-on-content authority (`.claude/skills/security-hygiene/`); (4) **adaptive reinforcement-based** → the reinforcement series (`src/reinforcement/` MA-6 + `src/eligibility/` MA-1 eligibility-trace decay + `src/temperature/` MD-4 annealed schedule). The "shed" step in the Root Theorem (2604.20874) accumulate-compress-rewrite-shed cycle is formally defined by this taxonomy: shedding can be passive, active, safety-triggered, or reinforcement-adaptive. The paper's reported numbers (+8.49% access efficiency, +29.2% signal-to-noise, 100% security-content elimination) become Safety Warden test targets. **HB-12 is deferred to v1.49.576+ backlog** per the tier-split directive — adoption now is policy-document-only at v1.49.575 (citation in §2 Subsystem 9 of impact.md), runtime instrumentation defers. Promotion trigger: bounded-learning theorem proof completion (carried forward from v1.49.572 Half B T1d) cites Root Theorem as the empirical anchor, and HB-12 lands as the operational taxonomy alongside.

D.13.2 §Target file paths

- `.planning/policies/bounded-learning-forgetting-taxonomy.md` — adopts the four-class taxonomy verbatim; maps each class to a GSD subsystem.
- `src/reinforcement/forgetting-class.ts` — class-tagging at the reinforcement-event emit point.
- `src/eligibility/decay-class.ts` — eligibility-trace decay receives a class tag (passive vs adaptive-reinforcement).
- `src/temperature/decay-rate.ts` — annealed schedule modulating decay rate per class (MD-4 hookup).
- Memory Arena M1–M13 demotion call sites — tag demotion events with `passive-decay` class.
- `done-retirement` skill — tags pipeline-retirement events with `active-deletion` class.
- `.claude/skills/security-hygiene/SKILL.md` — tags BLOCK-on-content events

with `safety-triggered` class.
- `.planning/policies/__tests__/taxonomy-coverage.test.ts`, `class-tag-emit.test.ts`, `safety-triggered-100pct.test.ts`, `signal-to-noise-improvement.test.ts`, `flag-off-byte-identical.test.ts`, `representation-audit-integration.test.ts`.

D.13.3 §Validation tests

1. `taxonomy-coverage.test.ts` — every forgetting code path in the repo emits an event tagged with one of the four classes.
2. `class-tag-emit.test.ts` — each class-tagged event records the class verbatim per the FSFM taxonomy.
3. `safety-triggered-100pct.test.ts` — safety-triggered class achieves 100% security-content elimination on a fixture with planted security-tagged content (paper's headline number; Safety Warden test target).
4. `signal-to-noise-improvement.test.ts` — synthetic benchmark records ≥some-threshold improvement in signal-to-noise ratio under the adaptive-reinforcement class (paper reports +29.2%; threshold to be calibrated).
5. `flag-off-byte-identical.test.ts` — forgetting code paths run unchanged with the flag off; class tags emit as no-ops.
6. `representation-audit-integration.test.ts` — composes with MD-6 representation-audit (collapse detection); audit consumes class tags.

D.13.4 §Default-off invariant

With `cs25-26-sweep.fsfm-taxonomy=false` (when promoted), forgetting code paths run as in v1.49.574 baseline; class tags emit as no-ops; no behavior change. The `flag-off-byte-identical.test.ts` fixture mirrors v1.49.574 `disabled-byte-identical`. At v1.49.575, only the policy document lands; no runtime tagging fires.

D.13.5 §Rollback criterion

- Behavior change with flag off: any forgetting code path that behaves differently with the flag off would regress the bounded-learning baseline.
- Taxonomy mismatch: any forgetting code path that does not map cleanly into one of the four classes — expand the taxonomy or document the gap.
- Hippocampal-indexing scope creep: per ADR Notes, the GSD adoption is taxonomic, not implementation. Any code path that imports the paper's specific hippocampal-indexing implementation is out of scope.

D.13.6 §Cross-references

- Source ADR: `.planning/missions/cs25-26-sweep/work/modules/M2/ADR-2604.20300.md`.
- xmod connection 2604.20300 ↔ 2604.20874 implements strong: Root Theorem accumulate-compress-rewrite-shed cycle's "shed" step is formally defined by this taxonomy.
- xmod connection 2604.20300 ↔ 2511.11439 extends strong: Retrofit's selective-forgetting + low-rank-sparse subspace pairs with the four-class taxonomy.
- xmod CC-4 bounded-tape × bounded-learning unified foundation.
- Prior milestones: v1.49.561 sensoria/umwelt/symbiosis (M6/M7/M8) — symbiosis teaching ledger uses adaptive-reinforcement decay; umwelt Markov-blanket boundary triggers safety-triggered forgetting on out-of-blanket content; v1.49.572 Bounded-Learning Theorem (T1d) — HB-12 lands alongside the proof comple-

tion; v1.49.561 trust-relationship (Stage 1) uses safety-triggered forgetting when trust degrades.

D.13.7 §Open questions for implementing engineer

- Class-tag enum vs string: enum (compile-time exhaustiveness) or string (forward-compat with future fifth class)?
- Adaptive-reinforcement decay-rate parameterisation: per-skill / per-task-type / per-(skill, task-type)? MD-5 already does per-(skill, task-type) — does HB-12 ride on the same indexing?
- Safety-triggered 100% elimination: how is "elimination" verified (Grove content-addressed records prove absence; what is the absence proof for in-memory state)?
- Hippocampal-indexing pilot: per ADR Notes, "if a future College of Knowledge mind-body experiment wants to pilot hippocampal indexing, this ADR is the citation seed" — is there a near-term pilot in scope?

E ADR Index (Phases 797–799)

This appendix indexes the 55 ADRs produced by Wave 1 (Phase 797 Module 1+2, Phase 798 Module 3+4, Phase 799 Module 5+6). Each row gives the arXiv ID, module, paper title, adoption decision (ADOPT/TRACK/REJECT), target subsystem (truncated where long), target milestone, and source ADR file path. Full ADR content is preserved at `.planning/missions/cs25-26-sweep/work/modules/M{1--6}/ADR-*.md`.

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2604.20874	M1	The Root Theorem of Context Engineering	ADOPT	`docs/substrate/bounded-learning-root-theorem.md` (NEW), *The Space Between* information-theory chapter, `CLAUDE.md` design-rationale section, every CAPCOM-gate policy document, fresh-context-subagent doc\{}ldots	v1.49.575 (Phase 805 substrate document + CS25-09 CLAUDE.md citation); v1.50 retrospective Half A (information-theory chapter)	M1/ADR-2604.20874.md
2604.21003	M1	The Last Harness You'll Ever Build	ADOPT	`src/skill-creator/role.md` (NEW), `gsd-mission-management-role.md`, `.claude/agents/` documentation, `.planning/PROJECT.md` bounded-learning-policy section	v1.49.575 (citation in Phase 801 + role-split spec); Half B Phase 808 ships HB-04 / CS25-16 implementation	M1/ADR-2604.21003.md
2604.21744	M1	Agentic AI-assisted coding offers a unique opportunity to instill epistemic grounding during software development (GROUNDINGmd)	ADOPT	`CLAUDE.md`, every `SKILL.md` template (cartridge authoring guide), `.claude/skills/security-hygiene/SKILL.md`, `src/cartridge/linter/structural-completeness.ts` (Module 3 spec)	v1.49.575 (citation + linter spec); v1.50 retrospective Half A (extended writeup)	M1/ADR-2604.21744.md
2604.21910	M1	From Research Question to Scientific Workflow: Leveraging Agentic AI for Science Automation	ADOPT	`CLAUDE.md`, `.college/README.md`, `docs/substrate/three-layer-architecture.md` (NEW), `docs/cartridge/SKILL.md` authoring guide	v1.49.575 (citation in Phase 801 report + CS25-09 CLAUDE.md edit); v1.50 retrospective Half A (extended writeup)	M1/ADR-2604.21910.md

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2509.18629	M2	HyperAdapt: Simple High-Rank Adaptation	ADOPT	`docs/silicon-layer/benchmark-suite/` (NEW), Silicon Layer benchmark spec, future Silicon Layer LoRA pipeline implementation, adapter-pooling experiment design	v1.49.575 (benchmark spec); future Silicon Layer benchmark suite	M2/ADR-2509.18629.md
2511.11439	M2	Retrofit: Continual Learning with Controlled Forgetting for Binary Security Detection	ADOPT	`docs/silicon-layer/bounded-update-learning/` (NEW), bounded-learning policy doc, future Silicon Layer LoRA pipeline	v1.50 proof companion; Silicon Layer benchmarks deferred to post-v1.50	M2/ADR-2511.11439.md
2604.20300	M2	FSFM: A Biologically-Inspired Framework for Selective Forgetting of Agent Memory	ADOPT	`.planning/policies/bounded-learning/forgetting-taxonomy.` (NEW), Safety Warden BLOCK paths, Memory Arena demotion logic, `done-retirement` skill, reinforcement-series decay parameters	v1.49.575 (taxonomy document); v1.50 proof companion (mathematical formalisation of shed)	M2/ADR-2604.20300.md
2604.20943	M2	SCM: Sleep-Consolidated Memory with Algorithmic Forgetting for LLMs	TRACK	`.college/experiments/sleep-wave-consolidation/` (NEW pilot); future v1.51 wave-architecture refactor if pilot succeeds	v1.49.575 (TRACK status logged); pilot post-v1.50; ADOPT decision in v1.51 if pilot succeeds	M2/ADR-2604.20943.md
2604.21330	M2	Teacher-Guided Routing for Sparse Vision Mixture-of-Experts	TRACK	`docs/chipset-architecture/future-work-note.md`, future learned-dispatch-layer design (post-v1.50)	v1.49.575 (future-work note); revisit if/when learned dispatch is proposed	M2/ADR-2604.21330.md
2604.21571	M2	Separable Expert Architecture: Composable Adapters and Deletable User Proxies	ADOPT	`docs/silicon-layer/separable-expert-architecture/` (NEW), Silicon Layer spec, Fox-Fiber/FoxCompute multi-tenant design, FoxData privacy whitepaper	v1.49.575 (architecture spec); future Silicon Layer pipeline implementation	M2/ADR-2604.21571.md
2604.21725	M2	AEL: Agent Evolving Learning for Open-Ended Environments	ADOPT	`src/skill-creator/auto-loop/` (NEW), reinforcement-series eligibility-trace integration	v1.49.577 bandit spec; Half B Phase 812 ships HB-07 / CS25-19 implementation	M2/ADR-2604.21725.md

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2604.21901	M2	GiVA: Gradient-Informed Bases for Vector-Based Adaptation	ADOPT	`docs/silicon-layer/benchmark-suite.md` (NEW), Silicon Layer benchmark spec, future Silicon Layer LoRA pipeline implementation, `src/learnable-k_h/` integration	v1.49.575 (benchmark spec); future Silicon Layer benchmark suite implementation	M2/ADR-2604.21901.md
2604.21905	M2	Low-Rank Adaptation Redux for Large Models	TRACK	`docs/silicon-layer/bibliography.md`, future Silicon Layer pipeline rank-selection documentation	v1.49.575 (bibliography entry); referenced as needed in future Silicon Layer milestones	M2/ADR-2604.21905.md
2604.21927	M2	Fine-Tuning Regimes Define Distinct Continual Learning Problems	ADOPT	`docs/silicon-layer/test-harness-spec.md` (NEW), `src/ab-harness/` ME-3 A/B harness, `src/representation-audit/` MD-6 audit trail, reinforcement-series test harness	v1.49.575 (test harness spec); v1.50 Silicon Layer benchmark suite (regime-fixed protocol enforced)	M2/ADR-2604.21927.md
2512.03048	M3	The Specification Trap: Why Static Value Alignment Alone Is Insufficient for Robust Alignment	ADOPT	`docs/substrate/specification-trap.md` + citations in The Space Between + College of Knowledge philosophy module	v1.49.575 (citation); v1.50 retrospective Half A (extended treatment)	M3/ADR-2512.03048.md
2601.18491	M3	AgentDoG: A Diagnostic Guardrail Framework for AI Agent Safety and Security	ADOPT	`src/safety/agentdog/` (new); integrates with existing Safety Warden BLOCK paths	v1.49.575 (HB-02, Phase 807)	M3/ADR-2601.18491.md
2604.20911	M3	Omission Constraints Decay While Commission Constraints Persist in Long-Context LLM Agents	ADOPT	`src/safety/std-calibration/` (new); calibration table at `.planning/safety/std-calibration.json`	v1.49.575 (HB-03, Phase 808)	M3/ADR-2604.20911.md
2604.20994	M3	Breaking MCP with Function Hijacking Attacks: Novel Threats for Function Calling and Agentic Models	ADOPT	`src/orchestration/planning-bridge/` + `docs/security/mcp-threat-model.md`	v1.49.575 (threat model design); v1.49.576+ (runtime binding implementation)	M3/ADR-2604.20994.md

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2604.21090	M3	Structural Quality Gaps in Practitioner AI Governance Prompts	ADOPT	src/cartridge/linter/structural-checks.ts + cartridge promotion pipeline at src/cartridge/	v1.49.575 (HB-05, Phase 810)	M3/ADR-2604.21090.md
2604.21131	M3	Cross-Session Threats in AI Agents: Benchmark, Evaluation, and Algorithms	ADOPT	src/safety/cross-session/ (architecture spec); docs/security/kill-chain-taxonomy.md	v1.49.575 (threat model + spec); v1.49.576+ (runtime correlator)	M3/ADR-2604.21131.md
2604.21477	M3	MCP Pitfall Lab: Exposing Developer Pitfalls in MCP Tool Server Security under Multi-Vector Attacks	ADOPT	src/orchestration/planning/bridge/ + src/cartridge/linter/mcp-pitfalls.ts	v1.49.575 (Phase 805)	M3/ADR-2604.21477.md
2604.21718	M3	Building a Precise Video Language with Human-AI Oversight (CHAI)	ADOPT	docs/capcom/ (terminology) + src/reinforcement/ (pre-X/post-X capture pipeline)	v1.49.575 (documentation + signal pipe)	M3/ADR-2604.21718.md
2508.21720	M4	PosterForest: Hierarchical Multi-Agent Collaboration for Scientific Poster Generation	ADOPT	src/skills/research-mission-generator/ (design) + docs/skills/mission-tree.md	v1.49.575 (design + spec); v1.49.576+ (v2 implementation)	M4/ADR-2508.21720.md
2510.26615	M4	SARA / SlideAgent: Hybrid RAG with Compression Vectors	ADOPT	src/college-of-knowledge/search/sara/ (design) + docs/college-of-knowledge/retrieval.md	v1.49.575 (design); v1.49.576+ (implementation)	M4/ADR-2510.26615.md
2511.00413	M4	Tree Training: Accelerating Agentic LLMs Training via Shared Prefix Reuse	ADOPT	docs/silicon-layer/tree-training.md (future-work notes) + v1.50 retrospective Half A	v1.49.575 (citation + future-work notes); future Silicon Layer milestone (implementation)	M4/ADR-2511.00413.md
2603.22823	M4	Empirical Comparison of Agent Communication Protocols for Task Orchestration	ADOPT	docs/architecture/dacp.md (citations) + .planning/dacp/comparative-benchmark-plan.md	v1.49.575 (methodology); v1.49.576+ (first comparative-benchmark run)	M4/ADR-2603.22823.md

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2604.20848	M4	MATRAG: Multi-Agent Transparent Retrieval-Augmented Generation	ADOPT	src/safety/capcom/transparency-scores/ (immediate) + src/orchestration/mission-crew/ (design)	v1.49.575 (transparency score); v1.49.576+ (Mission Crew Manifest implementation)	M4/ADR-2604.20848.md
2604.21255	M4	When Agents Look the Same: Quantifying Distillation-Induced Similarity in Tool-Use Behaviors	TRACK	src/chipset-validation/ (TRACK design) + docs/architecture/chipset.md (citation)	TRACK; backlog; promotion-trigger conditions noted	M4/ADR-2604.21255.md
2604.21282	M4	Strategic Heterogeneous Multi-Agent Architecture for Cost-Effective Code Vulnerability Detection	ADOPT	docs/architecture/chipset.md (citations) + v1.50 retrospective Half A	v1.49.575 (citations); v1.50 retrospective Half A (extended treatment)	M4/ADR-2604.21282.md
2604.21444	M4	HiCrew: Hierarchical Reasoning for Long-Form Video Understanding	ADOPT	src/orchestration/activation-profile/ (design) + src/college-of-knowledge/research/ (design)	v1.49.575 (pattern reference); v1.49.576+ (implementation as part of activation-profile and College-of-Knowledge work)	M4/ADR-2604.21444.md
2604.21590	M4	AgenticQwen: Training Small Agentic Language Models with Dual Data Flywheels	ADOPT	docs/architecture/dual-flywheel.md + v1.50 retrospective Half A	v1.49.575 (citation); v1.50 retrospective Half A (extended treatment)	M4/ADR-2604.21590.md
2604.21794	M4	Learning to Communicate: Toward End-to-End Optimization of Multi-Agent Language Systems (DiffMAS)	TRACK	docs/architecture/daemon.md (citation as upper-fidelity reference)	TRACK; promotion-trigger conditions noted above	M4/ADR-2604.21794.md
2604.21816	M4	Tool Attention Is All You Need: Dynamic Tool Gating and Lazy Schema Loading	ADOPT	src/orchestration/tool-attention/	v1.49.575 (HB-01, Phase 806)	M4/ADR-2604.21816.md
2511.23159	M5	AI for Software Engineering: From Probable to Provable	ADOPT	docs/substrate/the-space-between.md, docs/substrate/zfc-auditor.md (citation only)	v1.49.575	M5/ADR-2511.23159.md

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2603.01170	M5	ATLAS: AI-Assisted Threat-to-Assertion Learning for SoC Security Verification	TRACK	`src-tauri/src/mcp/security/` (forthcoming) design doc; planning-bridge MCP design	v1.49.576+ (after M3 MCP-taxonomy consolidates)	M5/ADR-2603.01170.md
2604.14709	M5	HWE-Bench: Benchmarking LLM Agents on Real-World Hardware Bug Repair Tasks	TRACK	Future `examples/chipsets/hardware-reflow/`; NASA Mission Series hardware-template doc	v1.49.576+ future NASA hardware mission	M5/ADR-2604.14709.md
2604.18792	M5	Tractable Verification of Model Transformations: A Cutoff-Theorem Approach for DSLTrans	ADOPT	`docs/substrate/zfc-auditor.md` (citation + framing); future `coprocessors/math/symbex/` integration sketch	v1.49.575	M5/ADR-2604.18792.md
2604.21187	M5	Doubly Saturated Ramsey Graphs: A Case Study in Computer-Assisted Mathematical Discovery	ADOPT	`docs/substrate/the-space-between.md`; `coprocessors/math/` integration sketch	v1.49.575	M5/ADR-2604.21187.md
2604.21505	M5	Assessing the Impact of Requirement Ambiguity on LLM-based Function-Level Code Generation (Orchid)	ADOPT	`tools/lint-skill.mjs`; `.claude/skills/skill-lint/SKILL.md` template; `src/cartridge/validation/`	v1.49.575 (Phase 81)	M5/ADR-2604.21505.md
2604.21570	M5	SpecSyn: LLM-based Synthesis and Refinement of Formal Specifications	TRACK	`docs/substrate/zfc-auditor.md` (design doc); future `src/cartridge/validation/spec-synth/`	v1.50 (auditor scaffold); reference now	M5/ADR-2604.21570.md
2604.21598	M5	DryRUN: On the Role of Public Tests in LLM-Driven Code Generation	TRACK	`src/cartridge/validation/` synthesizer plug-in; v1.50 proof-companion test-generator	v1.49.576+ (HB-09 backlog slot)	M5/ADR-2604.21598.md
2510.18731	M6	Mitigating Lost-in-Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards (RLAAR)	ADOPT	`docs/pedagogy/teacher-student-support/curriculum.md`; `src/reinforcement/` (abstention channel); `src/model-affinity/`	v1.49.575 (design doc); v1.49.576+ (channel wiring)	M6/ADR-2510.18731.md
2601.07262	M6	ColorBrowserAgent: Complex Long-Horizon Browser Agent with Adaptive Knowledge Evolution	ADOPT	`docs/capcom/feedback-capture.md`; future `src/capcom/feedback-capture/`; `src/output-structure/`	v1.49.575 (design doc); v1.49.576+ (implementation)	M6/ADR-2601.07262.md

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2601.10003	M6	SocraticKG: Knowledge Graph Construction via QA-Driven Fact Extraction	ADOPT	`.claude/skills/research-mission-generator/SKILL.md`; future `.claude/skills/college-kb-builder/SKILL.md`	v1.49.575	M6/ADR-2601.10003.md
2604.12710	M6	LASA: Language-Agnostic Semantic Alignment at the Semantic Bottleneck for LLM Safety	TRACK	Future `tests/cultural-sensitivity/`; cultural-sensitivity chip design	v1.49.576+ (focused experiment)	M6/ADR-2604.12710.md
2604.20487	M6	Knowledge Capsules: Structured Nonparametric Memory Units for LLMs	TRACK	`.college/` design doc; future `src/college/capsule/`	v1.51+ (after Silicon Layer kernel hooks)	M6/ADR-2604.20487.md
2604.20855	M6	Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis	ADOPT	`.claude/skills/research-mission-generator/`; future `coprocessors/research/`	v1.49.575 (v2 design); v1.49.576+ (adversarial-refinement phase)	M6/ADR-2604.20855.md
2604.20904	M6	Reinforcing Privacy Reasoning in LLMs via Normative Simulacra from Fiction	TRACK	`.planning/the-hundred-voices/`; `.planning/fox-companies/`	v1.50+ (cross-project collaboration phase)	M6/ADR-2604.20904.md
2604.20920	M6	Forget, Then Recall: Learnable Compression and Selective Unfolding via Gist Sparse Attention	TRACK	`.college/` design doc; future `src/orchestration/college/`	v1.50+ (College retrieval phase)	M6/ADR-2604.20920.md
2604.20996	M6	AFRILANGTUTOR: Advancing Language Tutoring and Culture Education in Low-Resource Languages with LLMs	TRACK	`.planning/missions/fox50-heritage/`; future `examples/chipsets/heritage-language/`	v1.50+ (gated on corpus-licensing consultation)	M6/ADR-2604.20996.md
2604.21308	M6	CI-Work: Benchmarking Contextual Integrity in Enterprise LLM Agents	ADOPT	`.planning/fox-companies/fiber/threat-model.md`; future `src/safety-warden/`	v1.49.575 (threat-model doc); v1.49.576+ (metric scaffolding)	M6/ADR-2604.21308.md
2604.21352	M6	CARE: Counselor-Aligned Response Engine for Online Mental-Health Support	TRACK	`.college/departments/mind-body/` design doc; Safety Warden interaction policy	v1.50+ (mind-body chip phase)	M6/ADR-2604.21352.md

arXiv ID	M	Title	Dec.	Target subsystem	Target milestone	Source ADR
2604.21380	M6	Conjecture and Inquiry: Quantifying Software Performance Requirements via Interactive RAG (IRAP)	TRACK	`.claude/skills/vision-to-mission/`; `docs/capcom/elicitation-policy.md`; `src/model-affinity/`	v1.49.576+ (HB-10 backlog slot); vision-to-mission discipline now	`M6/ADR-2604.21380.md`
2604.21751	M6	Why are all LLMs Obsessed with Japanese Culture? On the Hidden Cultural and Regional Biases of LLMs	ADOPT	Future `tests/cultural-sensitivity/croq-pacific-rim.csv`; cultural-sensitivity chip	v1.49.575 (data drafting); v1.49.576+ (harness wiring)	`M6/ADR-2604.21751.md`
2604.21829	M6	Black-Box Skill Stealing Attack from Proprietary LLM Agents	TRACK	`docs/skill-economy/d-hub-threat-model.md`; `src/memory/grove-formalism/`	v1.49.576+ (HB-11 backlog slot); threat-model doc may land earlier	`M6/ADR-2604.21829.md`